Module 2: Chapters

- Determinants
- Point and Straight Line
- Circle

DETERMINANT

1. INTRODUCTION :

If the equations $a_1x + b_1 = 0$, $a_2x + b_2 = 0$ are satisfied by the same value of x, then $a_1b_2 - a_2b_1 = 0$. The expression $a_1b_2 - a_2b_1$ is called a determinant of the second order, and is denoted by :

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$$

A determinant of second order consists of two rows and two columns.

Next consider the system of equations $a_1x + b_1y + c_1 = 0$, $a_2x + b_2y + c_2 = 0$, $a_3x + b_3y + c_3 = 0$

If these equations are satisfied by the same values of x and y, then on eliminating x and y we get.

$$a_1(b_2c_3 - b_3c_2) + b_1(c_2a_3 - c_3a_2) + c_1(a_2b_3 - a_3b_2) = 0$$

The expression on the left is called a determinant of the third order, and is denoted by

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

A determinant of third order consists of three rows and three columns.

2. VALUE OF A DETERMINANT :

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix} = a_1(b_2c_3 - b_3c_2) - b_1(a_2c_3 - a_3c_2) + c_1(a_2b_3 - a_3b_2)$$

Note : Sarrus diagram to get the value of determinant of order three :

–ve –ve –ve

$$D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \left|\begin{matrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{matrix}\right| \begin{matrix} a_1 & b_1 \\ a_2 & b_2 \\ a_3 & b_3 \end{matrix} = (a_1b_2c_3 + a_2b_3c_1 + a_3b_1c_2) - (a_3b_2c_1 + a_2b_1c_3 + a_1b_3c_2)$$

+ve +ve +ve

Note that the product of the terms in first bracket (i.e. $a_1a_2a_3b_1b_2b_3c_1c_2c_3$) is same as the product of the terms in second bracket.

Illustration 1 : The value of $\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix}$ is -

(A) 213 (B) – 231 (C) 231 (D) 39

Solution :

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix} = 1\begin{vmatrix} 3 & 6 \\ -7 & 9 \end{vmatrix} - 2\begin{vmatrix} -4 & 6 \\ 2 & 9 \end{vmatrix} + 3\begin{vmatrix} -4 & 3 \\ 2 & -7 \end{vmatrix}$$

$= (27 + 42) - 2(-36 - 12) + 3(28 - 6) = 231$

Alternative : By sarrus diagram

$$\begin{vmatrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{vmatrix} = \left|\begin{matrix} 1 & 2 & 3 \\ -4 & 3 & 6 \\ 2 & -7 & 9 \end{matrix}\right| \begin{matrix} 1 & 2 \\ -4 & 3 \\ 2 & -7 \end{matrix}$$

$= (27 + 24 + 84) - (18 - 42 - 72) = 135 - (18 - 114) = 231$ **Ans. (C)**

3. MINORS & COFACTORS :

The minor of a given element of determinant is the determinant obtained by deleting the row & the column in which the given element stands.

For example, the minor of a_1 in $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ is $\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix}$ & the minor of b_2 is $\begin{vmatrix} a_1 & c_1 \\ a_3 & c_3 \end{vmatrix}$.

Hence a determinant of order three will have "9 minors".

If M_{ij} represents the minor of the element belonging to i^{th} row and j^{th} column then the cofactor of that element is given by : $C_{ij} = (-1)^{i+j}.\, M_{ij}$

Illustration 2 : Find the minors and cofactors of elements '–3', '5', '–1' & '7' in the determinant $\begin{vmatrix} 2 & -3 & 1 \\ 4 & 0 & 5 \\ -1 & 6 & 7 \end{vmatrix}$

Solution : Minor of $-3 = \begin{vmatrix} 4 & 5 \\ -1 & 7 \end{vmatrix} = 33$; Cofactor of $-3 = -33$

Minor of $5 = \begin{vmatrix} 2 & -3 \\ -1 & 6 \end{vmatrix} = 9$; Cofactor of $5 = -9$

Minor of $-1 = \begin{vmatrix} -3 & 1 \\ 0 & 5 \end{vmatrix} = -15$; Cofactor of $-1 = -15$

Minor of $7 = \begin{vmatrix} 2 & -3 \\ 4 & 0 \end{vmatrix} = 12$; Cofactor of $7 = 12$

4. EXPANSION OF A DETERMINANT IN TERMS OF THE ELEMENTS OF ANY ROW OR COLUMN:

Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

(i) The sum of the product of elements of any row (column) with their corresponding cofactors is always equal to the value of the determinant.

D can be expressed in any of the six forms :

$a_1A_1 + b_1B_1 + c_1C_1$, $\quad a_1A_1 + a_2A_2 + a_3A_3$,

$a_2A_2 + b_2B_2 + c_2C_2$, $\quad b_1B_1 + b_2B_2 + b_3B_3$,

$a_3A_3 + b_3B_3 + c_3C_3$, $\quad c_1C_1 + c_2C_2 + c_3C_3$,

where A_i, B_i & C_i $(i = 1,2,3)$ denote cofactors of a_i, b_i & c_i respectively.

(ii) The sum of the product of elements of any row (column) with the cofactors of other row (column) is always equal to zero.

Hence,

$a_2A_1 + b_2B_1 + c_2C_1 = 0$,

$b_1A_1 + b_2A_2 + b_3A_3 = 0$ and so on.

where A_i, B_i & C_i $(i = 1,2,3)$ denote cofactors of a_i, b_i & c_i respectively.

Do yourself -1 :

1. Find minors & cofactors of elements '6', '5', '0' & '4' of the determinant $\begin{vmatrix} 2 & 1 & 3 \\ 6 & 5 & 7 \\ 3 & 0 & 4 \end{vmatrix}$.

2. Calculate the value of the determinant $\begin{vmatrix} 5 & -3 & 7 \\ -2 & 4 & -8 \\ 9 & 3 & -10 \end{vmatrix}$

3. The value of the determinant $\begin{vmatrix} a & b & 0 \\ 0 & a & b \\ b & 0 & a \end{vmatrix}$ is equal to -

(A) $a^3 - b^3$ (B) $a^3 + b^3$ (C) 0 (D) none of these

4. Find the value of 'k', if $\begin{vmatrix} 1 & 2 & 0 \\ 2 & 3 & 1 \\ 3 & k & 2 \end{vmatrix} = 4$

5. Which of the following is correct?

(A) $\begin{vmatrix} 2 & 4 \\ -5 & -1 \end{vmatrix} = 17$ (B) $\begin{vmatrix} 3 & -1 & -2 \\ 0 & 0 & -1 \\ 3 & -5 & 0 \end{vmatrix} = 12$

(C) $\begin{vmatrix} 3 & -4 & 5 \\ 1 & 1 & -2 \\ 2 & 3 & 1 \end{vmatrix} = 46$ (D) $\begin{vmatrix} 2 & -1 & -2 \\ 0 & 2 & -1 \\ 3 & -5 & 0 \end{vmatrix} = 4$

Comprehension (For Q.6 to 8)

Consider the determinant, $\Delta = \begin{vmatrix} p & q & r \\ x & y & z \\ \ell & m & n \end{vmatrix}$

M_{ij} denotes the minor of an element in i^{th} row and j^{th} column

C_{ij} denotes the cofactor of an element in i^{th} row and j^{th} column

6. The value of $p.C_{21} + q.C_{22} + r.C_{23}$ is

(A) 0 (B) $-\Delta$ (C) Δ (D) None of these

7. The value of $x.C_{21} + y.C_{22} + z.C_{23}$ is

(A) 0 (B) $-\Delta$ (C) Δ (D) None of these

8. The value of $q.M_{12} - y.M_{22} + m.M_{32}$ is

(A) 0 (B) $-\Delta$

(C) $x.M_{21} - y.M_{22} + z.M_{23}$ (D) Both (B) & (C)

9. If the minor of three-one element (i.e. M_{31}) in the determinant $\begin{vmatrix} 0 & 1 & \sec\alpha \\ \tan\alpha & -\sec\alpha & \tan\alpha \\ 1 & 0 & 1 \end{vmatrix}$ is 1 then find the value of $\alpha.(0 \le \alpha \le \pi)$

(A) 0 (B) $\frac{3\pi}{4}$ (C) π (D) All of these

10. If $f(x) = \begin{vmatrix} x-1 & x+1 \\ x-2 & x+2 \end{vmatrix}$, then find the value of $\begin{vmatrix} f(-1) & f(-2) \\ f(1) & f(2) \end{vmatrix}$.

(A) 0 (B) 1 (C) 3 (D) 4

5. PROPERTIES OF DETERMINANTS :

(a) The value of a determinant remains unaltered, if the rows & columns are inter-changed,

e.g. if $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

(b) If any two rows (or columns) of a determinant be interchanged, the value of determinant is changed in sign only. e.g.

Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ & $D_1 = \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$. Then $D_1 = -D$.

(c) If all the elements of a row (or column) are zero, then the value of the determinant is zero.

(d) If all the elements of any row (or column) are multiplied by the same number, then the determinant is multiplied by that number.

e.g. If $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D_1 = \begin{vmatrix} Ka_1 & Kb_1 & Kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$. Then $D_1 = KD$

(e) If all the elements of a row (or column) are proportional (or identical) to the element of any other row, then the determinant vanishes, i.e. its value is zero.

e.g. If $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix} \Rightarrow D = 0$; If $D_1 = \begin{vmatrix} a_1 & b_1 & c_1 \\ ka_1 & kb_1 & kc_1 \\ a_3 & b_3 & c_3 \end{vmatrix} \Rightarrow D_1 = 0$

Illustration 3 : Prove that $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

Solution : $D = \begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix} = \begin{vmatrix} a & x & p \\ b & y & q \\ c & z & r \end{vmatrix}$ (By interchanging rows & columns)

$= -\begin{vmatrix} x & a & p \\ y & b & q \\ z & c & r \end{vmatrix}$ $(C_1 \leftrightarrow C_2)$

$= \begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$ $(R_1 \leftrightarrow R_2)$

Illustration 4 : Find the value of the determinant $\begin{vmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix}$

Solution : $D = \begin{vmatrix} a^2 & ab & ac \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix} = a\begin{vmatrix} a & b & c \\ ab & b^2 & bc \\ ac & bc & c^2 \end{vmatrix} = abc\begin{vmatrix} a & b & c \\ a & b & c \\ a & b & c \end{vmatrix} = 0$

Since all rows are same, hence value of the determinant is zero.

Do yourself -2 :

1. Without expanding the determinant prove that $\begin{vmatrix} a & p & \ell \\ b & q & m \\ c & r & n \end{vmatrix} + \begin{vmatrix} r & n & c \\ q & m & b \\ p & \ell & a \end{vmatrix} = 0$

2. If $D = \begin{vmatrix} \alpha & \beta \\ \gamma & \delta \end{vmatrix}$, then $\begin{vmatrix} 2\alpha & 2\beta \\ 2\gamma & 2\delta \end{vmatrix}$ is equal to -

(A) D (B) 2D (C) 4D (D) 16D

3. Find the value of $\begin{vmatrix} 53 & 106 & 159 \\ 52 & 65 & 91 \\ 102 & 153 & 221 \end{vmatrix}$.

4. Find the value of $\begin{vmatrix} 13 & 3 & 23 \\ 30 & 7 & 53 \\ 39 & 9 & 70 \end{vmatrix}$

(A) 1 (B) 3 (C) 5 (D) 8

5. The determinant $\begin{vmatrix} a & b & c \\ x & y & z \\ p & q & r \end{vmatrix}$ is same as:

(A) $\begin{vmatrix} a & b & p \\ y & b & q \\ z & c & r \end{vmatrix}$ (B) $\begin{vmatrix} x & b & q \\ y & a & p \\ z & c & r \end{vmatrix}$ (C) $\begin{vmatrix} x & z & y \\ p & r & q \\ a & c & b \end{vmatrix}$ (D) $\begin{vmatrix} y & b & q \\ x & a & p \\ z & c & r \end{vmatrix}$

(f) If each element of any row (or column) is expressed as a sum of two (or more) terms, then the determinant can be expressed as the sum of two (or more) determinants.

e.g. $\begin{vmatrix} a_1 + x & b_1 + y & c_1 + z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

Note that : If $D_r = \begin{vmatrix} f(r) & g(r) & h(r) \\ a & b & c \\ a_1 & b_1 & c_1 \end{vmatrix}$

where $r \in N$ and a, b, c, a_1, b_1, c_1 are constants, then

$$\sum_{r=1}^{n} D_r = \begin{vmatrix} \sum_{r=1}^{n} f(r) & \sum_{r=1}^{n} g(r) & \sum_{r=1}^{n} h(r) \\ a & b & c \\ a_1 & b_1 & c_1 \end{vmatrix}$$

(g) **Row - column operation :** The value of a determinant remains unaltered under a column (C_i) operation of the form $C_i \to C_i + \alpha C_j + \beta C_k$ $(j, k \neq i)$ or row (R_i) operation of the form $R_i \to R_i + \alpha R_j + \beta R_k$ $(j, k \neq i)$. In other words, the value of a determinant is not altered by adding the elements of any row (or column) to the same multiples of the corresponding elements of any other row (or column)

e.g. Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

$$D = \begin{vmatrix} a_1 + \alpha a_2 & b_1 + \alpha b_2 & c_1 + \alpha c_2 \\ a_2 & b_2 & c_2 \\ a_3 + \beta a_2 & b_3 + \beta b_2 & c_3 + \beta c_2 \end{vmatrix} \quad (R_1 \to R_1 + \alpha R_2; R_3 \to R_3 + \beta R_2)$$

Note :

(i) By using the operation $R_i \to xR_i + yR_j + zR_k$ $(j, k \neq i)$, the value of the determinant becomes x times the original one.

(ii) While applying this property **ATLEAST ONE ROW (OR COLUMN)** must remain unchanged.

Illustration 5 : If $D_r = \begin{vmatrix} r & r^3 & 2 \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix}$, find $\sum_{r=0}^{n} D_r$.

Solution :

$$\sum_{r=0}^{n} D_r = \begin{vmatrix} \sum_{r=0}^{n} r & \sum_{r=0}^{n} r^3 & \sum_{r=0}^{n} 2 \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix} = \begin{vmatrix} \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \\ n & n^3 & 2n \\ \frac{n(n+1)}{2} & \left(\frac{n(n+1)}{2}\right)^2 & 2(n+1) \end{vmatrix} = 0$$ **Ans.**

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Illustration 6 : If $\begin{vmatrix} 3^2+k & 4^2 & 3^2+3+k \\ 4^2+k & 5^2 & 4^2+4+k \\ 5^2+k & 6^2 & 5^2+5+k \end{vmatrix} = 0$, then the value of k is-

(A) 2 (B) 1 (C) –1 (D) 0

Solution : Applying $(C_3 \to C_3 - C_1)$

$$D = \begin{vmatrix} 3^2+k & 4^2 & 3 \\ 4^2+k & 5^2 & 4 \\ 5^2+k & 6^2 & 5 \end{vmatrix} = 0$$

$$\Rightarrow \begin{vmatrix} 9+k & 16 & 3 \\ 7 & 9 & 1 \\ 9 & 11 & 1 \end{vmatrix} = 0 \qquad (R_3 \to R_3 - R_2;\ R_2 \to R_2 - R_1)$$

$\Rightarrow$ k – 1 = 0 $\Rightarrow$ k = 1 **Ans. (B)**

Do yourself - 3 :

1. Solve for x : $\begin{vmatrix} x & 2 & 0 \\ 2+x & 5 & -1 \\ 5-x & 1 & 2 \end{vmatrix} = 0$

2. If $D_r = \begin{vmatrix} 2r & 1 & n \\ 1 & -2 & 3 \\ 3 & 2 & 1 \end{vmatrix}$, then find the value of $\sum_{r=1}^{n} D_r$.

3. If A + B + C = π, then the value of the determinant $D = \begin{vmatrix} \sin^2 A & \cot A & 1 \\ \sin^2 B & \cot B & 1 \\ \sin^2 C & \cot C & 1 \end{vmatrix}$ is equal to

(A) 1 (B) –1 (C) 0 (D) none of these

4. The value of the determinant $\begin{vmatrix} \sin^2 13° & \sin^2 77° & \tan 135° \\ \sin^2 77° & \tan 135° & \sin^2 13° \\ \tan 135° & \sin^2 13° & \sin^2 77° \end{vmatrix}$ is equal to

(A) –1 (B) 0 (C) 3 (D) 7

5. If $D_r = \begin{vmatrix} 2^{r-1} & 2\left(3^{r-1}\right) & 4\left(5^{r-1}\right) \\ x & y & z \\ 2^n-1 & 3^n-1 & 5^n-1 \end{vmatrix}$ then the value of $\sum_{r=1}^{n} D_r$ is

(h) **Factor theorem :** If the elements of a determinant D are rational integral functions of x and two rows (or columns) become identical when x = a then (x – a) is a factor of D.

Note that if r rows become identical when a is substituted for x, then $(x-a)^{r-1}$ is a factor of D.

Illustration 7 : Prove that $\begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = m(x-a)(x-b)$

Solution : Using factor theorem,

Put x = a

$$D = \begin{vmatrix} a & a & a \\ m & m & m \\ b & a & b \end{vmatrix} = 0$$

Since R_1 and R_2 are proportional which makes D = 0, therefore (x – a) is a factor of D.

Similarly, by putting x = b, D becomes zero, therefore (x – b) is a factor of D.

$$D = \begin{vmatrix} a & a & x \\ m & m & m \\ b & x & b \end{vmatrix} = \lambda(x-a)(x-b) \qquad \text{..........(i)}$$

To get the value of λ, put x = 0 in equation (i)

$$\begin{vmatrix} a & a & 0 \\ m & m & m \\ b & 0 & b \end{vmatrix} = \lambda ab$$

$amb = \lambda ab \Rightarrow \lambda = m$

$\therefore \quad D = m(x-a)(x-b)$

Do yourself - 4 :

1. Without expanding the determinant prove that $\begin{vmatrix} 1 & a & bc \\ 1 & b & ca \\ 1 & c & ab \end{vmatrix} = (a-b)(b-c)(c-a)$

2. Using factor theorem, find the solution set of the equation $\begin{vmatrix} 1 & 4 & 20 \\ 1 & -2 & 5 \\ 1 & 2x & 5x^2 \end{vmatrix} = 0$

3. $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix}$ is equals to

(A) (a – b) (b – c) (c – a) (ab + bc + ca) (B) (a – b) (b – c) (c – a)

(C) (ab + bc + ca) (D) None of these

4. $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix}$ is equals to

(A) (a – b) (b – c) (c – a) (ab + bc + ca)
(B) (a – b) (b – c) (c – a)
(C) (a – b) (b – c) (c – a) (a + b + c)
(D) None of these

5. $\begin{vmatrix} a & b+c & a^2 \\ b & c+a & b^2 \\ c & a+b & c^2 \end{vmatrix}$ is equals to

(A) (a – b) (b – c) (c – a)
(B) –(a + b + c) (a – b) (b – c) (c – a)
(C) (a – b) (b – c) (c – a) (a + b + c)
(D) None of these

6. MULTIPLICATION OF TWO DETERMINANTS :

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} l_1 & m_1 \\ l_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1 l_1 + b_1 l_2 & a_1 m_1 + b_1 m_2 \\ a_2 l_1 + b_2 l_2 & a_2 m_1 + b_2 m_2 \end{vmatrix}$$

Similarly two determinants of order three are multiplied.

(a) Here we have multiplied row by column. We can also multiply row by row, column by row and column by column.

(b) If D_1 is the determinant formed by replacing the elements of determinant D of order n by their corresponding cofactors then $D_1 = D^{n-1}$

Illustration 8 : Let α & β be the roots of equation $ax^2 + bx + c = 0$ and $S_n = \alpha^n + \beta^n$ for $n \ge 1$. Evaluate the value of the determinant $\begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix}$.

Solution :

$$D = \begin{vmatrix} 3 & 1+S_1 & 1+S_2 \\ 1+S_1 & 1+S_2 & 1+S_3 \\ 1+S_2 & 1+S_3 & 1+S_4 \end{vmatrix} = \begin{vmatrix} 1+1+1 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix} \times \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \alpha^2 & \beta^2 \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \end{vmatrix}^2 = [(1-\alpha)(1-\beta)(\alpha-\beta)]^2$$

$D = (\alpha - \beta)^2 (\alpha + \beta - \alpha\beta - 1)^2$

$\because$ α & β are roots of the equation $ax^2 + bx + c = 0$

$$\Rightarrow \quad \alpha+\beta = \frac{-b}{a} \quad \& \quad \alpha\beta = \frac{c}{a} \quad \Rightarrow \quad |\alpha-\beta| = \frac{\sqrt{b^2-4ac}}{|a|}$$

$$D = \frac{(b^2-4ac)}{a^2}\left(\frac{a+b+c}{a}\right)^2 = \frac{(b^2-4ac)(a+b+c)^2}{a^4}$$ **Ans.**

Do yourself - 5 :

1. If the determinant $D=\begin{vmatrix} 1 & 1 & 1 \\ \alpha+\beta & \alpha^2+\beta^2 & 2\alpha\beta \\ \alpha+\beta & 2\alpha\beta & \alpha^2+\beta^2 \end{vmatrix}$ and $D_1=\begin{vmatrix} 1 & 0 & 0 \\ 0 & \alpha & \beta \\ 0 & \beta & \alpha \end{vmatrix}$, then find the determinant D_2 such that $D_2=\dfrac{D}{D_1}$.

2. If $D_1=\begin{vmatrix} ab^2-ac^2 & bc^2-a^2b & a^2c-b^2c \\ ac-ab & ab-bc & bc-ac \\ c-b & a-c & b-a \end{vmatrix}$ & $D_2=\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix}$, then D_1D_2 is equal to -

(A) 0 (B) D_1^2 (C) D_2^2 (D) D_2^3

3. Factorize : $\begin{vmatrix} 3 & a+b+c & a^3+b^3+c^3 \\ a+b+c & a^2+b^2+c^2 & a^4+b^4+c^4 \\ a^2+b^2+c^2 & a^3+b^3+c^3 & a^5+b^5+c^5 \end{vmatrix}$

(A) $[(a-b)^2(b-c)^2(c-a)^2(a+b+c)]$ (B) $[(a-b)(b-c)(c-a)(a+b+c)]$
(C) $[(a-b)^2(b-c)^2(c-a)^2]$ (D) None of these

4. If $\begin{vmatrix} 1 & x & x^2 \\ x & x^2 & 1 \\ x^2 & 1 & x \end{vmatrix}=3$, then find the value of $\begin{vmatrix} x^3-1 & 0 & x-x^4 \\ 0 & x-x^4 & x^3-1 \\ x-x^4 & x^3-1 & 0 \end{vmatrix}$

(A) 5 (B) 8 (C) 9 (D) 3

5. $\begin{vmatrix} a^2+b^2+c^2 & bc+ca+ab & bc+ca+ab \\ bc+ca+ab & a^2+b^2+c^2 & bc+ca+ab \\ bc+ca+ab & bc+ca+ab & a^2+b^2+c^2 \end{vmatrix}$ is always

(A) non-negative (B) 0 (C) negative (D) Can't say

7. SPECIAL DETERMINANTS :

(a) Cyclic Determinant :

The elements of the rows (or columns) are in cyclic arrangement.

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}=-(a^3+b^3+c^3-3abc)=-(a+b+c)(a^2+b^2+c^2-ab-bc-ac)$$

$$=-\frac{1}{2}(a+b+c)\times\{(a-b)^2+(b-c)^2+(c-a)^2\}$$

$= -(a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$, where ω,ω^2 are cube roots of unity

(b) Other Important Determinants :

(i) $\begin{vmatrix} 0 & b & -c \\ -b & 0 & a \\ c & -a & 0 \end{vmatrix} = 0$

(ii) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ bc & ac & ab \end{vmatrix} = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$

(iii) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(a+b+c)$

(iv) $\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca)$

(v) $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^4 & b^4 & c^4 \end{vmatrix} = (a-b)(b-c)(c-a)(a^2+b^2+c^2-ab-bc-ca)$

Illustration 9 : Prove that $\begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix} = -(1-\alpha^3)^2$.

Solution : This is a cyclic determinant.

$$\Rightarrow \begin{vmatrix} 1 & \alpha & \alpha^2 \\ \alpha & \alpha^2 & 1 \\ \alpha^2 & 1 & \alpha \end{vmatrix} = -(1+\alpha+\alpha^2)(1+\alpha^2+\alpha^4-\alpha-\alpha^2-\alpha^3)$$

$= -(1+\alpha+\alpha^2)(-\alpha+1-\alpha^3+\alpha^4)$

$= -(1+\alpha+\alpha^2)(1-\alpha)^2(1+\alpha+\alpha^2)$

$= -(1-\alpha)^2(1+\alpha+\alpha^2)^2 = -(1-\alpha^3)^2$

Do yourself - 6 :

1. The value of the determinant $\begin{vmatrix} ka & k^2+a^2 & 1 \\ kb & k^2+b^2 & 1 \\ kc & k^2+c^2 & 1 \end{vmatrix}$ is

(A) $k(a+b)(b+c)(c+a)$ (B) $kabc(a^2+b^2+c^2)$

(C) $k(a-b)(b-c)(c-a)$ (D) $k(a+b-c)(b+c-a)(c+a-b)$

2. Find the value of the determinant $\begin{vmatrix} a^2+b^2 & a^2-c^2 & a^2-c^2 \\ -a^2 & 0 & c^2-a^2 \\ b^2 & -c^2 & b^2 \end{vmatrix}$.

3. Prove that $\begin{vmatrix} a & b & c \\ bc & ca & ab \\ b+c & c+a & a+b \end{vmatrix} = (a+b+c)(a-b)(b-c)(c-a)$.

4. Let a, b, c be such that $b(a+c) \neq 0$. If $\begin{vmatrix} a & a+1 & a-1 \\ -b & b+1 & b-1 \\ c & c-1 & c+1 \end{vmatrix} + \begin{vmatrix} a+1 & b+1 & c-1 \\ a-1 & b-1 & c+1 \\ (-1)^{n+2}a & (-1)^{n+1}b & (-1)^{n}c \end{vmatrix} = 0$, then the value of n is :-

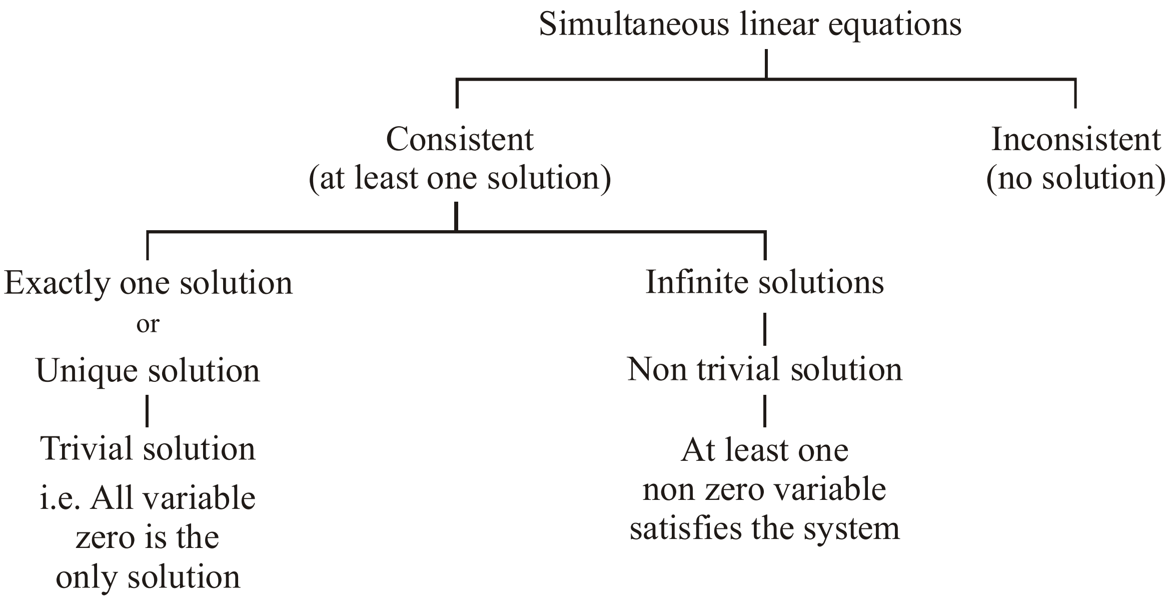
(A) Any odd integer (B) Any integer

(C) Zero (D) Any even integer

5. The product of all the values of α for which equations $(a-\alpha)x + by + c = 0$, $cx + (a-\alpha)y + b = 0$ and $bx + cy + a - \alpha = 0$ are consistent, when $a,b,c > 0$, is -

(A) zero (B) positive (C) negative (D) one

8. CRAMER'S RULE (SYSTEM OF LINEAR EQUATIONS) :



(a) Equations involving two variables :

(i) Consistent Equations : Definite & unique solution (Intersecting lines)

(ii) Inconsistent Equations : No solution (Parallel lines)

(iii) Dependent Equations : Infinite solutions (Identical lines)

Let, $a_1x + b_1y + c_1 = 0$

$a_2x + b_2y + c_2 = 0$ then :

(1) $\frac{a_1}{a_2} \neq \frac{b_1}{b_2}$ $\Rightarrow$ Given equations are consistent with unique solution

(2) $\frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$ $\Rightarrow$ Given equations are inconsistent

(3) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$ $\Rightarrow$ Given equations are consistent with infinite solutions

(b) Equations Involving Three variables :

Let $a_1x + b_1y + c_1z = d_1$ .... (i)

$a_2x + b_2y + c_2z = d_2$ .... (ii)

$a_3x + b_3y + c_3z = d_3$ .... (iii)

Then, $x = \frac{D_1}{D}$, $y = \frac{D_2}{D}$, $z = \frac{D_3}{D}$.

Where $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$; $D_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$; $D_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$ & $D_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$

Note :

(i) If $D \neq 0$ and atleast one of $D_1, D_2, D_3 \neq 0$, then the given system of equations is consistent and has unique non trivial solution.

(ii) If $D \neq 0$ & $D_1 = D_2 = D_3 = 0$, then the given system of equations is consistent and has trivial solution only.

(iii) If $D = 0$ but atleast one of D_1, D_2, D_3 is not zero then the equations are inconsistent and have no solution.

(iv) If $D = D_1 = D_2 = D_3 = 0$, then the given system of equations may have infinite or no solution.

Note that In case $\left.\begin{matrix} a_1x + b_1y + c_1z = d_1 \\ a_1x + b_1y + c_1z = d_2 \\ a_1x + b_1y + c_1z = d_3 \end{matrix}\right\}$ (Atleast two of d_1, d_2 & d_3 are not equal)

$D = D_1 = D_2 = D_3 = 0$. But these three equations represent three parallel planes. Hence the system is inconsistent.

(c) Homogeneous system of linear equations :

If x, y, z are not all zero, the condition for

$a_1x + b_1y + c_1z = 0$
$a_2x + b_2y + c_2z = 0$
$a_3x + b_3y + c_3z = 0$

to be consistent in x, y, z is that $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0.$

Remember that if a given system of linear equations have **Only Zero** Solution for all its variables then the given equations are said to have **TRIVIAL SOLUTION**.

Illustration 10 : Find the nature of solution for the given system of equations :
$x + 2y + 3z = 1;\ \ 2x + 3y + 4z = 3;\ \ 3x + 4y + 5z = 0$

Solution :
$$D = \begin{vmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix} = 0$$

$$\text{Now, } D_1 = \begin{vmatrix} 1 & 2 & 3 \\ 3 & 3 & 4 \\ 0 & 4 & 5 \end{vmatrix} = 5$$

$\because$ $D = 0$ but $D_1 \neq 0$

Hence no solution. **Ans.**

Illustration 11 : Find the value of λ, if the following equations are consistent :
$x + y - 3 = 0;\ (1 + \lambda)x + (2 + \lambda)y - 8 = 0;\ x - (1 + \lambda)y + (2 + \lambda) = 0$

Solution : The given equations in two unknowns are consistent, then $\Delta = 0$

$$\text{i.e. } \begin{vmatrix} 1 & 1 & -3 \\ 1+\lambda & 2+\lambda & -8 \\ 1 & -(1+\lambda) & 2+\lambda \end{vmatrix} = 0$$

Applying $C_2 \to C_2 - C_1$ and $C_3 \to C_3 + 3C_1$

$$\therefore \begin{vmatrix} 1 & 0 & 0 \\ 1+\lambda & 1 & 3\lambda-5 \\ 1 & -2-\lambda & 5+\lambda \end{vmatrix} = 0$$

$\Rightarrow$ $(5+\lambda) - (3\lambda-5)(-2-\lambda) = 0$ $\Rightarrow$ $3\lambda^2 + 2\lambda - 5 = 0$

$\therefore$ $\lambda = 1, -5/3$

Illustration 12 : If the system of equations $x + \lambda y + 1 = 0$, $\lambda x + y + 1 = 0$ & $x + y + \lambda = 0$. is consistent, then find the value of λ.

Solution : For consistency of the given system of equations

$$D = \begin{vmatrix} 1 & \lambda & 1 \\ \lambda & 1 & 1 \\ 1 & 1 & \lambda \end{vmatrix} = 0$$

$\Rightarrow$ $3\lambda = 1 + 1 + \lambda^3$ or $\lambda^3 - 3\lambda + 2 = 0$

$\Rightarrow$ $(\lambda-1)^2(\lambda + 2) = 0 \Rightarrow \lambda = 1$ or $\lambda = -2$ **Ans.**

Do yourself -7 :

1. Find nature of solution for given system of equations

$2x + y + z = 3$; $x + 2y + z = 4$; $3x + z = 2$

2. If the system of equations $x + y + z = 2$, $2x + y - z = 3$ & $3x + 2y + kz = 4$ has a unique solution, then

(A) $k \neq 0$ (B) $-1 < k < 1$ (C) $-2 < k < 1$ (D) $k = 0$

3. The system of equations $\lambda x + y + z = 0$, $-x + \lambda y + z = 0$ & $-x - y + \lambda z = 0$ has a non-trivial solution, then possible values of λ are -

(A) 0 (B) 1 (C) –3 (D) $\sqrt{3}$

4. Consider the system of linear equations : $x_1 + 2x_2 + x_3 = 3$, $2x_1 + 3x_2 + x_3 = 3$, $3x_1 + 5x_2 + 2x_3 = 1$ The system has

(A) Infinite number of solutions (B) Exactly 3 solutions

(C) A unique solution (D) No solution

5. The number of values of k for which the linear equations $4x + ky + 2z = 0$, $kx + 4y + z = 0$, $2x + 2y + z = 0$ possess a non-zero solution is :

(A) 1 (B) zero (C) 3 (D) 2

6. If the trivial solution is the only solution of the system of equations $x - ky + z = 0$, $kx + 3y - kz = 0$, $3x + y - z = 0$ Then the set of all values of k is :

(A) $\{2, -3\}$ (B) $R - \{2, -3\}$ (C) $R - \{2\}$ (D) $R - \{-3\}$

7. Solve the following system of equations $x + y + z = 5$, $2x + 2y + 2z = 7$, $3x + 3y + 3z = 6$

(A) Unique solution (B) infinite solution (C) No solution (D) None of these

8. The number of values of k for which the system of equations

$(k + 1)x + 8y = 4k$

$kx + (k + 3)y = 3k - 1$

has infinitely many solutions is

(A) 0 (B) 1 (C) 2 (D) Infinite

9. The values of 'λ' for which the system of equations

$(1 - \lambda)x + 3y - 4z = 0$

$x - (3 + \lambda)y + 5z = 0$

$3x + y - \lambda z = 0$

possesses non-trivial solution, is/are

(A) –1 (B) 0 (C) 1 (D) 2

EXERCISE (O-1)

Straight Objective Type

1. $\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix}$ equals -

(A) $x^2y^2z^2$ (B) $4x^2y^2z^2$ (C) xyz (D) $4xyz$

DT0001

2. If $\begin{vmatrix} 1 & 3 & 4 \\ 1 & x-1 & 2x+2 \\ 2 & 5 & 9 \end{vmatrix} = 0$, then x is equal to-

(A) 2 (B) 1 (C) 4 (D) 0

DT0002

3. If $px^4 + qx^3 + rx^2 + sx + t = \begin{vmatrix} x^2+3x & x-1 & x+3 \\ x+1 & 2-x & x-3 \\ x-3 & x+4 & 3x \end{vmatrix}$ then t is equal to -

(A) 33 (B) 0 (C) 21 (D) none

DT0003

4. There are two numbers x making the value of the determinant $\begin{vmatrix} 1 & -2 & 5 \\ 2 & x & -1 \\ 0 & 4 & 2x \end{vmatrix}$ equal to 86. The sum of these two numbers, is-

(A) –4 (B) 5 (C) –3 (D) 9

DT0004

5. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_2, B_2, C_2 are respectively cofactors of a_2, b_2, c_2 then $a_1A_2 + b_1B_2 + c_1C_2$ is equal to-

(A) $-\Delta$ (B) 0 (C) Δ (D) none of these

DT0005

6. If in the determinant $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, A_1, B_1, C_1 etc. be the co-factors of a_1, b_1, c_1 etc., then which of the following relations is incorrect-

(A) $a_1A_1 + b_1B_1 + c_1C_1 = \Delta$

(B) $a_2A_2 + b_2B_2 + c_2C_2 = \Delta$

(C) $a_3A_3 + b_3B_3 + c_3C_3 = \Delta$

(D) $a_1A_2 + b_1B_2 + c_1C_2 = \Delta$ **DT0006**

7. If $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and A_1, B_1, C_1 denote the co-factors of a_1, b_1, c_1 respectively, then the value of the determinant $\begin{vmatrix} A_1 & B_1 & C_1 \\ A_2 & B_2 & C_2 \\ A_3 & B_3 & C_3 \end{vmatrix}$ is -

(A) Δ (B) Δ^2 (C) Δ^3 (D) 0

DT0007

8. If a, b, c are in AP, then $\begin{vmatrix} x+1 & x+2 & x+a \\ x+2 & x+3 & x+b \\ x+3 & x+4 & x+c \end{vmatrix}$ equals -

(A) a + b + c (B) x + a + b + c (C) 0 (D) none of these

DT0008

9. For positive numbers x, y and z, the numerical value of the determinant $\begin{vmatrix} 1 & \log_x y & \log_x z \\ \log_y x & 1 & \log_y z \\ \log_z x & \log_z y & 1 \end{vmatrix}$ is-

(A) 0 (B) log xyz (C) log(x + y + z) (D) logx logy logz

DT0009

10. Let a determinant is given by $A = \begin{vmatrix} a & b & c \\ p & q & r \\ x & y & z \end{vmatrix}$ and suppose A = 6. If $B = \begin{vmatrix} p+x & q+y & r+z \\ a+x & b+y & c+z \\ a+p & b+q & c+r \end{vmatrix}$ then

(A) B = 6 (B) B = – 6 (C) B = 12 (D) B = – 12

DT0010

11. The value of an odd order determinant in which $a_{ij} + a_{ji} = 0 \ \forall$ i, j is -

(A) perfect square (B) negative (C) ± 1 (D) 0

DT0011

12. If $S_r = \begin{vmatrix} 2r & x & n(n+1) \\ 6r^2-1 & y & n^2(2n+3) \\ 4r^3-2nr & z & n^3(n+1) \end{vmatrix}$, then $\sum_{r=1}^{n} S_r$ does not depend on -

(A) x (B) y (C) n (D) all of these

DT0012

13. If a, b, c > 0 and x, y, z $\in$ R, then the determinant $\begin{vmatrix} (a^x+a^{-x})^2 & (a^x-a^{-x})^2 & 1 \\ (b^y+b^{-y})^2 & (b^y-b^{-y})^2 & 1 \\ (c^z+c^{-z})^2 & (c^z-c^{-z})^2 & 1 \end{vmatrix}$ is equal to -

(A) $a^x b^y c^x$ (B) $a^{-x} b^{-y} c^{-z}$ (C) $a^{2x} b^{2y} c^{2z}$ (D) zero

DT0013

14. If a, b, c are sides of a scalene triangle, then the value of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$ is :

(A) non-negative (B) negative (C) positive (D) non-positive

DT0014

15. The value of k for which the set of equations $3x+ky-2z=0$, $x+ky+3z=0$ and $2x+3y-4z=0$ has a non-trivial solution is-

(A) 15 (B) 16 (C) 31/2 (D) 33/2

DT0015

16. If the system of linear equations

$$x_1 + 2x_2 + 3x_3 = 6$$
$$x_1 + 3x_2 + 5x_3 = 9$$
$$2x_1 + 5x_2 + ax_3 = b$$

is consistent and has infinite number of solutions, then :-

(A) $a \in R - \{8\}$ and $b \in R - \{15\}$ (B) a = 8, b can be any real number

(C) a = 8, b = 15 (D) b = 15, a can be any real number

DT0016

17. Consider the system of equations : $x + ay = 0$, $y + az = 0$ and $z + ax = 0$. Then the set of all real values of 'a' for which the system has a unique solution is :

(A) $\{1, -1\}$ (B) $R - \{-1\}$ (C) $\{1, 0, -1\}$ (D) $R - \{1\}$

DT0017

18. Let a, b, c be any real numbers. Suppose that there are real numbers x, y, z not all zero such that $x = cy + bz$, $y = az + cx$ and $z = bx + ay$, then $a^2 + b^2 + c^2 + 2abc$ is equal to

(A) 2 (B) –1 (C) 0 (D) 1

DT0018

19. Let $f(x) = \begin{vmatrix} 1+\sin^2 x & \cos^2 x & 4\sin 2x \\ \sin^2 x & 1+\cos^2 x & 4\sin 2x \\ \sin^2 x & \cos^2 x & 1+4\sin 2x \end{vmatrix}$, then the maximum value of f(x) , is-

(A) 2 (B) 4 (C) 6 (D) 8

DT0019

20. If the determinant $\begin{vmatrix} a+p & \ell+x & u+f \\ b+q & m+y & v+g \\ c+r & n+z & w+h \end{vmatrix}$ splits into exactly K determinants of order 3, each element of which contains only one term, then the value of K, is-

(A) 6 (B) 8 (C) 9 (D) 12

DT0020

21. Let $D_1 = \begin{vmatrix} a & b & a+b \\ c & d & c+d \\ a & b & a-b \end{vmatrix}$ and $D_2 = \begin{vmatrix} a & c & a+c \\ b & d & b+d \\ a & c & a+b+c \end{vmatrix}$ then the value of $\frac{D_1}{D_2}$ where $b \neq 0$ and $ad \neq bc$, is

(A) – 2 (B) 0 (C) – 2b (D) 2b

DT0021

22. If $a^2 + b^2 + c^2 = -2$ and $f(x) = \begin{vmatrix} 1+a^2x & (1+b^2)x & (1+c^2)x \\ (1+a^2)x & 1+b^2x & (1+c^2)x \\ (1+a^2)x & (1+b^2)x & 1+c^2x \end{vmatrix}$ then $f(x)$ is a polynomial of degree-

(A) 0 (B) 1 (C) 2 (D) 3

DT0022

23. The number of real values of x satisfying $\begin{vmatrix} x & 3x+2 & 2x-1 \\ 2x-1 & 4x & 3x+1 \\ 7x-2 & 17x+6 & 12x-1 \end{vmatrix} = 0$ is -

(A) 3 (B) 0 (C) 1 (D) infinite

DT0023

24. The determinant $\begin{vmatrix} \cos(\theta+\phi) & -\sin(\theta+\phi) & \cos 2\phi \\ \sin\theta & \cos\theta & \sin\phi \\ -\cos\theta & \sin\theta & \cos\phi \end{vmatrix}$ is -

(A) 0 (B) independent of θ

(C) independent of ϕ (D) independent of θ & ϕ both

DT0024

25. If the system of equation, $a^2x - ay = 1 - a$ & $bx + (3 - 2b)y = 3 + a$ possess a unique solution $x = 1, y = 1$ than :

(A) a = 1; b = –1 (B) a = –1, b = 1 (C) a = 0, b = 0 (D) none

DT0025

EXERCISE (O-2)

Multiple Correct Answer Type

1. The determinant $\begin{vmatrix} a^2 & a^2-(b-c)^2 & bc \\ b^2 & b^2-(c-a)^2 & ca \\ c^2 & c^2-(a-b)^2 & ab \end{vmatrix}$ is divisible by -

(A) $a + b + c$
(B) $(a + b)(b + c)(c + a)$
(C) $a^2 + b^2 + c^2$
(D) $(a - b)(b - c)(c - a)$

DT0026

2. The value of θ lying between $-\frac{\pi}{4}$ & $\frac{\pi}{2}$ and $0 \le A \le \frac{\pi}{2}$ and satisfying the equation

$$\begin{vmatrix} 1+\sin^2 A & \cos^2 A & 2\sin 4\theta \\ \sin^2 A & 1+\cos^2 A & 2\sin 4\theta \\ \sin^2 A & \cos^2 A & 1+2\sin 4\theta \end{vmatrix} = 0 \text{ are -}$$

(A) $A = \frac{\pi}{4},\quad \theta = -\frac{\pi}{8}$
(B) $A = \frac{3\pi}{8} = \theta$
(C) $A = \frac{\pi}{5},\quad \theta = -\frac{\pi}{8}$
(D) $A = \frac{\pi}{6},\quad \theta = \frac{3\pi}{8}$

DT0027

3. Which of the following determinant(s) vanish(es) ?

(A) $\begin{vmatrix} 1 & bc & bc(b+c) \\ 1 & ca & ca(c+a) \\ 1 & ab & ab(a+b) \end{vmatrix}$

(B) $\begin{vmatrix} 1 & ab & \frac{1}{a}+\frac{1}{b} \\ 1 & bc & \frac{1}{b}+\frac{1}{c} \\ 1 & ca & \frac{1}{c}+\frac{1}{a} \end{vmatrix}$

(C) $\begin{vmatrix} 0 & a-b & a-c \\ b-a & 0 & b-c \\ c-a & c-b & 0 \end{vmatrix}$

(D) $\begin{vmatrix} \log_x xyz & \log_x y & \log_x z \\ \log_y xyz & 1 & \log_y z \\ \log_z xyz & \log_z y & 1 \end{vmatrix}$

DT0028

4. The determinant $\begin{vmatrix} a & b & a\alpha+b \\ b & c & b\alpha+c \\ a\alpha+b & b\alpha+c & 0 \end{vmatrix}$ is equal to zero, if -

(A) a, b, c are in AP
(B) a, b, c are in GP
(C) α is a root of the equation $ax^2+bx+c=0$
(D) $(x-\alpha)$ is a factor of $ax^2 + 2bx + c$

DT0029

5. System of linear equations in x,y,z

$$2x + y + z = 1$$
$$x - 2y + z = 2$$
$$3x - y + 2z = 3 \text{ have infinite solutions which}$$

(A) can be written as $(-3\lambda - 1, \lambda, 5\lambda + 3)\ \forall\ \lambda \in R$

(B) can be written as $(3\lambda - 1, -\lambda, -5\lambda + 3)\ \forall\ \lambda \in R$

(C) are such that every solution satisfy $x - 3y + 1 = 0$

(D) are such that none of them satisfy $5x + 3z = 1$

DT0030

6. System of equation $x + y + az = b$, $2x + 3y = 2a$ & $3x + 4y + a^2z = ab + 2$ has

(A) unique solution when $a \neq 0, b \in R$

(B) no solution when $a = 0, b = 1$

(C) infinite solution when $a = 0, b = 2$

(D) infinite solution when $a = 1, b \in R$

DT0031

7. If $\begin{vmatrix} a_1 + b_1x & a_1x + b_1 & c_1 \\ a_2 + b_2x & a_2x + b_2 & c_2 \\ a_3 + b_3x & a_3x + b_3 & c_3 \end{vmatrix} = 0$, then possible conditions is/are -

(A) $x = 1\ \forall\ a_i, b_i$, where $1 \leq i, j \leq 3$

(B) $x = -1\ \forall\ a_i, b_i$, where $1 \leq i, j \leq 3$

(C) $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$

(D) $x = \pm 2\ \forall\ a_i, b_i$, where $1 \leq i, j \leq 3$

DT0095

8. $\begin{vmatrix} x^2 & (y+z)^2 & yz \\ y^2 & (x+z)^2 & zx \\ z^2 & (x+y)^2 & xy \end{vmatrix}$ is divisible by -

(A) $x^2 + y^2 + z^2$

(B) $x - y$

(C) $x - y - z$

(D) $x + y + z$

DT0096

9. If $\begin{vmatrix} \frac{a}{b} + \frac{a}{c} & \frac{b}{c} + \frac{b}{a} & 1 \\ \frac{b}{a} + \frac{b}{c} & \frac{c}{a} + \frac{c}{b} & 1 \\ \frac{c}{a} + \frac{c}{b} & \frac{a}{b} + \frac{a}{c} & 1 \end{vmatrix} = 0$, where $a, b, c \in R^+$, then which of the following is necessarily true -

(A) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$

(B) $a^2 + b^2 + c^2 = ab + bc + ac$

(C) $a = b = c$

(D) $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 1$

DT0097

10. The system of linear equations $x + y + z = 6$, $x + 2y + 3z = 14$ and $2x + 5y + pz = q$ have -

(A) infinitely many solution when $p = 8, q = 36$ (B) unique solution when $p \neq 8, q \neq 36$

(C) no solution when $p = 8, q \neq 36$ (D) atleast one solution for $q = 36, p \in R$

DT0098

11. If a, b, c are in A.P. and α, β, γ are positive real numbers in G.P., then the equation $\begin{vmatrix} x+a & x^2+\log\alpha & k \\ x+b & x^2+\log\beta & k \\ x+c & x^2+\log\gamma & k \end{vmatrix} = 0$:-

(A) is an identity (B) has a root $x = 1$

(C) has a root $x = 0$ (D) has real & identical roots

DT0099

12. If α, β, γ satisfy the equation $\begin{vmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{vmatrix} = 0$, then

(A) $\alpha + \beta + \gamma = 0$ (B) $\alpha^2 + \beta^2 + \gamma^2 = 6$ (C) $\alpha^3 + \beta^3 + \gamma^3 = -6$ (D) $\alpha^4 + \beta^4 + \gamma^4 = 18$

DT0100

13. If the system of equations $x + y - 3 = 0$, $(1 + K)x + (2 + K)y - 8 = 0$ & $x - (1 + K)y + (2 + K) = 0$ is consistent then the value of K may be -

(A) 1 (B) $\frac{3}{5}$ (C) $-\frac{5}{3}$ (D) 2

DT0101

14. If system of equation $a_1x + b_1y = c_1$ & $a_2x + b_2y = c_2$ (where $a_1,b_1,c_1, a_2,b_2,c_2 \neq 0$) has infinite solutions, then-

(A) $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$

(B) $\frac{a_1+a_2}{a_1-a_2} = \frac{b_1+b_2}{b_1-b_2} = \frac{c_1+c_2}{c_1-c_2}$

(C) the quadratic equations $a_1x^2 + b_1x + c_1 = 0$ & $a_2x^2 + b_2x + c_2 = 0$ have no common root

(D) system of equation $a_1^2 a_2x + b_1^2b_2y = c_1^2c_2$ & $a_1a_2^2x + b_1b_2^2 y = c_1c_2^2$ will also have infinite number of solutions

DT0102

15. If $\ell < m < n$, then $\begin{vmatrix} 1 & \ell & \ell^4 \\ 1 & m & m^4 \\ 1 & n & n^4 \end{vmatrix}$ will always be greater than -

(A) 3 (B) 6 (C) –1 (D) 0

DT0103

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question 1 to 3

Let x, y, z $\in R^+$ & $D = \begin{vmatrix} x & x^3 & x^4-1 \\ y & y^3 & y^4-1 \\ z & z^3 & z^4-1 \end{vmatrix}$

On the basis of above information, answer the following questions :

1. If $x \neq y \neq z$ & x, y, z are in GP and D = 0, then y is equal to -

(A) 1 (B) 2 (C) 4 (D) none of these

DT0104

2. If x, y, z are the roots of $t^3 - 21t^2 + bt - 343 = 0$, $b \in R$, then D is equal to-

(A) 1 (B) 0 (C) dependent on x, y, z (D) data inadequate

DT0105

3. If $x \neq y \neq z$ & x, y, z are in A.P. and D = 0, then $2xy^2z + x^2z^2$ is equal to-

(A) 1 (B) 2 (C) 3 (D) none of these

DT0106

Paragraph for Question 4 to 6

Let $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ \alpha & \beta & \gamma \end{vmatrix} = t$, where t is an even prime number & α, β, γ are the integral roots of the equation

$x^3 - 14x^2 + Px - 36 = 0$

On the basis of above information answer the following :

4. The value of P is-

(A) a rational number (B) a prime number

(C) an odd natural number (D) an even natural number

DT0107

5. The value of $\alpha(\beta^2+\gamma^2)+\beta(\gamma^2+\alpha^2)+\gamma(\alpha^2+\beta^2)$ is divisible by -

(A) 17 (B) 34 (C) 51 (D) 68

DT0108

6. Which of following statement is/are false -

(A) t is divisible by $(\alpha - \beta)$ (B) t is divisible by $(\beta - \gamma)$

(C) t is divisible by $(\gamma - \alpha)$ (D) $(\gamma - \alpha)$ is divisible by t

DT0109

Paragraph for Question 7 & 8

Let $f(x) = \begin{vmatrix} 1 & x-1 & x \\ 2(x-1) & (x-1)(x-2) & x(x-1) \\ 3(x-1)(x-2) & (x-1)(x-2)(x-3) & x(x-1)(x-2) \end{vmatrix}$ & $D_r = \begin{vmatrix} 1 & 10 & 2r \\ 70 & 17 & 3r+1 \\ 1 & 1 & 1 \end{vmatrix}$.

On the basis of above information, answer the following questions :

7. The value of $f(50) - D_5$ is -

DT0110

8. $\sum_{r=1}^{10} D_r$ is

DT0111

Matrix Match Type

9. Consider a system of linear equations $a_ix + b_iy + c_iz = d_i$(where $a_i,b_i,c_i \neq 0$ and i = 1,2,3) & (α,β,γ) is its unique solution, then match list-I with list-II

List-I		List-II	
(I)	If $a_i = k$, $d_i = k^2$, $(k \neq 0)$ and $\alpha + \beta + \gamma = 2$, then k is	(P)	1
(II)	If $a_i = d_i = k \neq 0$, then $\alpha + \beta + \gamma$ is	(Q)	2
(III)	If $a_i = k > 0$, $d_i = k + 1$, then $\alpha + \beta + \gamma$ can be	(R)	0
(IV)	If $a_i = k < 0$, $d_i = k + 1$, then $\alpha + \beta + \gamma$ can be	(S)	3
		(T)	–1

DT0032

(A) I $\rightarrow$ P,Q; II $\rightarrow$ R; III $\rightarrow$ Q,S; IV $\rightarrow$ T

(B) I $\rightarrow$ P; II $\rightarrow$ Q; III $\rightarrow$ R,S; IV $\rightarrow$ T

(C) I $\rightarrow$ Q; II $\rightarrow$ P,R; III $\rightarrow$ S; IV $\rightarrow$ T

(D) I $\rightarrow$ Q; II $\rightarrow$ P; III $\rightarrow$ Q,S; IV $\rightarrow$ R,T

10. Match the following for the system of linear equations

$\lambda x + y + z = 1$, $x + \lambda y + z = \lambda$, $x + y + \lambda z = \lambda^2$

Column-I		Column-II	
(A)	$\lambda = 1$	(P)	unique solution
(B)	$\lambda \neq 1$	(Q)	infinite solutions
(C)	$\lambda \neq 1, \lambda \neq -2$	(R)	no solution
(D)	$\lambda = -2$	(S)	finite many solutions

DT0112

EXERCISE (O-4)

Numerical Grid Type

1. Let a, b, c are the solutions of the cubic $x^3 - 5x^2 + 3x - 1 = 0$, then find the value of the determinant

$$\begin{vmatrix} a & b & c \\ a-b & b-c & c-a \\ b+c & c+a & a+b \end{vmatrix}.$$

DT0040

2. If $\Delta(x) = \begin{vmatrix} 0 & 2x-2 & 2x+8 \\ x-1 & 4 & x^2+7 \\ 0 & 0 & x+4 \end{vmatrix}$ and $f(x) = \sum_{j=1}^{3}\sum_{i=1}^{3} a_{ij}c_{ij}$, where a_{ij} is the element of i^{th} and j^{th} column in $\Delta(x)$ and c_{ij} is the cofactor a_{ij} $\forall$ i and j, then find the greatest value of f(x), where $x \in [-3, 18]$

DT0041

3. If the equations $a(y + z) = x$, $b(z + x) = y$, $c(x + y) = z$ (where $a,b,c \neq -1$)have nontrivial solutions, then find the value of $\frac{1}{1+a} + \frac{1}{1+b} + \frac{1}{1+c}$.

DT0053

4. Find the sum of all positive integral values of a for which every solution to the system of equation $x + ay = 3$ and $ax + 4y = 6$ satisfy the inequalities $x > 1$, $y > 0$.

DT0062

5. For a determinant Δ of order 3, the element a_{ij} is defined as $a_{ij} = \tan^{-1}(\tan(i - j))$ $\forall$ i, j, then the value of Δ is equal to (where 'i' represents row and 'j' represents column)

DT0113

6. The number of triplets (α, β, γ) satisfing the following constraints

$2\alpha - \beta + 3\gamma = 4$

$\alpha + \beta - 3\gamma = -1$

$5\alpha - \beta + 3\gamma = 7$

$\alpha\beta\gamma \leq 0$

& $\alpha, \beta, \gamma \in I$

DT0114

7. Let a, b, c, ℓ, m, n $\in$ R such that $a\ell + bm + cn = 0$, $b\ell + cm + an = 0$, $c\ell + am + bn = 0$. If a, b & c are distinct & $f(x) = ax^3 + bx^2 + cx + 5$, then the value of $f(1)$ is

DT0115

node06\B0BC-BD\Kota\JEE(Advanced)\Nurture\Maths\Module\4-Determinant-St.Line-Circle\Eng\01_Determinant\02-Determinant.p65

8. If $\begin{vmatrix} \sin^2 2x & \cos 2x & 4\sin^2\frac{x}{2} \\ \tan^2\frac{x}{2} & \cos^2 x & -\sin^2 x \\ -2\cos 4x & \tan^2\frac{x}{2} & \sin 4x \end{vmatrix} = a_0 + a_1(\cos x) + a_2(\cos^2 x) + ... + a_n(\cos^n x)$, then a_0 is -

DT0116

9. If x, y, z are distinct digits $(0 \le x, y, z \le 9)$ & the minimum possible value of $\begin{vmatrix} z & 9y & x \\ z & y & 9x \\ 9z & y & x \end{vmatrix}$ is λ

then $\frac{\lambda}{83700}$ is

(where 9x, 9y & 9z are two digits number)

DT0117

10. $\begin{vmatrix} a & b-c & c+b \\ a+c & b & c-a \\ a-b & b+a & c \end{vmatrix} = \begin{vmatrix} b-c & b & b+a \\ a & -a-c & b-a \\ c+b & c-a & c \end{vmatrix}$ and $a > 0, b > 0, c > 0$, then the minimum value of $\frac{c^2}{ab}$ is

DT0118

EXERCISE (JM)

1. The number of values of k, for which the system of equations : **[JEE(Main)-2013]**

$(k + 1)x + 8y = 4k$, $kx + (k + 3)y = 3k - 1$ has no solution, is -

(1) infinite (2) 1 (3) 2 (4) 3

DT0069

2. If $\alpha, \beta \neq 0$, and $f(n) = \alpha^n + \beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1-\alpha)^2 (1-\beta)^2 (\alpha-\beta)^2$, then

K is equal to : **[JEE(Main)-2014]**

(1) $\alpha\beta$ (2) $\frac{1}{\alpha\beta}$ (3) 1 (4) -1

DT0070

3. The set of all values of λ for which the system of linear equations :

$2x_1 - 2x_2 + x_3 = \lambda x_1$, $2x_1 - 3x_2 + 2x_3 = \lambda x_2$, $-x_1 + 2x_2 = \lambda x_3$ has a non-trivial solution **[JEE(Main)-2015]**

(1) contains two elements (2) contains more than two elements

(3) is an empty set (4) is a singleton

DT0071

4. The system of linear equations $x + \lambda y - z = 0$, $\lambda x - y - z = 0$, $x + y - \lambda z = 0$ has a non-trivial solution for : **[JEE(Main)-2016]**

(1) exactly three values of λ. (2) infinitely many values of λ.

(3) exactly one value of λ. (4) exactly two values of λ.

DT0072

5. If S is the set of distinct values of 'b' for which the following system of linear equations

$$x + y + z = 1$$
$$x + ay + z = 1$$
$$ax + by + z = 0$$

has no solution, then S is : **[JEE(Main)-2017]**

(1) a singleton (2) an empty set

(3) an infinite set (4) a finite set containing two or more elements

DT0073

6. If $\begin{vmatrix} x-4 & 2x & 2x \\ 2x & x-4 & 2x \\ 2x & 2x & x-4 \end{vmatrix} = (A + Bx)(x - A)^2$, then the ordered pair (A, B) is equal to :

[JEE(Main)-2018]

(1) $(-4, 3)$ (2) $(-4, 5)$ (3) $(4, 5)$ (4) $(-4, -5)$

DT0074

7. If the system of linear equations
$$x + ky + 3z = 0$$
$$3x + ky - 2z = 0$$
$$2x + 4y - 3z = 0$$
has a non-zero solution (x, y, z), then $\frac{xz}{y^2}$ is equal to : **[JEE(Main)-2018]**

(1) 10 (2) – 30 (3) 30 (4) –10

DT0075

8. If the system of linear equations
$$x - 4y + 7z = g$$
$$3y - 5z = h$$
$$-2x + 5y - 9z = k$$
is consistent, then : **[JEE(Main) 2019]**

(1) $g + h + k = 0$ (2) $2g + h + k = 0$

(3) $g + h + 2k = 0$ (4) $g + 2h + k = 0$

DT0076

9. Let $d \in R$, and $A = \begin{bmatrix} -2 & 4+d & (\sin\theta)-2 \\ 1 & (\sin\theta)+2 & d \\ 5 & (2\sin\theta)-d & (-\sin\theta)+2+2d \end{bmatrix}$, $\theta \in [0,2\pi]$. If the minimum value of det(A) is 8, then a value of d is : **[JEE(Main) 2019]**

(1) –7 (2) $2\left(\sqrt{2}+2\right)$ (3) –5 (4) $2\left(\sqrt{2}+1\right)$

DT0077

10. Let $a_1, a_2, a_3, ..., a_{10}$ be in G.P. with $a_i > 0$ for $i = 1,2,....., 10$ and S be the set of pairs (r,k), r, k $\in$ N (the set of natural numbers) for which $\begin{vmatrix} \log_e a_1^r a_2^k & \log_e a_2^r a_3^k & \log_e a_3^r a_4^k \\ \log_e a_4^r a_5^k & \log_e a_5^r a_6^k & \log_e a_6^r a_7^k \\ \log_e a_7^r a_8^k & \log_e a_8^r a_9^k & \log_e a_9^r a_{10}^k \end{vmatrix} = 0$. Then the number of elements in S, is : **[JEE(Main) 2019]**

(1) Infinitely many (2) 4 (3) 10 (4) 2

DT0078

11. The set of all values of λ for which the system of linear equations.
$$x - 2y - 2z = \lambda x$$
$$x + 2y + z = \lambda y$$
$$-x - y = \lambda z$$
has a non-trivial solution. **[JEE(Main) 2019]**

(1) contains more than two elements

(2) is a singleton

(3) is an empty set

(4) contains exactly two elements **DT0079**

12. If the system of linear equations

$2x + 2ay + az = 0$

$2x + 3by + bz = 0$

$2x + 4cy + cz = 0,$ **[JEE(Main) 2020]**

where a, b, c $\in$ R are non-zero and distinct; has a non-zero solution, then :

(1) a, b, c are in A.P. (2) $a + b + c = 0$

(3) a, b, c are in G.P. (4) $\frac{1}{a}, \frac{1}{b}, \frac{1}{c}$ are in A.P. **DT0080**

13. The system of linear equations

$\lambda x + 2y + 2z = 5$

$2\lambda x + 3y + 5z = 8$

$4x + \lambda y + 6z = 10$ has **[JEE(Main) 2020]**

(1) infinitely many solutions when $\lambda = 2$ (2) a unique solution when $\lambda = -8$

(3) no solution when $\lambda = 8$ (4) no solution when $\lambda = 2$

DT0081

14. For which of the following ordered pairs (μ,δ), the system of linear equations

$x + 2y + 3z = 1$

$3x + 4y + 5z = \mu$

$4x + 4y + 4z = \delta$

is inconsistent ? **[JEE(Main) 2020]**

(1) (1,0) (2) (4,6) (3) (3,4) (4) (4,3)

DT0082

15. Let $a - 2b + c = 1$. If $f(x) = \begin{vmatrix} x+a & x+2 & x+1 \\ x+b & x+3 & x+2 \\ x+c & x+4 & x+3 \end{vmatrix}$, then : **[JEE(Main) 2020]**

(1) $f(-50) = 501$ (2) $f(-50) = -1$ (3) $f(50) = 1$ (4) $f(50) = -501$

DT0083

16. Let α,β,γ be the real roots of the equation, $x^3 + ax^2 + bx + c = 0$, (a,b,c $\in$ R and a,b $\neq$ 0). If the system of equations (in, u,v,w) given by $\alpha u + \beta v + \gamma w = 0$, $\beta u + \gamma v + \alpha w = 0$; $\gamma u + \alpha v + \beta w = 0$ has non-trivial solution, then the value of $\frac{a^2}{b}$ is **[JEE(Main) 2021]**

(1) 5 (2) 3 (3) 1 (4) 0

DT0119

17. Let the system of linear equations

$4x + \lambda y + 2z = 0$

$2x - y + z = 0$

$\mu x + 2y + 3z = 0$, $\lambda, \mu \in R$.

has a non-trivial solution. Then which of the following is true ? **[JEE(Main) 2021]**

(1) $\mu = 6, \lambda \in R$ (2) $\lambda = 2, \mu \in R$ (3) $= 3, \mu \in R$ (4) $\mu = -6, \lambda \in R$

DT0120

18. The number of real values λ, such that the system of linear equations

$2x - 3y + 5z = 9$

$x + 3y - z = -18$

$3x - y + (\lambda^2 - |\lambda|)z = 16$

has no solution, is :- **[JEE(Main) 2022]**

(A) 0 (B) 1 (C) 2 (D) 4

DT0121

19. If the system of equations

$x + y + z = 6$

$2x + 5y + \alpha z = \beta$

$x + 2y + 3z = 14$

has infinitely many solutions, then $\alpha + \beta$ is equal to : **[JEE(Main) 2022]**

(A) 8 (B) 36 (C) 44 (D) 48

DT0122

EXERCISE (JA)

1. The number of all possible values of θ, where $0 < \theta < \pi$, for which the system of equations

$$(y + z)\cos 3\theta = (xyz)\sin 3\theta$$

$$x\sin 3\theta = \frac{2\cos 3\theta}{y} + \frac{2\sin 3\theta}{z}$$

$$(xyz)\sin 3\theta = (y + 2z)\cos 3\theta + y\sin 3\theta$$

have a solution (x_0, y_0, z_0) with $y_0 z_0 \neq 0$, is **[JEE 2010, 3]**

DT0084

2. Which of the following values of α satisfy the equation $\begin{vmatrix} (1+\alpha)^2 & (1+2\alpha)^2 & (1+3\alpha)^2 \\ (2+\alpha)^2 & (2+2\alpha)^2 & (2+3\alpha)^2 \\ (3+\alpha)^2 & (3+2\alpha)^2 & (3+3\alpha)^2 \end{vmatrix} = -648\alpha$?

[JEE(Advanced)-2015, 4M, –2M]

(A) –4 (B) 9 (C) –9 (D) 4

DT0085

3. The total number of distinct $x \in \mathbb{R}$ for which $\begin{vmatrix} x & x^2 & 1+x^3 \\ 2x & 4x^2 & 1+8x^3 \\ 3x & 9x^2 & 1+27x^3 \end{vmatrix} = 10$ is

[JEE(Advanced)-2016, 3(0)]

DT0086

4. Let $a,\lambda,m \in \mathbb{R}$. Consider the system of linear equations

$$ax + 2y = \lambda$$
$$3x - 2y = \mu$$

Which of the following statement(s) is(are) correct ?

(A) If $a = -3$, then the system has infinitely many solutions for all values of λ and μ

(B) If $a \neq -3$, then the system has a unique solution for all values of λ and μ

(C) If $\lambda + \mu = 0$, then the system has infinitely many solutions for $a = -3$

(D) If $\lambda + \mu \neq 0$, then the system has no solution for $a = -3$ **[JEE(Advanced)-2016, 4(–2)]**

DT0087

5. Let p,q,r be nonzero real numbers that are, respectively, the 10^{th}, 100^{th} and 1000^{th} terms of a harmonic progression. Consider the system of linear equations

$$x + y + z = 1$$
$$10x + 100y + 1000z = 0$$
$$qr\,x + pr\,y + pq\,z = 0.$$

[JEE(Advanced)-2022]

	List-I		**List-II**
(I)	If $\frac{q}{r} = 10$, then the system of linear equations has	(P)	$x = 0,\ y = \frac{10}{9}, z = -\frac{1}{9}$ as a solution
(II)	If $\frac{p}{r} \neq 100$, then the system of linear equations has	(Q)	$x = \frac{10}{9},\ y = -\frac{1}{9}, z = 0$ as a solution
(III)	If $\frac{p}{q} \neq 10$, then the system of linear equations has	(R)	infinitely many solutions
(IV)	If $\frac{p}{q} = 10$, then the system of linear equations has	(S)	no solution
		(T)	at least one solution

The correct option is:

(A) (I) → (T); (II) → (R); (III) → (S); (IV) → (T)

(B) (I) → (Q); (II) → (S); (III) → (S); (IV) → (R)

(C) (I) → (Q); (II) → (R); (III) → (P); (IV) → (R)

(D) (I) → (T); (II) → (S); (III) → (P); (IV) → (T)

DT0123

ANSWER KEY

Do yourself -1

1. minors : 4, –1, –4, 4 ; cofactors : –4, –1, 4, 4 **2.** –98 **3.** B **4.** 0
5. C **6.** A **7.** C **8.** D **9.** D **10.** A

Do yourself -2

2. C **3.** 0 **4.** A **5.** D

Do yourself -3

1. 2 **2.** 0 **3.** C **4.** B **5.** 0.00

Do yourself -4

2. $x = -1, 2$ **3.** A **4.** C **5.** B

Do yourself -5

1. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & \alpha & \beta \\ 1 & \beta & \alpha \end{vmatrix}$ **2.** D **3.** A **4.** C **5.** A

Do yourself -6

1. C **2.** 0 **4.** A **5.** B

Do yourself -7

1. infinite solutions **2.** A **3.** A **4.** D **5.** D **6.** B
7. C **8.** B **9.** B

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	A	C	A	B	D	B	C	A	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	D	D	D	B	D	C	B	D	C	B
Que.	21	22	23	24	25					
Ans.	A	C	D	B	A					

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C,D	A,B,C,D	A,B,C,D	B,D	A,B,D	B,C,D	A,B,C	A,B,D	B,C	A,B,C,D
Que.	11	12	13	14	15					
Ans.	A,B,C	A,B,C,D	A,C	A,B,D	C,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	A	B	C	A,C	A,B	A,B,C	9.00	575.00	D
Que. 10	A	B	C	D					
	Q	P,R	P	R					

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	80	0	2	4	0	1	5	0	9	4

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	1	1	1	2	1	2	3	1
Que.	11	12	13	14	15	16	17	18	19	
Ans.	2	4	4	4	3	2	1	C	C	

EXERCISE # JEE-ADVANCED

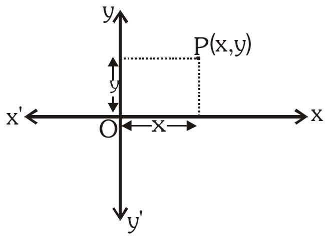
Que.	1	2	3	4	5
Ans.	3	B,C	2	B,C,D	B

POINT & STRAIGHT LINE

1. INTRODUCTION OF COORDINATE GEOMETRY :

Coordinate geometry is the combination of algebra and geometry. A systematic study of geometry by the use of algebra was first carried out by celebrated French philosopher and mathematician René Descartes. The resulting combination of analysis and geometry is referred as ***analytical geometry***.

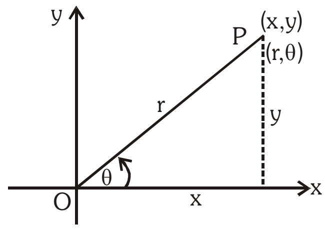
2. CARTESIAN CO-ORDINATES SYSTEM :



In two dimensional coordinate system, two lines are used; the lines are at right angles, forming a rectangular coordinate system. The horizontal axis is the x-axis and the vertical axis is y-axis. The point of intersection O is the origin of the coordinate system. Distances along the x-axis to the right of the origin are taken as positive, distances to the left as negative.

Distances along the y-axis above the origin are positive; distances below are negative. The position of a point anywhere in the plane can be specified by two numbers, the coordinates of the point, written as (x, y). The x-coordinate (or abscissa) is the distance of the point from the y-axis in a direction parallel to the x-axis (i.e. horizontally). The y-coordinate (or ordinate) is the distance from the x-axis in a direction parallel to the y-axis (vertically). The origin O is the point (0, 0).

3. POLAR CO-ORDINATES SYSTEM :



A coordinate system in which the position of a point is determined by the length of a line segment from a fixed origin together with the angle that the line segment makes with a fixed line. The origin is called the pole and the line segment is the radius vector (r).

The angle θ between the polar axis and the radius vector is called the vectorial angle. By convention, positive values of θ are measured in an anticlockwise sense, negative values in clockwise sense. The coordinates of the point are then specified as (r, θ).

If (x,y) are cartesian co-ordinates of a point P, then : $x = r\cos\theta$, $y = r\sin\theta$

and $r = \sqrt{x^2 + y^2}$, $\theta = \tan^{-1}\left(\dfrac{y}{x}\right)$

4. DISTANCE FORMULA AND ITS APPLICATIONS :

If $A(x_1,y_1)$ and $B(x_2,y_2)$ are two points, then $AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

Note :

(i) Three given points A,B and C are collinear, when sum of any two distances out of AB,BC, CA is equal to the remaining third otherwise the points will be the vertices of a triangle.

(ii) Let A,B,C & D be the four given points in a plane. Then the quadrilateral will be :

(a) Square if $AB = BC = CD = DA$ & $AC = BD$; $AC \perp BD$

(b) Rhombus if $AB = BC = CD = DA$ and $AC \neq BD$; $AC \perp BD$

(c) Parallelogram if $AB = DC$, $BC = AD$; $AC \neq BD$; $AC \not\perp BD$

(d) Rectangle if $AB = CD$, $BC = DA$, $AC = BD$; $AC \not\perp BD$

Illustration 1 : The number of points on x-axis which are at a distance $c(c < 3)$ from the point (2, 3) is

(A) 2 (B) 1 (C) infinite (D) no point

Solution : Let a point on x-axis is $(x_1, 0)$ then its distance from the point (2, 3)

$$= \sqrt{(x_1 - 2)^2 + 9} = c \quad \text{or} \quad (x_1 - 2)^2 = c^2 - 9$$

$$\therefore \; x_1 - 2 = \pm\sqrt{c^2 - 9} \;\; \text{since} \;\; c < 3 \Rightarrow c^2 - 9 < 0$$

$\therefore$ x_1 will be imaginary. **Ans. (D)**

Illustration 2 : The distance between the point $P(a\cos\alpha,\, a\sin\alpha)$ and $Q(a\cos\beta,\, a\sin\beta)$, where $a > 0$ & $\alpha > \beta$, is -

(A) $4a\sin\frac{\alpha-\beta}{2}$ (B) $2a\sin\frac{\alpha+\beta}{2}$ (C) $2a\sin\frac{\alpha-\beta}{2}$ (D) $2a\cos\frac{\alpha-\beta}{2}$

Solution :

$$d^2 = (a\cos\alpha - a\cos\beta)^2 + (a\sin\alpha - a\sin\beta)^2 = a^2(\cos\alpha - \cos\beta)^2 + a^2(\sin\alpha - \sin\beta)^2$$

$$= a^2\left\{2\sin\frac{\alpha+\beta}{2}\sin\frac{\beta-\alpha}{2}\right\}^2 + a^2\left\{2\cos\frac{\alpha+\beta}{2}\sin\frac{\alpha-\beta}{2}\right\}^2$$

$$= 4a^2\sin^2\frac{\alpha-\beta}{2}\left\{\sin^2\frac{\alpha+\beta}{2} + \cos^2\frac{\alpha+\beta}{2}\right\} = 4a^2\sin^2\frac{\alpha-\beta}{2} \Rightarrow d = 2a\sin\frac{\alpha-\beta}{2}$$

Ans. (C)

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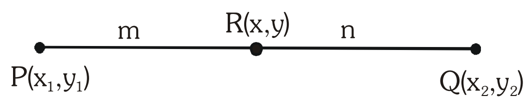
5. SECTION FORMULA :

The co-ordinates of a point dividing a line joining the points $P(x_1,y_1)$ and $Q(x_2,y_2)$ in the ratio m:n is given by :

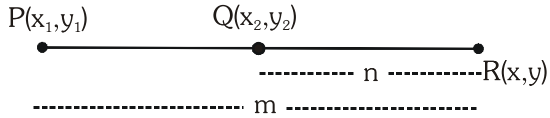
(a) For internal division : P - R - Q $\Rightarrow$ R divides line segment PQ, internally.

$$(x, y) \equiv \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right)$$



(b) For external division : R - P - Q or P - Q - R $\Rightarrow$ R divides line segment PQ, externally.

$$(x, y) \equiv \left(\frac{mx_2 - nx_1}{m - n}, \frac{my_2 - ny_1}{m - n}\right)$$



$\frac{\mathbf{(PR)}}{\mathbf{(QR)}} \mathbf{< 1}$ $\Rightarrow$ R lies on the left of P & $\frac{\mathbf{(PR)}}{\mathbf{(QR)}} \mathbf{> 1}$ $\Rightarrow$ R lies on the right of Q

(c) Harmonic conjugate : If P divides AB internally in the ratio m : n & Q divides AB externally in the ratio m : n then P & Q are said to be harmonic conjugate of each other w.r.t. AB. Mathematically ; $\frac{2}{AB} = \frac{1}{AP} + \frac{1}{AQ}$ i.e. AP, AB & AQ are in **H.P.**

Illustration 3 : Determine the ratio in which y – x + 2 = 0 divides the line joining (3, –1) and (8, 9).

Solution : Suppose the line y – x + 2 = 0 divides the line segment joining A(3, –1) and B(8, 9) in the ratio λ : 1 at a point P, then the co-ordinates of the point P are $\left(\frac{8\lambda + 3}{\lambda + 1}, \frac{9\lambda - 1}{\lambda + 1}\right)$

But P lies on y – x + 2 = 0 therefore $\left(\frac{9\lambda - 1}{\lambda + 1}\right) - \left(\frac{8\lambda + 3}{\lambda + 1}\right) + 2 = 0$

$\Rightarrow$ $9\lambda - 1 - 8\lambda - 3 + 2\lambda + 2 = 0$

$\Rightarrow$ $3\lambda - 2 = 0$ or $\lambda = \frac{2}{3}$

So, the required ratio is $\frac{2}{3}$: 1, i.e., 2 : 3 (internally) since here λ is positive.

Do yourself - 1 :

1. Find the distance between the points P(–3, 2) and Q(2, –1).

2. If the distance between the points P(–3, 5) and Q(–x, – 2) is $\sqrt{58}$, then find the value(s) of x.

3. A line segment is of the length 15 units and one end is at the point (3, 2), if the abscissa of the other end is 15, then find possible ordinates.

4. A triangle with vertices (4, 0), (–1, –1), (3, 5) is

(A) isosceles and right angled (B) isosceles but not right angled

(C) right angled but not isosceles (D) neither right angled nor isosceles

5. Find the value of x_1 if the distance between the points $(x_1, 2)$ and (3, 4) be 8.

(A) $3 \pm 2\sqrt{15}$ (B) $3 \pm \sqrt{15}$ (C) $2 \pm 3\sqrt{15}$ (D) $2 \pm \sqrt{15}$

6. The four points whose co-ordinates are (2,1),(1,4),(4,5),(5,2) form :

(A) a rectangle which is not a square (B) a trapezium which is not a parallelogram

(C) a square (D) a rhombus which is not a square

7. Find the co-ordinates of the point dividing the join of A(1, – 2) and B(4, 7) :

(a) Internally in the ratio 1 : 2 (b) Externally in the ratio of 2 : 1

8. In what ratio is the line joining A(8, 9) and B(– 7, 4) divided by

(a) the point (2, 7) (b) the x-axis (c) the y-axis.

9. If A and B are the points (–3,4) and (2,1), then the co-ordinates of the point C on AB produced such that AC = 2BC are :

(A) (2,4) (B) (3,7) (C) (7,–2) (D) $\left(-\frac{1}{2},\frac{5}{2}\right)$

10. Determine the ratio in which the point P(3 , 5) divides the join of A(1, 3) & B(7, 9). Find the harmonic conjugate of P w.r.t. A & B.

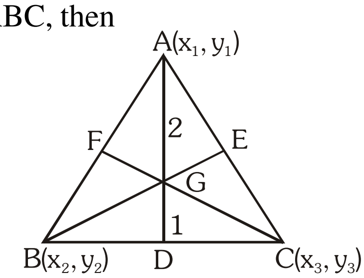
11. If the line 2x + y = k passes through the point which divides the line segment joining the points (1, 1) and (2, 4) in the ratio 3 : 2, then k equals :

(A) $\frac{11}{5}$ (B) $\frac{29}{5}$ (C) 5 (D) 6

6. CO-ORDINATES OF SOME PARTICULAR POINTS :

Let $A(x_1,y_1)$, $B(x_2,y_2)$ and $C(x_3,y_3)$ are vertices of any triangle ABC, then

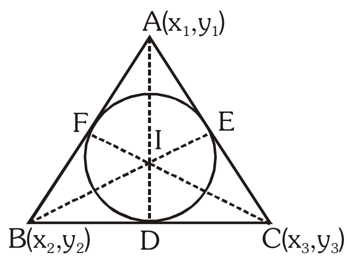
(a) Centroid :



The centroid is the point of intersection of the medians (line joining the mid point of sides and opposite vertices). Centroid divides each median in the ratio of 2 : 1.

Co-ordinates of centroid $G\left(\frac{x_1+x_2+x_3}{3},\frac{y_1+y_2+y_3}{3}\right)$

(b) Incenter :



The incenter is the point of intersection of internal bisectors of the angles of a triangle. Also it is a centre of the circle touching all the sides of a triangle.

Co-ordinates of incenter $I\left(\frac{ax_1+bx_2+cx_3}{a+b+c},\frac{ay_1+by_2+cy_3}{a+b+c}\right)$

where a, b, c are the sides of triangle ABC.

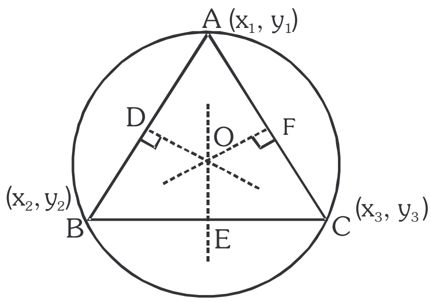
Note :

(i) Angle bisector divides the opposite sides in the ratio of remaining sides. e.g.

$$\frac{BD}{DC}=\frac{AB}{AC}=\frac{c}{b}$$

(ii) Incenter divides the angle bisectors in the ratio $(b+c):a$, $(c+a):b$, $(a+b):c$.

(c) Circumcenter :



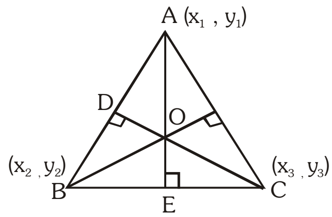
It is the point of intersection of perpendicular bisectors of the sides of a triangle. If O is the circumcenter of any triangle ABC, then $OA^2 = OB^2 = OC^2$. Also it is a centre of a circle touching all the vertices of a triangle.

Note :

(i) If the triangle is right angled, then its circumcenter is the mid point of hypotenuse.

(ii) Co-ordinates of circumcenter $\left(\dfrac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \dfrac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}\right)$

(d) Orthocenter :



It is the point of intersection of perpendiculars drawn from vertices on the opposite sides of a triangle and it can be obtained by solving the equation of any two altitudes.

Note :

(i) If a triangle is right angled, then orthocenter is the point where right angle is formed.

(ii) Co-ordinates of orthocenter $\left(\dfrac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \dfrac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C}\right)$

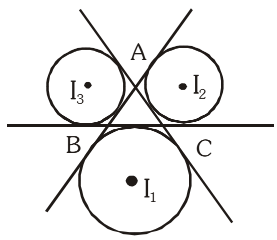
Remarks :

(i) If the triangle is equilateral, then centroid, incentre, orthocenter, circumcenter, coincide.

(ii) Orthocentre, centroid and circumcentre are always collinear and centroid divides the line joining orthocentre and circumcentre in the ratio 2 : 1

(iii) In an isosceles triangle centroid, orthocentre, incentre & circumcentre lie on the same line.

(e) Ex-centers :



The centre of a circle which touches side BC and the extended portions of sides AB and AC is called the ex-centre of ΔABC with respect to the vertex A. It is denoted by I_1 and its coordinates are $I_1\left(\dfrac{-ax_1 + bx_2 + cx_3}{-a+b+c}, \dfrac{-ay_1 + by_2 + cy_3}{-a+b+c}\right)$

Similarly ex-centers of ΔABC with respect to vertices B and C are denoted by I_2 and I_3 respectively , and

$$I_2\left(\frac{ax_1 - bx_2 + cx_3}{a-b+c}, \frac{ay_1 - by_2 + cy_3}{a-b+c}\right), I_3\left(\frac{ax_1 + bx_2 - cx_3}{a+b-c}, \frac{ay_1 + by_2 - cy_3}{a+b-c}\right)$$

Illustration 4 : If $\left(\frac{5}{3}, 3\right)$ is the centroid of a triangle and its two vertices are (0, 1) and (2, 3), then find its third vertex, circumcentre, circumradius & orthocentre.

Solution : Let the third vertex of triangle be (x, y), then

$\frac{5}{3} = \frac{x+0+2}{3} \Rightarrow x = 3$ and $3 = \frac{y+1+3}{3} \Rightarrow y = 5$. So third vertex is (3, 5).

Now three vertices are A(0, 1), B(2, 3) and C(3, 5)

Let circumcentre be P(h, k),

then AP = BP = CP = R (circumradius) $\Rightarrow$ $AP^2 = BP^2 = CP^2 = R^2$

$h^2 + (k-1)^2 = (h-2)^2 + (k-3)^2 = (h-3)^2 + (k-5)^2 = R^2$ .... (i)

from the first two equations, we have

$h + k = 3$ .... (ii)

from the first and third equation, we obtain

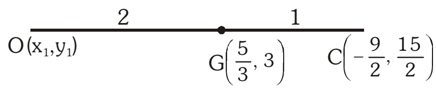
$6h + 8k = 33$ .... (iii)

On solving, (ii) & (iii), we get

$h = -\frac{9}{2}, \ k = \frac{15}{2}$

Substituting these values in (i), we have

$R = \frac{5}{2}\sqrt{10}$



Let $O(x_1, y_1)$ be the orthocentre, then $\frac{x_1 + 2\left(-\frac{9}{2}\right)}{3} = \frac{5}{3} \Rightarrow x_1 = 14, \frac{y_1 + 2\left(\frac{15}{2}\right)}{3} = 3$

$\Rightarrow$ $y_1 = -6$. Hence orthocentre of the triangle is (14, –6).

Illustration 5 : The vertices of a triangle are A(0, –6), B(–6, 0) and C(1,1) respectively, then coordinates of the ex-centre opposite to vertex A is :

(A) $\left(\frac{-3}{2}, \frac{-3}{2}\right)$ (B) $\left(-4, \frac{3}{2}\right)$ (C) $\left(\frac{-3}{2}, \frac{3}{2}\right)$ (D) (–4, 6)

Solution : $a = BC = \sqrt{(-6-1)^2 + (0-1)^2} = \sqrt{50} = 5\sqrt{2}$

$b = CA = \sqrt{(1-0)^2 + (1+6)^2} = \sqrt{50} = 5\sqrt{2}$

$c = AB = \sqrt{(0+6)^2 + (-6-0)^2} = \sqrt{72} = 6\sqrt{2}$

coordinates of ex-centre opposite to vertex A will be :

$$x = \frac{-ax_1 + bx_2 + cx_3}{-a+b+c} = \frac{-5\sqrt{2}.0 + 5\sqrt{2}(-6) + 6\sqrt{2}(1)}{-5\sqrt{2}+5\sqrt{2}+6\sqrt{2}} = \frac{-24\sqrt{2}}{6\sqrt{2}} = -4$$

$$y = \frac{-ay_1 + by_2 + cy_3}{-a+b+c} = \frac{-5\sqrt{2}(-6) + 5\sqrt{2}.0 + 6\sqrt{2}(1)}{-5\sqrt{2}+5\sqrt{2}+6\sqrt{2}} = \frac{36\sqrt{2}}{6\sqrt{2}} = 6$$

Hence coordinates of ex-centre is (–4, 6) **Ans. (D)**

Do yourself - 2 :

1. The coordinates of the vertices of a triangle are (1, 2), (3, 4) and (4, 6) :

(a) Find centroid of the triangle.

(b) Find circumcentre & the circumradius.

(c) Find orthocentre of the triangle.

2. A triangle has a vertex at (1, 2) and the mid points of the two sides through it are (–1, 1) and (2,3). Then the centroid of this triangle is :

(A) $\left(\frac{1}{3},1\right)$ (B) $\left(\frac{1}{3},2\right)$ (C) $\left(1,\frac{7}{3}\right)$ (D) $\left(\frac{1}{3},\frac{5}{3}\right)$

3. If a vertex of a triangle is (1, 1) and the mid points of two sides through this vertex are (–1, 2) and (3, 2) then the centroid of the triangle is

(A) $\left(-1,\frac{7}{3}\right)$ (B) $\left(\frac{-1}{3},\frac{7}{3}\right)$ (C) $\left(1,\frac{7}{3}\right)$ (D) $\left(\frac{1}{3},\frac{7}{3}\right)$

4. Coordinates of the vertices of a triangle ABC are (12,8), (–2,6) and (6,0) then the **correct** statement is -

(A) triangle is right but not isosceles

(B) triangle is isosceles but not right

(C) triangle is obtuse

(D) the product of the abscissa of the centroid, orthocenter and circumcenter is 160.

5. If the two vertices of a triangle are (7,2) and (1,6) and its centroid is (4,6) then the coordinate of the third vertex are (a,b). The value of (a + b), is-

(A) 13 (B) 14 (C) 15 (D) 16

6. The orthocenter of the triangle ABC is 'B' and the circumcenter is 'S' (a,b). If A is the origin then the co-ordinates of C are :

(A) (2a,2b) (B) $\left(\frac{a}{2},\frac{b}{2}\right)$ (C) $\left(\sqrt{a^2+b^2},0\right)$ (D) none

7. The medians of a triangle meet at (0,–3) and its two vertices are at (–1,4) and (5,2). Then the third vertex is at -

(A) (4,15) (B) (–4,–15) (C) (–4,15) (D) (4,–15)

8. Line $\frac{x}{6}+\frac{y}{8}=1$ intersects the x and y axes at M and N respectively. If the coordinates of the point P lying inside the triangle OMN (where 'O' is origin) are (a, b) such that the areas of the triangle POM, PON and PMN are equal. Find

(a) the coordinates of the point P and

(b) the radius of the circle escribed opposite to the angle N.

9. Let A (-3, 2) and B (-2,1) be the vertices of a triangle ABC. If the centroid of this triangle lies on the line $3x + 4y + 2 = 0$, then the vertex C lies on the line

(1) $4x + 3y + 5 = 0$ (2) $3x + 4y + 5 = 0$

(3) $3x + 4y + 3 = 0$ (4) $4x + 3y + 3 = 0$

7. AREA OF TRIANGLE :

Let $A(x_1,y_1)$, $B(x_2,y_2)$ and $C(x_3,y_3)$ are vertices of a triangle, then

$$\text{Area of } \Delta ABC = \left|\frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}\right| = \frac{1}{2}\left|[x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)]\right|$$

To remember the above formula, take the help of the following method :

$$=\frac{1}{2}\begin{bmatrix} x_1 & x_2 & x_3 & x_1 \\ y_1 & y_2 & y_3 & y_1 \end{bmatrix} = \left|\frac{1}{2}[(x_1y_2-x_2y_1)+(x_2y_3-x_3y_2)+(x_3y_1-x_1y_3)]\right|$$

Remarks :

(i) If the area of triangle joining three points is zero, then the points are collinear.

(ii) **Area of Equilateral triangle :** If altitude of any equilateral triangle is P, then its area = $\frac{P^2}{\sqrt{3}}$. If 'a' be the side of equilateral triangle, then its area $=\left(\frac{a^2\sqrt{3}}{4}\right)$.

(iii) Area of quadrilateral with given vertices $A(x_1, y_1)$, $B(x_2, y_2)$, $C(x_3, y_3)$, $D(x_4, y_4)$

$$\text{Area of quad. ABCD} = \frac{1}{2}\left|\begin{bmatrix} x_1 & x_2 & x_3 & x_4 & x_1 \\ y_1 & y_2 & y_3 & y_4 & y_1 \end{bmatrix}\right|$$

Note : Area of a polygon can be obtained by dividing the polygon into disjoined triangles and then adding their areas.

Illustration 6 : If the vertices of a triangle are (1, 2), (4, –6) and (3, 5) then its area is

(A) $\frac{25}{2}$ sq. units (B) 12 sq. units (C) 5 sq. units (D) 25 sq. units

Solution : $\Delta=\frac{1}{2}\left[1(-6-5)+4(5-2)+3(2+6)\right]=\frac{1}{2}[-11+12+24]=\frac{25}{2}\text{square units}$

Ans. (A)

Illustration 7 : The point A divides the join of the points (–5, 1) and (3, 5) in the ratio k:1 and coordinates of points B and C are (1, 5) and (7, –2) respectively. If the area of ΔABC be 2 units, then k equals -

(A) 7, 9 (B) 6, 7 (C) $7, \frac{31}{9}$ (D) $9, \frac{31}{9}$

Solution : $A\equiv\left(\frac{3k-5}{k+1}, \frac{5k+1}{k+1}\right)$

Area of $\Delta ABC=2$ units $\Rightarrow\frac{1}{2}\left[\frac{3k-5}{k+1}(5+2)+1\left(-2-\frac{5k+1}{k+1}\right)+7\left(\frac{5k+1}{k+1}-5\right)\right]=\pm2$

$\Rightarrow 14k-66=\pm4(k+1)\Rightarrow k=7 \text{ or } \frac{31}{9}$ **Ans. (C)**

Illustration 8 : Prove that the co-ordinates of the vertices of an equilateral triangle can not all be rational.

Solution : Let A(x_1 y_1), B(x_2, y_2) and C(x_3, y_3) be the vertices of a triangle ABC. If possible let x_1, y_1, x_2, y_2, x_3, y_3 be all rational.

Now area of $\Delta ABC=\frac{1}{2}|x_1(y_2-y_3)+x_2(y_3-y_1)+x_3(y_1-y_2)|=$ Rational .... (i)

Since ΔABC is equilateral

$\therefore$ Area of $\Delta ABC=\frac{\sqrt{3}}{4}(\text{side})^2=\frac{\sqrt{3}}{4}(AB)^2=\frac{\sqrt{3}}{4}\{(x_1-x_2)^2+(y_1-y_2)^2\}=$ Irrational ...... (ii)

From (i) and (ii),

Rational = Irrational

which is contradiction.

Hence x_1, y_1, x_2, y_2, x_3, y_3 cannot all be rational.

8. CONDITIONS FOR COLLINEARITY OF THREE GIVEN POINTS :

Three given points A (x_1, y_1), B (x_2, y_2), C (x_3, y_3) are collinear if any one of the following conditions are satisfied.

(a) Area of triangle ABC is zero i.e. $\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} = 0$

(b) Slope of AB = slope of BC = slope of AC. i.e. $\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_3 - y_2}{x_3 - x_2} = \frac{y_3 - y_1}{x_3 - x_1}$

(c) Find the equation of line passing through 2 given points, if the third point satisfies the given equation of the line, then three points are collinear.

Do yourself - 3 :

1. Find the area of the triangle whose vertices are A(1,1), B(7, – 3) and C(12, 2)

2. Find the area of the quadrilateral whose vertices are A(1,1) B(7, – 3), C(12,2) and D(7, 21)

3. Prove that the points A(a, b + c), B(b, c + a) and C(c, a + b) are collinear (By determinant method)

4. Let A(h, k), B(1, 1) and C(2, 1) be the vertices of a right angled triangle with AC as its hypotenuse. If the area of the triangle is 1 square unit, then the set of values which 'k' can take is given by

(A) {–1, 3} (B) {–3, –2} (C) {1, 3} (D) {0, 2}

5. In a triangle ABC, coodinates of A are (1, 2) and the equations of the medians through B and C are x + y = 5 and x = 4 respectively. Then area of ΔABC (in sq. units) is

(A) 5 (B) 9 (C) 12 (D) 4

6. The point A divides the join of P(–5, 1) & Q(3, 5) in the ratio K : 1. Find the two values of K for which the area of triangle ABC, where B is (1, 5) & C is (7, –2), is equal to 2 units in magnitude.

7. The area of a triangle is 5. Two of its vertices are (2, 1) & (3, –2). The third vertex lies on y = x + 3. Find the third vertex.

9. LOCUS :

The locus of a moving point is the path traced out by that point under one or more geometrical conditions.

(a) Equation of Locus :

The equation to a locus is the relation which exists between the coordinates of any point on the path, and which holds for no other point except those lying on the path.

(b) Procedure for finding the equation of the locus of a point :

(i) If we are finding the equation of the locus of a point P, assign coordinates (h, k) to P.

(ii) Express the given condition as equations in terms of the known quantities to facilitate calculations. We sometimes include some unknown quantities known as parameters.

(iii) Eliminate the parameters, so that the eliminate contains only h, k and known quantities.

(iv) Replace h by x, and k by y, in the eliminate. The resulting equation would be the equation of the locus of P.

Illustration 9 : The ends of the rod of length ℓ moves on two mutually perpendicular lines, find the locus of the point on the rod which divides it in the ratio $m_1 : m_2$. Considering perpendicular lines a coordinate axis, m_1 from x-axis and m_2 from y-axis.

(A) $m_1^2x^2 + m_2^2y^2 = \dfrac{\ell^2}{(m_1+m_2)^2}$ (B) $(m_2x)^2 + (m_1y)^2 = \left(\dfrac{m_1m_2\ell}{m_1+m_2}\right)^2$

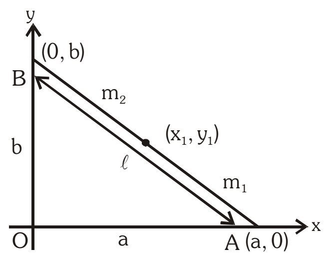
(C) $(m_1x)^2 + (m_2y)^2 = \left(\dfrac{m_1m_2\ell}{m_1+m_2}\right)^2$ (D) none of these

Solution : Let (h,k) be the point that divide the rod $AB = \ell$, in the ratio $m_1 : m_2$, and $OA = a,\ OB = b$ say

$\therefore \quad a^2 + b^2 = \ell^2$ ..... (i)

Now $h = \left(\dfrac{m_2a}{m_1+m_2}\right) \Rightarrow a = \left(\dfrac{m_1+m_2}{m_2}\right)h$

$k = \left(\dfrac{m_1b}{m_1+m_2}\right) \Rightarrow b = \left(\dfrac{m_1+m_2}{m_1}\right)k$



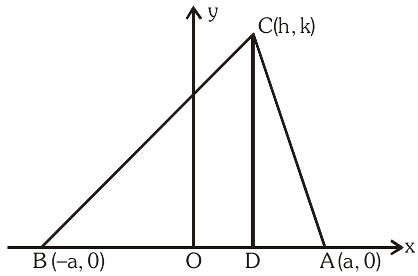
putting these values in (i) $\dfrac{(m_1+m_2)^2}{m_2^2}h^2 + \dfrac{(m_1+m_2)^2}{m_1^2}k^2 = \ell^2$

$\therefore$ Locus of (h,k) is $m_1^2x^2 + m_2^2y^2 = \left(\dfrac{m_1m_2\ell}{m_1+m_2}\right)^2$ **Ans. (C)**

Illustration 10 : A(a, 0) and B(–a, 0) are two fixed points of ΔABC. If its vertex C moves in such a way that cotA + cotB = λ, where λ is a constant, then the locus of the point C is -

(A) $y\lambda = 2a$ (B) $y = \lambda a$ (C) $ya = 2\lambda$ (D) none of these

Solution : Given that coordinates of two fixed points A and B are (a, 0) and (–a, 0) respectively. Let variable point C is (h, k). From the adjoining figure



$$\cot A = \frac{DA}{CD} = \frac{a-h}{k}$$

$$\cot B = \frac{BD}{CD} = \frac{a+h}{k}$$

But cotA + cotB = λ, so we have

$$\frac{a-h}{k} + \frac{a+h}{k} = \lambda \Rightarrow \frac{2a}{k} = \lambda$$

Hence locus of C is $y\lambda = 2a$ **Ans. (A)**

Do yourself - 4 :

1. Find the locus of a variable point which is at a distance of 2 units from the y-axis.

2. Find the locus of a point which is equidistant from both the axes.

3. A point P moves on the line $2x - 3y + 4 = 0$. If Q(1,4) and R(3,–2) are fixed points, then the locus of the centroid of ΔPQR is a line :

(A) parallel to x-axis (B) with slope $\frac{2}{3}$

(C) with slope $\frac{3}{2}$ (D) parallel to y-axis

4. Let O(0, 0) and A(0, 1) be two fixed points. Then the locus of a point P such that the perimeter of ΔAOP is 4, is :

(A) $8x^2 - 9y^2 + 9y = 18$ (B) $9x^2 + 8y^2 - 8y = 16$

(C) $8x^2 + 9y^2 - 9y = 18$ (D) $9x^2 - 8y^2 + 8y = 16$

5. If the equation of the locus of a point equidistant from the point (a_1, b_1) and (a_2, b_2) is $(a_1 - b_2)x + (a_1 - b_2)y + c = 0$, then the value of 'c' is

(A) $\sqrt{a_1^2 + b_1^2 - a_2^2 - b_2^2}$ (B) $\frac{1}{2}a_2^2 + b_2^2 - a_1^2 - b_1^2$

(C) $a_1^2 - a_2^2 + b_1^2 - b_2^2$ (D) $\frac{1}{2}\left(a_1^2 + a_2^2 + b_1^2 + b_2^2\right)$

6. Locus of centroid of the triangle whose vertices are (acost, asint), (bsint, –bcost) and (1, 0), where t is a parameter, is

(A) $(3x + 1)^2 + (3y)^2 = a^2 - b^2$ (B) $(3x - 1)^2 + (3y)^2 = a^2 - b^2$

(C) $(3x - 1)^2 + (3y)^2 = a^2 + b^2$ (D) $(3x + 1)^2 + (3y)^2 = a^2 + b^2$

7. Find the equation to the locus of a point which moves so that its distance from the axis of x is three times its distance from the axis of y.

8. Find the equation to the locus of a point which moves so that the sum of the squares of its distances from the axes is equal to 3.

9. Find the equation to the locus of a point which moves so that its distance from the point (3, 0) is three times its distance from (0, 2).

10. Find the equation to the locus of a point which moves so that its distance from the axis of x is always one half its distance from the origin.

10. STRAIGHT LINE :

Introduction : A relation between x and y which is satisfied by co-ordinates of every point lying on a line is called equation of the straight line. Here, remember that every one degree equation in variable x and y always represents a straight line i.e. $\mathbf{ax + by + c = 0}$ **; a & b** $\mathbf{\neq 0}$ simultaneously.

(a) Equation of a line parallel to x-axis at a distance 'a' is $\mathbf{y = a}$ or $\mathbf{y = -a}$.

(b) Equation of x-axis is $\mathbf{y = 0}$.

(c) Equation of a line parallel to y-axis at a distance 'b' is $\mathbf{x = b}$ or $\mathbf{x = -b}$.

(d) Equation of y-axis is $\mathbf{x = 0}$.

Illustration 11 : Prove that every first degree equation in x, y represents a straight line.

Solution : Let ax + by + c = 0 be a first degree equation in x, y

where a, b, c are constants.

Let $P(x_1, y_1)$ & $Q(x_2, y_2)$ be any two points on the curve represented by ax + by + c = 0.
Then $ax_1 + by_1 + c = 0$ and $ax_2 + by_2 + c = 0$

Let R be any point on the line segment joining P & Q

Suppose R divides PQ in the ratio λ : 1. Then, the coordinates of R are

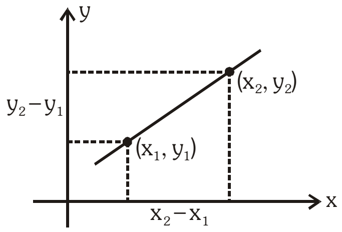
$$\left(\frac{\lambda x_2 + x_1}{\lambda + 1}, \frac{\lambda y_2 + y_1}{\lambda + 1}\right)$$

We have $a\left(\frac{\lambda x_2 + x_1}{\lambda + 1}\right) + b\left(\frac{\lambda y_2 + y_1}{\lambda + 1}\right) + c = \lambda\ 0 + 0 = 0$

$\therefore R\left(\frac{\lambda x_2 + x_1}{\lambda + 1}, \frac{\lambda y_2 + y_1}{\lambda + 1}\right)$ lies on the curve represented by ax + by + c = 0. Thus every point on the line segment joining P & Q lies on ax + by + c = 0.

Hence ax + by + c = 0 represents a straight line.

11. SLOPE OF LINE :



If a given line makes an angle $\theta(0° \le \theta < 180°,\ \theta \ne 90°)$ with the positive direction of x-axis, then slope of this line will be $\tan\theta$ and is usually denoted by the letter **m** i.e. $\mathbf{m = \tan\theta}$. If $A(x_1, y_1)$ and $B(x_2, y_2)$ & $x_1 \ne x_2$ then slope of line AB $= \dfrac{y_2 - y_1}{x_2 - x_1}$

Remark :

(i) If $\theta = 90°$, **m does not exist** and line is parallel to **y-axis.**

(ii) If $\theta = 0°$, **m = 0** and the line is parallel to **x-axis.**

(iii) Let m_1 and m_2 be slopes of two given lines (none of them is parallel to y-axis)

(a) If lines are parallel, $\mathbf{m_1 = m_2}$ and vice-versa.

(b) If lines are perpendicular, $\mathbf{m_1 m_2 = -1}$ and vice-versa

12. STANDARD FORMS OF EQUATIONS OF A STRAIGHT LINE :

(a) **Slope Intercept form :** Let m be the slope of a line and c its intercept on y-axis. Then the equation of this straight line is written as : $y = mx + c$

If the line passes through origin, its equation is written as $y = mx$

(b) **Point Slope form :** If m be the slope of a line and it passes through a point (x_1, y_1), then its equation is written as : $y - y_1 = m(x - x_1)$

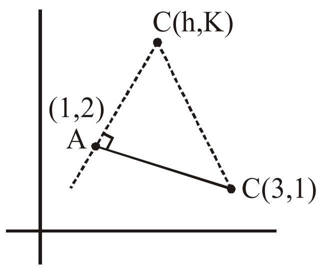
(c) **Two point form :** Equation of a line passing through two points (x_1, y_1) and (x_2, y_2) is written as :

$$y - y_1 = \frac{y_2 - y_1}{x_2 - x_1}(x - x_1) \quad \text{or} \quad \begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0$$

Illustration 12 : A triangle ABC lying in the first quadrant has two vertices as A(1, 2) and B(3, 1). If $\angle BAC = 90°$, and ar(ΔABC) $= 5\sqrt{5}$ sq. units, then the abscissa of the vertex C is :

(A) $2+\sqrt{5}$ (B) $1+\sqrt{5}$ (C) $1+2\sqrt{5}$ (D) $2\sqrt{5}-1$

Solution : $\left(\dfrac{K-2}{h-1}\right)\left(\dfrac{1-2}{3-1}\right) = -1 \Rightarrow K = 2h \quad ...(1)$



$\sqrt{5}\,|h-1| = 10$

$\because\ [\Delta ABC] = 5\sqrt{5}$

$$\Rightarrow \frac{1}{2}\left(\sqrt{5}\right)\sqrt{(h-1)^2+(K-2)^2} = 5\sqrt{5} \quad ....(2)$$

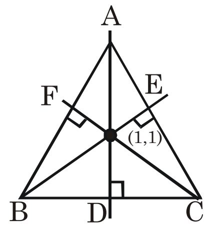
$\Rightarrow$ h = $2\sqrt{5}+1$ (h > 0) **Ans. (C)**

Illustration 13 : Let the equations of two sides of a triangle be $3x - 2y + 6 = 0$ and $4x + 5y - 20 = 0$. If the orthocentre of this triangle is at (1,1), then the equation of its third side is :

(A) $122y - 26x - 1675 = 0$ (B) $26x + 61y + 1675 = 0$

(C) $122y + 26x + 1675 = 0$ (D) $26x - 122y - 1675 = 0$

Solution : Equation of AB is



$3x - 2y + 6 = 0$

equation of AC is

$4x + 5y - 20 = 0$

Equation of BE is

$2x + 3y - 5 = 0$

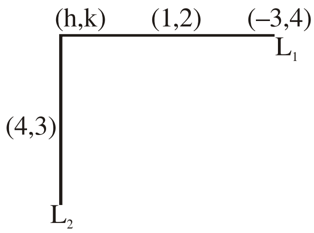
Equation of CF is $5x - 4y - 1 = 0$

$\Rightarrow$ Equation of BC is $26x - 122y = 1675$ **Ans. (D)**

Illustration 14 : Suppose that the points (h,k), (1,2) and (–3,4) lie on the line L_1. If a line L_2 passing through the points (h,k) and (4,3) is perpendicular to L_1, then $\frac{k}{h}$ equals :

(A) 3 (B) $-\frac{1}{7}$ (C) $\frac{1}{3}$ (D) 0

Solution : equation of L_1 is



$$y = -\frac{1}{2}x + \frac{5}{2} \quad ...(1)$$

equation of L_2 is

$$y = 2x - 5 \quad ...(2)$$

by (1) and (2)

$x = 3$

$y = 1 \Rightarrow h = 3, k = 1$

$$\frac{k}{h} = \frac{1}{3}$$ **Ans. (C)**

Do yourself - 5 :

1. If the perpendicular bisector of the line segment joining the points P (1, 4) and Q (k, 3) has y-intercept equal to –4, then a value of k is :-

(A) $\sqrt{15}$ (B) –2 (C) $\sqrt{14}$ (D) –4

2. If in a parallelogram ABDC, the coordinates of A, B and C are respectively (1, 2), (3, 4) and (2, 5), then the equation of the diagonal AD is:-

(A) $5x + 3y - 11 = 0$ (B) $3x - 5y + 7 = 0$

(C) $3x + 5y - 13 = 0$ (D) $5x - 3y + 1 = 0$

3. If the straight line, $2x-3y+17 = 0$ is perpendicular to the line passing through the points (7, 17) and (15, β), then β equals :-

(A) –5 (B) $-\frac{35}{3}$ (C) $\frac{35}{3}$ (D) 5

4. If the two lines $x + (a - 1)y = 1$ and $2x + a^2y = 1(a \in R - \{0, 1\})$ are perpendicular, then the distance of their point of intersection from the origin is :-

(A) $\frac{2}{5}$ (B) $\frac{2}{\sqrt{5}}$ (C) $\frac{\sqrt{2}}{5}$ (D) $\sqrt{\frac{2}{5}}$

5. A point on the straight line, $3x + 5y = 15$ which is equidistant from the coordinate axes will lie only in :

(A) 1^{st} and 2^{nd} quadrants (B) 4^{th} quadrant

(C) 1^{st}, 2^{nd} and 4^{th} quadrant (D) 1^{st} quadrant

6. The circumcentre of a triangle lies at the origin and its centroid is the mid point of the line segment joining the points $(a^2 + 1, a^2 + 1)$ and $(2a, -2a)$, $a \neq 0$. Then for any a, the orthocentre of this triangle lies on the line :

(A) $y - (a^2 + 1)x = 0$ (B) $y + x = 0$

(C) $(a - 1)^2x - (a + 1)^2y = 0$ (D) $y - 2ax = 0$

7. If a line intercepted between the coordinate axes is trisected at a point A (4, 3), which is nearer to x-axis, then its equation is :-

(A) $3x + 8y = 36$ (B) $4x - 3y = 7$

(C) $x + 3y = 13$ (D) $3x + 2y = 18$

8. Given three points P, Q, R with P(5, 3) and R lies on the x-axis. If equation of RQ is $x - 2y = 2$ and PQ is parallel to the x-axis, then the centroid of ΔPQR lies on the line:

(A) $x - 2y + 1 = 0$ (B) $5x - 2y = 0$

(C) $2x + y - 9 = 0$ (D) $2x - 5y = 0$

9. If the straight lines $x + 3y = 4$, $3x + y = 4$ and $x + y = 0$ form a triangle, then the triangle is:-

(A) Scalene (B) Equilateral

(C) Isosceles (D) Right angled isosceles

10. If A(2, –3) and B(–2, 1) are two vertices of a triangle and third vertex moves on the line $2x + 3y = 9$, then the locus of the centroid of the triangle is :

(A) $2x - 3y = 1$ (B) $x - y = 1$ (C) $2x + 3y = 1$ (D) $2x + 3y = 3$

11. Let A(2, –3) and B(–2, 3) be vertices of a triangle ABC. If the centroid of this triangle moves on the line 2x + 3y = 1, then the locus of the vertex C is the line

(A) 3x – 2y = 3 (B) 2x – 3y = 7 (C) 3x + 2y = 5 (D) 2x + 3y = 9

12. If two vertices of a triangle are (5, –1) and (–2, 3) and its orthocentre is at (0, 0), then the third vertex is

(A) (4,–7) (B) (–4, –7) (C) (–4, 7) (D) (4, 7)

13. If two vertical poles 20 m and 80 m high stand apart on a horizontal plane, then the height (in m) of the point of intersection of the lines joining the top of each pole to the foot of other is :-

(A) 16 (B) 15 (C) 50 (D) 18

14. The perpendicular bisector of the line segment joining P (1, 4) and Q (k, 3) has y-intercept –4. Then a possible value of k is

(A) 1 (B) 2 (C) –2 (D) –4

15. If x_1, x_2, x_3 and y_1, y_2, y_3 are both in G.P. with the same common ratio, then the points (x_1, y_1), (x_2, y_2) and (x_3, y_3)

(A) are vertices of a triangle (B) lie on a straight line

(C) lie of an ellipse (D) lie on a circle

16. The diagonals of a parallelogram PQRS are along the lines x + 3y = 4 and 6x – 2y = 7. Then PQRS must be a :

(A) rectangle (B) square

(C) cyclic quadrilateral (D) rhombus

17. Number of lines that can be drawn through the point(4,–5) so that its distance from (–2,3) will be equal to 12 is equal to-

(A) 0 (B) 1 (C) 2 (D) 3

18. A line is such that its segment between the straight lines 5x – y – 4 = 0 and 3x + 4y – 4 = 0 is bisected at the point (1, 5). Obtain the equation.

19. The points (1, 3) & (5, 1) are two opposite vertices of a rectangle. The other two vertices lie on the line y = 2x + c. Find c & the remaining vertices.

20. Consider a triangle ABC with sides AB and AC having the equations $L_1 = 0$ and $L_2 = 0$. Let the centroid, orthocentre and circumcentre of the ΔABC are G, H and S respectively. L = 0 denotes the equation of sides BC.

(a) If $L_1 : 2x – y = 0$ and $L_2 : x + y = 3$ and G(2, 3) then find the slope of the line L = 0.

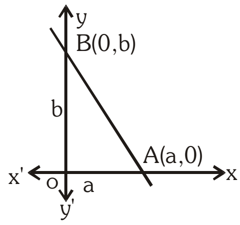
(b) If $L_1 : 2x + y = 0$ and $L_2 : x – y + 2 = 0$ and H(2, 3) then find the y-intercept of L = 0.

(c) If $L_1 : x + y – 1= 0$ and $L_2 : 2x – y + 4 = 0$ and S(2, 1) then find the x-intercept of the line L = 0.

21. Two equal sides of an isosceles triangle are given by the equations 7x − y + 3 = 0 and x + y − 3 = 0 & its third side passes through the point (1, −10). Determine the equation of the third side.

22. Two consecutive sides of a parallelogram are 4x + 5y = 0 & 7x + 2y = 0. If the equation to one diagonal is 11x + 7y = 9, find the equation to the other diagonal.

(d) Intercept form : If a and b are the intercepts made by a line on the axes of x and y, its equation is written as : $\frac{x}{a}+\frac{y}{b}=1$



(i) Length of intercept of line between the coordinate axes $=\sqrt{a^2+b^2}$

(ii) Area of triangle AOB $=\frac{1}{2}OA.OB=\left|\frac{1}{2}ab\right|$

Illustration 15 : The equation of the lines which passes through the point (3, 4) and the sum of its intercepts on the axes is 14 is -

(A) $4x - 3y = 24$, $x - y = 7$ (B) $4x + 3y = 24$, $x + y = 7$

(C) $4x + 3y + 24 = 0$, $x + y + 7=0$ (D) $4x - 3y + 24 = 0$, $x - y + 7 = 0$

Solution : Let the equation of the line be $\frac{x}{a}+\frac{y}{b}=1$ .....(i)

This passes through (3 , 4) , therefore $\frac{3}{a}+\frac{4}{b}=1$ .....(ii)

It is given that $a + b = 14 \Rightarrow b = 14 - a$. Putting $b = 14 - a$ in (ii) , we get

$\frac{3}{a}+\frac{4}{14-a}=1 \Rightarrow a^2 - 13a + 42 = 0 \Rightarrow (a - 7)(a - 6) = 0 \Rightarrow a = 7 , 6$

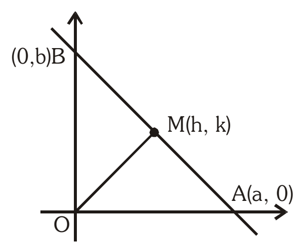
For a = 7 , b = 14 – 7 = 7 and for a = 6, b = 14 – 6 = 8

Putting the values of a and b in (i) , we get the equations of the lines

$\frac{x}{7}+\frac{y}{7}=1$ and $\frac{x}{6}+\frac{y}{8}=1$ or $x+ y = 7$ and $4x + 3y = 24$ **Ans. (B)**

Illustration 16 : Two points A and B move on the positive direction of x-axis and y-axis respectively, such that OA + OB = K. Show that the locus of the foot of the perpendicular from the origin O on the line AB is $(x + y)(x^2 + y^2) = Kxy$.

Solution : Let the equation of AB be $\frac{x}{a}+\frac{y}{b}=1$ .... (i)



given, $a + b = K$ .... (ii)

now, $m_{AB} \times m_{OM} = -1 \Rightarrow ah = bk$ .... (iii)

from (ii) and (iii),

$a=\frac{kK}{h+k}$ and $b=\frac{hK}{h+k}$

$\therefore$ from (i) $\dfrac{x(h+k)}{k.K}+\dfrac{y(h+k)}{h.K}=1$

as it passes through (h, k)

$\dfrac{h(h+k)}{k.K}+\dfrac{k(h+k)}{h.K}=1 \Rightarrow \quad (h + k)(h^2 + k^2) = Khk$

$\therefore$ locus of (h, k) is $(x + y)(x^2 + y^2) = Kxy$.

(e) **Normal form :** If p is the length of perpendicular on a line from the origin, and α the angle which this perpendicular makes with positive x-axis, then the equation of this line is written as : $\mathbf{x\cos\alpha + y\sin\alpha = p}$ (p is always positive) where $\mathbf{0 \le \alpha < 2\pi}$.

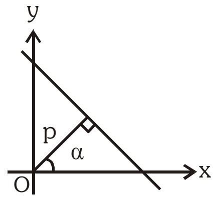
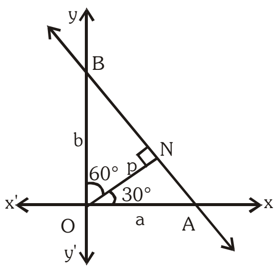


Illustration 17 : Find the equation of the straight line on which the perpendicular from origin makes an angle 30° with positive x-axis and which forms a triangle of area $\left(\dfrac{50}{\sqrt{3}}\right)$ sq. units with the co-ordinates axes.

Solution : $\angle NOA = 30°$

Let ON = p > 0, OA = a, OB = b



In ΔONA, $\cos 30° = \dfrac{ON}{OA}=\dfrac{p}{a} \Rightarrow \dfrac{\sqrt{3}}{2}=\dfrac{p}{a}$

or $a = \dfrac{2p}{\sqrt{3}}$

and in ΔONB, $\cos 60° = \dfrac{ON}{OB}=\dfrac{p}{b} \Rightarrow \dfrac{1}{2}=\dfrac{p}{b}$

or $b = 2p$

$\because$ Area of $\Delta OAB = \dfrac{1}{2}\ ab = \dfrac{1}{2}\left(\dfrac{2p}{\sqrt{3}}\right)(2p)=\dfrac{2p^2}{\sqrt{3}}$

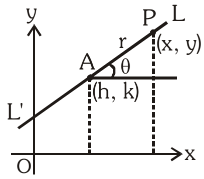
$\therefore \quad \dfrac{2p^2}{\sqrt{3}}=\dfrac{50}{\sqrt{3}} \quad \Rightarrow \quad p^2 = 25$

or $p = 5$

$\therefore$ Using $x\cos\alpha + y\sin\alpha = p$, the equation of the line AB is $x\cos 30° + y \sin 30° = 5$

or $x\sqrt{3} + y = 10$

(f) Parametric form : To find the equation of a straight line which passes through a given point A(h, k) and makes a given angle θ with the positive direction of the x-axis. P(x, y) is any point on the line LAL'.



Let AP = r, then $\mathbf{x - h = r\cos\theta,\ y - k = r\sin\theta}$ & $\frac{x-h}{\cos\theta} = \frac{y-k}{\sin\theta} = \mathbf{r}$ is the equation of the straight line LAL'.

Any point P on the line will be of the form $(h + r\cos\theta, k + r\sin\theta)$, where $|r|$ gives the distance of the point P from the fixed point (h, k).

Illustration 18 : Equation of a line which passes through point A(2, 3) and makes an angle of 45° with x axis. If this line meet the line x + y + 1 = 0 at point P then distance AP is -

(A) $2\sqrt{3}$ (B) $3\sqrt{2}$ (C) $5\sqrt{2}$ (D) $2\sqrt{5}$

Solution : Here $x_1 = 2$, $y_1 = 3$ and $\theta = 45°$ hence $\frac{x-2}{\cos 45°} = \frac{y-3}{\sin 45°} = r$

from first two parts $\Rightarrow$ $x - 2 = y - 3 \Rightarrow x - y + 1 = 0$

Co-ordinate of point P on this line is $\left(2 + \frac{r}{\sqrt{2}}, 3 + \frac{r}{\sqrt{2}}\right)$.

If this point is on line x + y + 1 = 0 then

$\left(2 + \frac{r}{\sqrt{2}}\right) + \left(3 + \frac{r}{\sqrt{2}}\right) + 1 = 0 \quad \Rightarrow r = -3\sqrt{2} \quad ; \quad |r| = 3\sqrt{2}$ **Ans. (B)**

Illustration 19 : A variable line is drawn through O, to cut two fixed straight lines L_1 and L_2 in A_1 and A_2, respectively. A point A is taken on the variable line such that $\frac{m+n}{OA} = \frac{m}{OA_1} + \frac{n}{OA_2}$.

Show that the locus of A is a straight line passing through the point of intersection of L_1 and L_2 where O is being the origin.

Solution : Let the variable line passing through the origin is $\frac{x}{\cos\theta} = \frac{y}{\sin\theta} = r_i$ ...... (i)

Let the equation of the line L_1 is $p_1x + q_1y = 1$ ...... (ii)

Equation of the line L_2 is $p_2x + q_2y = 1$ ...... (iii)

the variable line intersects the line (ii) at A_1 and (iii) at A_2.

Let $OA_1 = r_1$.

Then $A_1 = (r_1\cos\theta, r_1\sin\theta) \Rightarrow A_1$ lies on L_1

$$\Rightarrow \quad r_1 = OA_1 = \frac{1}{p_1\cos\theta + q_1\sin\theta}$$

Similarly, $r_2 = OA_2 = \dfrac{1}{p_2\cos\theta + q_2\sin\theta}$

Let OA = r

Let co-ordinate of A are (h, k) $\Rightarrow$ $(h, k) \equiv (r\cos\theta, r\sin\theta)$

Now $\dfrac{m+n}{r} = \dfrac{m}{OA_1} + \dfrac{n}{OA_2} \Rightarrow \dfrac{m+n}{r} = \dfrac{m}{r_1} + \dfrac{n}{r_2}$

$$\Rightarrow \quad m + n = m(p_1r\cos\theta + q_1r\sin\theta) + n(p_2r\cos\theta + q_2r\sin\theta)$$

$$\Rightarrow \quad (p_1h + q_1k - 1) + \frac{n}{m}(p_2h + q_2k - 1) = 0$$

Therefore, locus of A is $(p_1x+q_1y - 1) + \dfrac{n}{m}(p_2x + q_2y - 1) = 0$

$$\Rightarrow \quad L_1 + \lambda L_2 = 0 \text{ where } \lambda = \frac{n}{m}.$$

This is the equation of the line passing through the intersection of L_1 and L_2.

Illustration 20 : A straight line through P(–2, –3) cuts the pair of straight lines $x^2 + 3y^2 + 4xy - 8x - 6y - 9 = 0$ in Q and R. Find the equation of the line if PQ. PR = 20.

Solution : Let line be $\dfrac{x+2}{\cos\theta} = \dfrac{y+3}{\sin\theta} = r$

$\Rightarrow \quad x = r\cos\theta - 2,\ y = r\sin\theta - 3$ .... (i)

Now, $x^2 + 3y^2 + 4xy - 8x - 6y - 9 = 0$ .... (ii)

Taking intersection of (i) with (ii) and considering terms of r^2 and constant (as we need PQ . PR = $r_1 . r_2$ = product of the roots)

$r^2(\cos^2\theta + 3\sin^2\theta + 4\sin\theta\cos\theta)$ + (some terms)r + 80 = 0

$$\therefore \quad r_1.r_2 = PQ.\ PR = \frac{80}{\cos^2\theta + 4\sin\theta\cos\theta + 3\sin^2\theta}$$

$\therefore \quad \cos^2\theta + 4\sin\theta\cos\theta + 3\sin^2\theta = 4$ $\quad (\because$ PQ . PR = 20)

$\therefore \quad \sin^2\theta - 4\sin\theta\cos\theta + 3\cos^2\theta = 0$

$\Rightarrow \quad (\sin\theta - \cos\theta)(\sin\theta - 3\cos\theta) = 0$

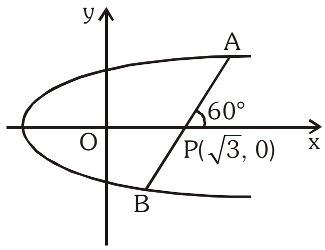
$\therefore \quad \tan\theta = 1, \tan\theta = 3$

hence equation of the line is $y + 3 = 1(x + 2) \Rightarrow x - y = 1$

and $y + 3 = 3(x + 2) \Rightarrow 3x - y + 3 = 0$.

Illustration 21 : If the line $y - \sqrt{3}x + 3 = 0$ cuts the parabola $y^2 = x + 2$ at A and B, then find the value of PA.PB {where $P \equiv (\sqrt{3}, 0)$}

Solution : Slope of line $y - \sqrt{3}x + 3 = 0$ is $\sqrt{3}$.



If line makes an angle θ with x-axis, then $\tan\theta = \sqrt{3}$

$\therefore \quad \theta = 60°$

$$\frac{x-\sqrt{3}}{\cos 60°} = \frac{y-0}{\sin 60°} = r \Rightarrow \left(\sqrt{3} + \frac{r}{2}, \frac{r\sqrt{3}}{2}\right)$$ be a point on the parabola $y^2 = x + 2$

then $\frac{3}{4}r^2 = \sqrt{3} + \frac{r}{2} + 2 \Rightarrow 3r^2 - 2r - 4(2 + \sqrt{3}) = 0$

$$\therefore \quad PA.PB = r_1r_2 = \left|\frac{-4\left(2+\sqrt{3}\right)}{3}\right| = \frac{4\left(2+\sqrt{3}\right)}{3}$$

Do yourself - 6 :

1. Reduce the line $2x - 3y + 5 = 0$,

(a) In slope- intercept form and hence find slope & Y-intercept

(b) In intercept form and hence find intercepts on the axes.

(c) In normal form and hence find perpendicular distance from the origin and angle made by the perpendicular with the positive x-axis.

2. Find distance of point A (2, 3) measured parallel to the line $x - y = 5$ from the line $2x + y + 6 = 0$.

3. If a straight line passing thourgh the point P(–3, 4) is such that its intercepted portion between the coordinate axes is bisected at P, then its equation is :

(A) $x - y + 7 = 0$ (B) $3x - 4y + 25 = 0$

(C) $4x + 3y = 0$ (D) $4x - 3y + 24 = 0$

4. A square of each side 2, lies above the x-axis and has one vertex at the origin. If one of the sides passing through the origin makes an angle 30° with the positive direction of the x-axis, then the sum of the x-coordinates of the vertices of the square is :

(A) $\sqrt{3} - 2$ (B) $\sqrt{3} - 1$ (C) $2\sqrt{3} - 1$ (D) $2\sqrt{3} - 2$

5. A straight line through the point A(3, 4) is such that its intercept between the axes is bisected at A. Its equation is

(A) $x + y = 7$ (B) $3x - 4y + 7 = 0$

(C) $4x + 3y = 24$ (D) $3x + 4y = 25$

6. The equation of the straight lines passing through the point (4, 3) and making intercepts on the co-ordinates axes whose sum is –1 is

(A) $\frac{x}{2}-\frac{y}{3}=1$ and $\frac{x}{-2}+\frac{y}{1}=1$ (B) $\frac{x}{2}-\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

(C) $\frac{x}{2}+\frac{y}{3}=1$ and $\frac{x}{2}+\frac{y}{1}=1$ (D) $\frac{x}{2}+\frac{y}{3}=-1$ and $\frac{x}{-2}+\frac{y}{1}=-1$

7. Locus of mid point of the portion between the axes of $x\cos\alpha + y\sin\alpha = p$ where p is constant is

(A) $x^2+y^2=\frac{4}{p^2}$ (B) $x^2+y^2=4p^2$ (C) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{2}{p^2}$ (D) $\frac{1}{x^2}+\frac{1}{y^2}=\frac{4}{p^2}$

8. Area of the quadrilateral formed by the lines $|x| + |y| = 2$ is :
(A) 8 (B) 6 (C) 4 (D) none

9. If the x intercept of the line $y = mx + 2$ is greater than 1/2 then the gradient of the line lies in the interval-
(A) (–1,0) (B) (–1/4,0) (C) $(-\infty,-4)$ (D) (–4,0)

10. A line is drawn through the point (1, 2) to meet the coordinate axes at P and Q such that it forms a triangle OPQ, where O is the origin. If the area of the triangle OPQ is least, then the slope of the line PQ is :

(1) $-\frac{1}{2}$ (2) $-\frac{1}{4}$ (3) –4 (4) –2

11. If the x-intercept of some line L is double as that of the line, $3x + 4y = 12$ and the y-intercept of L is half as that of the same line, then the slope of L is :-
(1) –3 (2) –3/2 (3) –3/8 (4) –3/16

(g) **General form :** We know that a first degree equation in x and y, $ax + by + c = 0$ always represents a straight line. This form is known as general form of straight line.

(i) Slope of this line $=\frac{-a}{b}=-\frac{\text{coeff. of } x}{\text{coeff. of } y}$

(ii) Intercept by this line on x-axis $=-\frac{c}{a}$ and intercept by this line on y-axis $=-\frac{c}{b}$

(iii) To change the general form of a line to normal form, first take c to right hand side and make it positive, then divide the whole equation by $\sqrt{a^2+b^2}$.

13. EQUATION OF LINES PARALLEL AND PERPENDICULAR TO A GIVEN LINE :

(a) Equation of line parallel to line $ax + by + c = 0$

$\mathbf{ax + by + \lambda = 0}$

(b) Equation of line perpendicular to line $ax + by + c = 0$

$\mathbf{bx - ay + k = 0}$

Here λ, k, are parameters and their values are obtained with the help of additional information given in the problem.

14. ANGLE BETWEEN TWO LINES :

(a) If θ be the angle between two lines : $y = m_1x + c_1$ and $y = m_2x + c_2$, then $\tan\theta = \pm\left(\dfrac{m_1 - m_2}{1+m_1m_2}\right)$

Note :

(i) There are two angles formed between two lines but usually the acute angle is taken as the angle between the lines. So we shall find θ from the above formula only by taking positive value of $\tan\theta$.

(ii) Let m_1, m_2, m_3 are the slopes of three lines $L_1 = 0$; $L_2 = 0$; $L_3 = 0$ where $m_1 > m_2 > m_3$ then the interior angles of the Δ ABC found by these formulas are given by,

$$\tan A = \frac{m_1 - m_2}{1 + m_1 m_2}\ ;\ \tan B = \frac{m_2 - m_3}{1 + m_2 m_3}\ \ \&\ \tan C = \frac{m_3 - m_1}{1 + m_3 m_1}$$

(b) If equation of lines are $a_1x+b_1y+c_1=0$ and $a_2x+b_2y+c_2=0$, then these lines are -

(i) Parallel $\Leftrightarrow$ $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} \neq \dfrac{c_1}{c_2}$

(ii) Perpendicular $\Leftrightarrow$ $a_1a_2+b_1b_2=0$

(iii) Coincident $\Leftrightarrow$ $\dfrac{a_1}{a_2} = \dfrac{b_1}{b_2} = \dfrac{c_1}{c_2}$

Illustration 22 : If $x + 4y - 5 = 0$ and $4x + ky + 7 = 0$ are two perpendicular lines then k is -

(A) 3 (B) 4 (C) –1 (D) –4

Solution : $m_1 = -\dfrac{1}{4}$ $m_2 = -\dfrac{4}{k}$

Two lines are perpendicular if $m_1 m_2 = -1$

$\Rightarrow \left(-\dfrac{1}{4}\right)\times\left(-\dfrac{4}{k}\right) = -1 \Rightarrow k = -1$ **Ans. (C)**

Illustration 23 : A line L passes through the points (1, 1) and (0, 2) and another line M which is perpendicular to L passes through the point (0, –1/2).The area of the triangle formed by these lines with y-axis is -

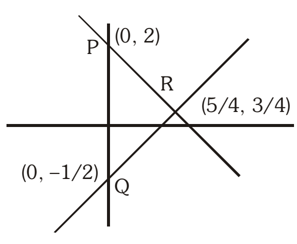
(A) 25/8 (B) 25/16 (C) 25/4 (D) 25/32

Solution : Equation of the line L is $y - 1 = \dfrac{-1}{1}(x - 1) \Rightarrow y = -x + 2$

Equation of the line M is $y = x - 1/2$.

If these lines meet y-axis at P and Q, then PQ = 5/2.

Also x-coordinate of their point of intersection R = 5/4



$\therefore$ area of the $\Delta PQR = \dfrac{1}{2}\left(\dfrac{5}{2}\times\dfrac{5}{4}\right) = 25/16.$ **Ans. (B)**

Illustration 24 : If the straight line $3x + 4y + 5 - k(x + y + 3) = 0$ is parallel to y-axis, then the value of k is -

(A) 1 (B) 2 (C) 3 (D) 4

Solution : A straight line is parallel to y-axis, if its y - coefficient is zero, i.e. $4 - k = 0$ i.e. $k = 4$

Ans. (D)

15. STRAIGHT LINE MAKING A GIVEN ANGLE WITH A LINE :

Equation of line passing through a point (x_1,y_1) and making an angle α, with the line $y=mx+c$ is written as :

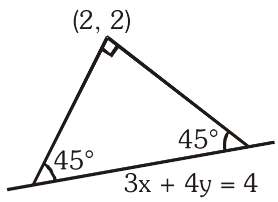
$$y - y_1 = \frac{m \pm \tan\alpha}{1 \mp m\tan\alpha}(x - x_1)$$

Illustration 25 : Find the equation to the sides of an isosceles right-angled triangle, the equation of whose hypotenuse is $3x + 4y = 4$ and the opposite vertex is the point $(2, 2)$.

Solution : The problem can be restated as :

Find the equations of the straight lines passing through the given point $(2, 2)$ and making equal angles of $45°$ with the given straight line $3x + 4y - 4 = 0$. Slope of the line $3x + 4y - 4 = 0$ is $m_1 = -3/4$.

$$\Rightarrow \quad \tan 45° = \pm\frac{m - m_1}{1 + m_1 m}, \text{ i.e., } 1 = \pm\frac{m + 3/4}{1 - \frac{3}{4}m}$$



$m_A = \frac{1}{7}$, and $m_B = -7$

Hence the required equations of the two lines are

$y - 2 = m_A(x - 2)$ and $y - 2 = m_B(x - 2)$

$\Rightarrow \quad 7y - x - 12 = 0$ and $7x + y = 16$ **Ans.**

Do yourself - 7 :

1. Find the angle between the lines $3x + y - 7 = 0$ and $x + 2y - 9 = 0$.

2. Find the line passing through the point $(2, 3)$ and perpendicular to the straight line $4x - 3y = 10$.

3. Find the equation of the line which has positive y-intercept 4 units and is parallel to the line $2x - 3y - 7 = 0$. Also find the point where it cuts the x-axis.

4. Classify the following pairs of lines as coincident, parallel or intersecting :

(a) $x + 2y - 3 = 0$ & $-3x - 6y + 9 = 0$

(b) $x + 2y + 1 = 0$ & $2x + 4y + 3 = 0$

(c) $3x - 2y + 5 = 0$ & $2x + y - 5 = 0$

5. Consider the straight lines

$L_1 : x - y = 1$

$L_2 : x + y = 1$

$L_3 : 2x + 2y = 5$

$L_4 : 2x - 2y = 7$

The correct statement is :

(A) $L_1 \perp L_2$, $L_2 \parallel L_3$, L_1 intersects L_4
(B) $L_1 \perp L_2$, $L_1 \parallel L_3$, L_1 intersects L_2
(C) $L_1 \perp L_2$, $L_1 \perp L_3$, L_2 intersects L_4
(D) $L_1 \parallel L_4$, $L_2 \parallel L_3$, L_2 intersects L_3

6. A straight line L through the point (3,–2) is inclined at an angle of 60° to the line $\sqrt{3}x + y = 1$. If L also intersects the x-axis, then the equation of L is :-

(A) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$
(B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$
(C) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$
(D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

7. If a, b, c $\in$ R and 1 is a root of the equation $ax^2 + bx + c = 0$, then the curve $y = 4ax^2 + 3bx + 2c$, $a \neq 0$ intersects x-axis at

(A) no point
(B) exactly two distinct point
(C) exactly one point
(D) two distinct points whose coordinates are always rational numbers

8. The point of intersection of the lines $(a^3 + 3)x + ay + a - 3 = 0$ and $(a^5 + 2)x + (a + 2)y + 2a + 3 = 0$ (a real) lies on the y-axis for :-

(A) No value of a
(B) Exactly two values of a
(C) More than two values of a
(D) Exactly one value of a

9. The lines $p(p^2 + 1)x - y + q = 0$ and $(p^2 + 1)^2x + (p^2 + 1)y + 2q = 0$ are perpendicular to a common line for :

(A) exactly one values of p
(B) exactly two values of p
(C) more than two values of p
(D) no value of p

10. A straight line L is perpendicular to the line $5x - y = 1$. The area of the triangle formed by the line L & the coordinate axes is 5. Find the equation of the line.

16. LENGTH OF PERPENDICULAR FROM A POINT ON A LINE :

Length of perpendicular from a point (x_1,y_1) on the line $ax + by + c = 0$ is $\left|\dfrac{ax_1 + by_1 + c}{\sqrt{a^2 + b^2}}\right|$

In particular, the length of the perpendicular from the origin on the line $ax + by + c = 0$ is $P = \dfrac{|c|}{\sqrt{a^2 + b^2}}$

Illustration 26 : If the algebraic sum of perpendiculars from n given points on a variable straight line is zero then prove that the variable straight line passes through a fixed point.

Solution : Let n given points be (x_i, y_i) where i = 1, 2.... n and the variable straight line is $ax + by + c = 0$.

Given that $\sum_{i=1}^{n}\left(\frac{ax_i + by_i + c}{\sqrt{a^2+b^2}}\right) = 0 \Rightarrow a\Sigma x_i + b\Sigma y_i + cn = 0 \Rightarrow a\frac{\Sigma x_i}{n} + b\frac{\Sigma y_i}{n} + c = 0$.

Hence the variable straight line always passes through the fixed point $\left(\frac{\Sigma x_i}{n}, \frac{\Sigma y_i}{n}\right)$. **Ans.**

Illustration 27 : Prove that no line can be drawn through the point (4, –5) so that its distance from (–2, 3) will be equal to 12.

Solution : Suppose, if possible.

Equation of line through (4, –5) with slope m is $y + 5 = m(x - 4)$

$\Rightarrow \quad mx - y - 4m - 5 = 0$

Then $\frac{|m(-2) - 3 - 4m - 5|}{\sqrt{m^2+1}} = 12$

$\Rightarrow \quad |-6m - 8| = 12\sqrt{(m^2+1)}$

On squaring, $\quad (6m + 8)^2 = 144(m^2 + 1)$

$\Rightarrow \quad 4(3m + 4)^2 = 144(m^2 + 1) \Rightarrow (3m + 4)^2 = 36(m^2 + 1)$

$\Rightarrow \quad 27m^2 - 24m + 20 = 0 \qquad$ .... (i)

Since the discriminant of (i) is $(-24)^2 - 4.27.20 = -1584$ which is negative, there is no real value of m. Hence no such line is possible.

17. DISTANCE BETWEEN TWO PARALLEL LINES :

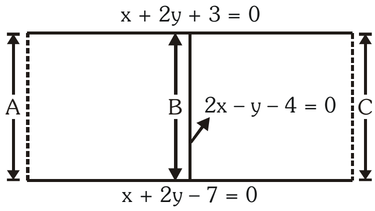
(a) The distance between two parallel lines $ax + by + c_1=0$ and $ax+by+c_2=0$ is $= \frac{|c_1 - c_2|}{\sqrt{a^2+b^2}}$

(**Note :** The coefficients of x & y in both equations should be same)

(b) The area of the parallelogram $= \frac{p_1 p_2}{\sin\theta}$, where p_1 & p_2 are distances between two pairs of opposite sides & θ is the angle between any two adjacent sides. Note that area of the parallelogram bounded by the lines $y = m_1x + c_1$, $y = m_1x + c_2$ and $y = m_2x + d_1$, $y = m_2x + d_2$ is given by $\left|\frac{(c_1 - c_2)(d_1 - d_2)}{m_1 - m_2}\right|$.

Illustration 28 : Three lines $x + 2y + 3 = 0$, $x + 2y - 7 = 0$ and $2x - y - 4 = 0$ form 3 sides of two squares. Find the equation of remaining sides of these squares.

Solution : Distance between the two parallel lines is $\frac{|7+3|}{\sqrt{5}} = 2\sqrt{5}$.



The equations of sides A and C are of the form
$2x - y + k = 0$.
Since distance between sides A and B
= distance between sides B and C

$$\Rightarrow \quad \frac{|k-(-4)|}{\sqrt{5}} = 2\sqrt{5} \quad \Rightarrow \quad \frac{k+4}{\sqrt{5}} = \pm 2\sqrt{5} \quad \Rightarrow \quad k = 6, -14.$$

Hence the fourth sides of the two squares are (i) $2x - y + 6 = 0$ (ii) $2x - y - 14 = 0$.

Ans.

Do yourself - 8 :

1. Find the distances between the following pair of parallel lines :

(a) $3x + 4y = 13$, $3x + 4y = 3$

(b) $3x - 4y + 9 = 0$, $6x - 8y - 15 = 0$

2. Find the points on the x-axis such that their perpendicular distance from the line $\frac{x}{a} + \frac{y}{b} = 1$ is 'a', $a, b > 0$.

3. Show that the area of the parallelogram formed by the lines

$2x - 3y + a = 0$, $3x - 2y - a = 0$, $2x - 3y + 3a = 0$ and $3x - 2y - 2a = 0$ is $\frac{2a^2}{5}$ square units.

4. If the line, $2x - y + 3 = 0$ is at a distance $\frac{1}{\sqrt{5}}$ and $\frac{2}{\sqrt{5}}$ from the lines $4x - 2y + \alpha = 0$ and $6x - 3y + \beta = 0$, respectively, then the sum of all possible values of α and β is _____

5. A straight line through origin O meets the lines $3y = 10 - 4x$ and $8x + 6y + 5 = 0$ at points A and B respectively. Then O divides the segment AB in the ratio :

(A) 2 : 3 (B) 1 : 2 (C) 4 : 1 (D) 3 : 4

6. The point (2, 1) is translated parallel to the line $L : x - y = 4$ by $2\sqrt{3}$ units. If the new point Q lies in the third quadrant, then the equation of the line passing through Q and perpendicular to L is :

(A) $x + y = 3 - 3\sqrt{6}$ (B) $x + y = 3 - 2\sqrt{6}$

(C) $x + y = 2 - \sqrt{6}$ (4) $2x + 2y = 1 - \sqrt{6}$

7. If a line L is perpendicular to the line $5x - y = 1$, and the area of the triangle formed by the line L and the coordinate axes is 5, then the distance of line L from the line $x + 5y = 0$ is:-

(1) $\frac{7}{\sqrt{5}}$ (2) $\frac{5}{\sqrt{13}}$ (3) $\frac{7}{\sqrt{13}}$ (4) $\frac{5}{\sqrt{7}}$

8. The base of an equilateral triangle is along the line given by 3x + 4y = 9. If a vertex of the triangle is (1, 2), then the length of a side of the triangle is :

(A) $\frac{4\sqrt{3}}{15}$ (B) $\frac{4\sqrt{3}}{5}$ (C) $\frac{2\sqrt{3}}{15}$ (D) $\frac{2\sqrt{3}}{5}$

9. The shortest distance between the line y – x = 1 and the curve $x = y^2$ is :-

(A) $\frac{3\sqrt{2}}{5}$ (B) $\frac{\sqrt{3}}{4}$ (C) $\frac{3\sqrt{2}}{8}$ (D) $\frac{2\sqrt{3}}{8}$

18. POSITION OF TWO POINTS WITH RESPECT TO A GIVEN LINE :

Let the given line be ax + by + c = 0 and $P(x_1, y_1)$, $Q(x_2, y_2)$ be two points. If the expressions $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ have the same signs, then both the points P and Q lie on the same side of the line ax + by + c = 0. If the quantities $ax_1 + by_1 + c$ and $ax_2 + by_2 + c$ have opposite signs, then they lie on the opposite sides of the line.

19. CONCURRENCY OF LINES :

(a) Three lines $a_1x + b_1y + c_1 = 0$; $a_2x + b_2y + c_2 = 0$ and $a_3x + b_3y + c_3 = 0$ are concurrent,

if $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_2 & c_3 \end{vmatrix} = 0$

(b) To test the concurrency of three lines, first find out the point of intersection of any two of the three lines. If this point lies on the remaining line (i.e. coordinates of the point satisfy the equation of the line) then the three lines are concurrent otherwise not concurrent.

Illustration 29 : If the lines ax + by + p = 0, xcosα + ysinα – p = 0 (p ≠ 0) and xsinα – ycosα = 0 are concurrent and the first two lines include an angle $\frac{\pi}{4}$, then $a^2 + b^2$ is equal to -

(A) 1 (B) 2 (C) 4 (D) p^2

Solution : Since the given lines are concurrent,

$$\begin{vmatrix} a & b & p \\ \cos\alpha & \sin\alpha & -p \\ \sin\alpha & -\cos\alpha & 0 \end{vmatrix} = 0$$

$\Rightarrow \quad a\cos\alpha + b\sin\alpha + 1 = 0$ .... (i)

As ax + by + p = 0 and $x\cos\alpha + y\sin\alpha - p = 0$ include an angle $\frac{\pi}{4}$.

$$\pm\tan\frac{\pi}{4}=\frac{-\frac{a}{b}+\frac{\cos\alpha}{\sin\alpha}}{1+\frac{a}{b}\frac{\cos\alpha}{\sin\alpha}}$$

$\Rightarrow$ $-a\sin\alpha + b\cos\alpha = \pm(b\sin\alpha + a\cos\alpha)$

$\Rightarrow$ $-a\sin\alpha + b\cos\alpha = \pm 1$ [from (i)] .... (ii)

Squaring and adding (i) & (ii), we get

$a^2 + b^2 = 2.$ **Ans. (B)**

Do yourself - 9 :

1. Examine the positions of the points (3, 4) and (2, – 6) w.r.t. $3x - 4y = 8$

2. If (2, 9), (– 2, 1) and (1, –3) are the vertices of a triangle, then prove that the origin lies inside the triangle.

3. Find the equation of the line joining the point (2, – 9) and the point of intersection of lines $2x + 5y - 8 = 0$ and $3x - 4y - 35 = 0$.

4. Find the value of λ, if the lines $3x - 4y - 13 = 0$, $8x - 11y - 33 = 0$ and $2x - 3y + \lambda = 0$ are concurrent.

5. The set of all possible values of θ in the interval $(0, \pi)$ for which the points (1, 2) and $(\sin\theta, \cos\theta)$ lie on the same side of the line $x + y = 1$ is :

(A) $\left(0,\frac{\pi}{4}\right)$ (B) $\left(0,\frac{3\pi}{4}\right)$ (C) $\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$ (D) $\left(0,\frac{\pi}{2}\right)$

6. If the point (1, a) lies in between the straight lines $x + y = 1$ and $2(x + y) = 3$ then a lies in interval :-

(A) $\left(1,\frac{3}{2}\right)$ (B) $\left(0,\frac{1}{2}\right)$ (C) $(-\infty, 0)$ (D) $\left(\frac{3}{2},\infty\right)$

7. If (a, a^2) falls inside the angle made by the lines $y=\frac{x}{2}$, $x > 0$ and $y = 3x$, $x > 0$, then a belong to

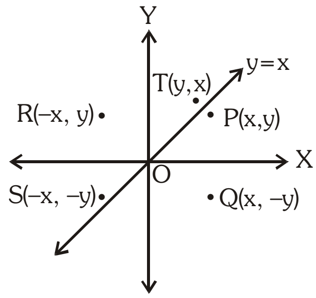
(A) $\left(0,\frac{1}{2}\right)$ (B) $(3, \infty)$ (C) $\left(\frac{1}{2},3\right)$ (D) $\left(-3,-\frac{1}{2}\right)$

8. If the three distinct lines $x + 2ay + a = 0$, $x + 3by + b = 0$ and $x + 4ay + a = 0$ are concurrent, then the point (a, b) lies on a :-

(1) circle (2) straight line (3) parabola (4) hyperbola

20. REFLECTION OF A POINT :

Let P(x, y) be any point, then its image with respect to



(a) x-axis is Q(x, –y)

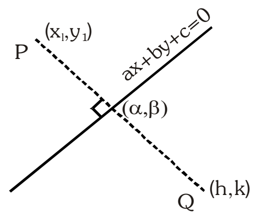
(b) y-axis is R(–x, y)

(c) origin is S(–x,–y)

(d) line y = x is T(y, x)

(e) Reflection of a point about any arbitrary line : The image (h,k) of a point $P(x_1, y_1)$ about the line ax + by + c = 0 is given by following formula.

$$\frac{h-x_1}{a}=\frac{k-y_1}{b}=-2\frac{(ax_1+by_1+c)}{a^2+b^2}$$

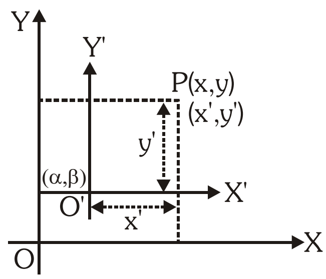


and the foot of perpendicular (α,β) from a point (x_1, y_1) on the line ax + by + c = 0 is given by following formula.

$$\frac{\alpha-x_1}{a}=\frac{\beta-y_1}{b}=-\frac{ax_1+by_1+c}{a^2+b^2}$$

21. TRANSFORMATION OF AXES

(a) Shifting of origin without rotation of axes :



Let P (x, y) with respect to axes OX and OY.

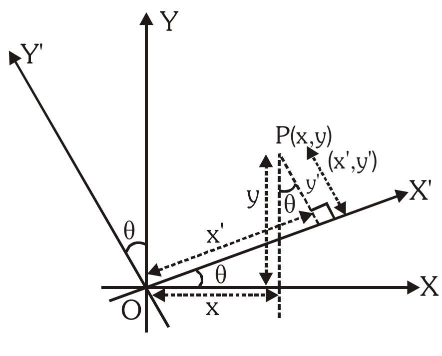
Let O' (α, β) is new origin with respect to axes OX and OY and let P (x', y') with respect to axes O'X' and O'Y' , where OX and O'X' are parallel and OY and O'Y' are parallel.

Then **x = x' + α, y = y' + β**

or **x' = x – α, y' = y – β**

Thus if origin is shifted to point (α, β) without rotation of axes, then new equation of curve can be obtained by putting x + α in place of x and y + β in place of y.

(b) Rotation of axes without shifting the origin :



Let O be the origin. Let P (x, y) with respect to axes OX and OY and let P (x', y') with respect to axes OX' and OY' where ∠X'OX = ∠YOY' = θ, where θ is measured in anticlockwise direction.

then x = x' cos θ – y' sin θ

y = x' sin θ + y' cos θ

and x' = x cos θ + y sin θ

y' = –x sin θ + y cos θ

The above relation between (x, y) and (x', y') can be easily obtained with the help of following table

New \ Old	x ↓	y ↓
x' →	cos θ	sin θ
y' →	−sin θ	cos θ

Illustration 30 : Through what angle should the axes be rotated so that the equation $9x^2 - 2\sqrt{3}\,xy + 7y^2 = 10$ may be changed to $3x^2 + 5y^2 = 5$?

Solution : Let angle be θ then replacing (x, y) by (x cosθ – y sinθ, x sinθ + y cosθ)

then $9x^2 - 2\sqrt{3}xy + 7y^2 = 10$ becomes

$$9(x\cos\theta - y\sin\theta)^2 - 2\sqrt{3}\left(x\cos\theta - y\sin\theta\right)\left(x\sin\theta + y\cos\theta\right) + 7(x\sin\theta + y\cos\theta)^2 = 10$$

$$\Rightarrow \quad x^2(9\cos^2\theta - 2\sqrt{3}\sin\theta\cos\theta + 7\sin^2\theta) + 2xy(-9\sin\theta\cos\theta - \sqrt{3}\cos 2\theta + 7\sin\theta\cos\theta)$$

$$+ y^2(9\cos^2\theta + 2\sqrt{3}\sin\theta\cos\theta + 7\cos^2\theta) = 10$$

On comparing with $3x^2 + 5y^2 = 5$ (coefficient of xy = 0)

We get $\quad -9\sin\theta\cos\theta - \sqrt{3}\cos 2\theta + 7\sin\theta\cos\theta = 0$

or $\quad \sin 2\theta = -\sqrt{3}\cos 2\theta \quad$ or $\quad \tan 2\theta = -\sqrt{3} = \tan(180° - 60°)$

or $\quad 2\theta = 120° \quad \therefore \quad \theta = 60°$

Do yourself - 10 :

1. The point (4, 1) undergoes the following transformations, then the match the correct alternatives:

Column-I		Column-II
(A) Reflection about x-axis is	(p)	(4, –1)
(B) Reflection about y-axis is	(q)	(–4, –1)
(C) Reflection about origin is	(r)	$\left(-\frac{12}{25}, -\frac{59}{25}\right)$
(D) Reflection about the line y = x is	(s)	(–4, 1)
(E) Reflection about the line 4x + 3y – 5 = 0 is	(t)	(1, 4)

2. To what point must the origin be shifted, so that the coordinates of the point (4, 5) become (– 3, 9).

3. If the axes be turned through an angle $\tan^{-1}2$ (in anticlockwise direction),what does the equation $4xy - 3x^2 = a^2$ become ?

4. Let L denote the line in the xy-plane with x and y intercepts as 3 and 1 respectively. Then the image of the point (–1, –4) in this line is :

(A) $\left(\frac{8}{5}, \frac{29}{5}\right)$ (B) $\left(\frac{29}{5}, \frac{11}{5}\right)$ (C) $\left(\frac{11}{5}, \frac{28}{5}\right)$ (D) $\left(\frac{29}{5}, \frac{8}{5}\right)$

5. A ray of light is incident along a line which meets another line, $7x - y + 1 = 0$, at the point (0, 1). The ray is then reflected from this point along the line, $y + 2x = 1$. Then the equation of the line of incidence of the ray of light is :-

(A) $41x - 25y + 25 = 0$ (B) $41x + 25y - 25 = 0$

(C) $41x + 38y - 38 = 0$ (D) $41x - 38y + 38 = 0$

6. Let L be the line passing through the point P(1, 2) such that its intercepted segment between the co-ordinate axes is bisected at P. If L_1 is the line perpendicular to L and passing through the point (–2, 1), then the point of intersection of L and L_1 is :

(A) $\left(\frac{3}{10}, \frac{17}{5}\right)$ (B) $\left(\frac{11}{20}, \frac{29}{10}\right)$ (C) $\left(\frac{4}{5}, \frac{12}{5}\right)$ (D) $\left(\frac{3}{5}, \frac{23}{10}\right)$

7. A ray of light coming from the point $\left(2, 2\sqrt{3}\right)$ is incident at an angle 30° on the line x=l at the point A. The ray gets reflected on the line x = 1 and meets x-axis at the point B. Then, the line AB passes through the point:

(A) $\left(3, -\frac{1}{\sqrt{3}}\right)$ (B) $\left(3, -\sqrt{3}\right)$ (C) $\left(4, -\frac{\sqrt{3}}{2}\right)$ (D) $\left(4, -\sqrt{3}\right)$

8. The foot of the perpendicular drawn from the origin, on the line, $3x + y = \lambda(\lambda \neq 0)$ is P. If the line meets x-axis at A and y–axis at B, then the ratio BP : PA is :-

(A) 9 : 1 (B) 1 : 3 (C) 3 : 1 (D) 1 : 9

9. A ray of light passing through the point A(1,2) is reflected at a point B on the x-axis and then passes through (5,3). Then the equation of AB is :

(A) $5x + 4y = 13$ (B) $5x - 4y = -3$ (C) $4x + 5y = 14$ (D) $4x - 5y = -6$

10. The equations of the perpendicular bisectors of the sides AB & AC of a triangle ABC are $x - y + 5 = 0$ & $x + 2y = 0$, respectively. If the point A is (1, –2) find the equation of the line BC.

11. Let P be the point (3, 2). Let Q be the reflection of P about the x-axis. Let R be the reflection of Q about the line $y = -x$ and let S be the reflection of R through the origin. PQRS is a convex quadrilateral. Find the area of PQRS.

12. A ray of light along $x + \sqrt{3}y = \sqrt{3}$ gets reflected upon reaching x-axis, the equation of the reflected ray is :

(A) $y = x + \sqrt{3}$ (B) $\sqrt{3}y = x - \sqrt{3}$ (C) $y = \sqrt{3}x - \sqrt{3}$ (D) $\sqrt{3}y = x - 1$

13. If the image of point P(2, 3) in a line L is Q (4, 5) then, the image of point R (0, 0) in the same line is :

(A) (4, 5) (B) (2, 2) (C) (3, 4) (D) (7, 7)

14. If a circle of radius R passes through the origin O and intersects the coordinate axes at A and B, then the locus of the foot of perpendicular from O on AB is :

(A) $(x^2 + y^2)^2 = 4Rx^2y^2$ (B) $(x^2 + y^2)(x + y) = R^2xy$

(C) $(x^2 + y^2)^3 = 4R^2x^2y^2$ (D) $(x^2 + y^2)^2 = 4R^2x^2y^2$

22. EQUATION OF BISECTORS OF ANGLES BETWEEN TWO LINES :

If equation of two intersecting lines are $a_1x+b_1y+c_1=0$ and $a_2x + b_2y+c_2=0$, then equation of bisectors of the angles between these lines are written as :

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}} = \pm\frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}} \qquad \text{..........(i)}$$

(a) Equation of bisector of angle containing origin :

If the equation of the lines are written with constant terms $\mathbf{c_1}$ and $\mathbf{c_2}$ positive, then the equation of the bisectors of the angle containing the origin is obtained by taking positive sign in (i)

(b) Equation of bisector of acute/obtuse angles :

To find the equation of the bisector of the acute or obtuse angle :

(i) let ϕ be the angle between one of the two bisectors and one of two given lines. Then if $\mathbf{\tan\phi < 1}$ i.e. $\phi < 45°$ i.e. $2\phi < 90°$, the angle bisector will be bisector of acute angle.

(ii) See whether the constant terms c_1 and c_2 in the two equation are +ve or not. If not then multiply both sides of given equation by –1 to make the constant terms positive.

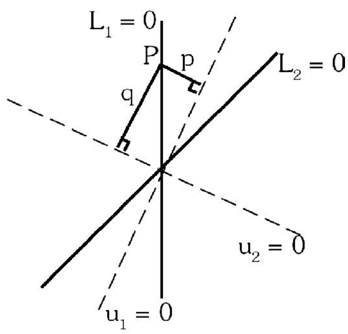
Determine the sign of $\mathbf{a_1a_2 + b_1b_2}$

If sign of $a_1a_2 + b_1b_2$	For obtuse angle bisector	For acute angle bisector
+	use + sign in eq. (1)	use – sign in eq. (1)
–	use – sign in eq. (1)	use + sign in eq. (1)

i.e. if $a_1a_2 + b_1b_2 > 0$, then the bisector corresponding to + sign gives obtuse angle bisector

$$\frac{a_1x+b_1y+c_1}{\sqrt{a_1^2+b_1^2}} = \frac{a_2x+b_2y+c_2}{\sqrt{a_2^2+b_2^2}}$$

(iii) Another way of identifying an acute and obtuse angle bisector is as follows :



Let $L_1 = 0$ & $L_2 = 0$ are the given lines & $u_1 = 0$ and $u_2 = 0$ are the bisectors between $L_1 = 0$ & $L_2 = 0$. Take a point P on any one of the lines $L_1 = 0$ or $L_2 = 0$ and drop perpendicular on $u_1 = 0$ & $u_2 = 0$ as shown . If,

$|p| < |q| \Rightarrow u_1$ is the acute angle bisector .

$|p| > |q| \Rightarrow u_1$ is the obtuse angle bisector .

$|p| = |q| \Rightarrow$ the lines L_1 & L_2 are perpendicular .

Note : Equation of straight lines passing through $P(x_1, y_1)$ & equally inclined with the lines $a_1x + b_1y + c_1 = 0$ & $a_2x + b_2y + c_2 = 0$ are those which are parallel to the bisectors between these two lines & passing through the point P.

Illustration 31 : For the straight lines 4x + 3y – 6 = 0 and 5x + 12y + 9 = 0, find the equation of the

(i) bisector of the obtuse angle between them.

(ii) bisector of the acute angle between them.

(iii) bisector of the angle which contains origin.

(iv) bisector of the angle which contains (1, 2).

Solution : Equations of bisectors of the angles between the given lines are

$$\frac{4x+3y-6}{\sqrt{4^2+3^2}}=\pm\frac{5x+12y+9}{\sqrt{5^2+12^2}} \Rightarrow 9x-7y-41=0 \text{ and } 7x+9y-3=0$$

If θ is the acute angle between the line 4x + 3y – 6 = 0 and the bisector

$$9x-7y-41=0, \text{ then } \tan\theta=\left|\frac{-\frac{4}{3}-\frac{9}{7}}{1+\left(\frac{-4}{3}\right)\frac{9}{7}}\right|=\frac{11}{3}>1$$

Hence

(i) bisector of the obtuse angle is 9x – 7y – 41 = 0

(ii) bisector of the acute angle is 7x + 9y – 3 = 0

(iii) bisector of the angle which contains origin

$$\frac{-4x-3y+6}{\sqrt{(-4)^2+(-3)^2}}=\frac{5x+12y+9}{\sqrt{5^2+12^2}} \Rightarrow 7x+9y-3=0$$

(iv) $L_1(1, 2) = 4 \times 1 + 3\times 2 - 6 = 4 > 0$

$L_2(1, 2) = 5 \times 1 + 12 \times 2 + 9 = 38 > 0$

+ve sign will give the required bisector, $\dfrac{4x+3y-6}{5}=+\dfrac{5x+12y+9}{13}$

⇒ 9x – 7y – 41 = 0.

Alternative :

Making c_1 and c_2 positive in the given equation, we get –4x – 3y + 6 = 0 and 5x + 12y + 9 = 0

Since $a_1a_2 + b_1b_2 = -20 - 36 = -56 < 0$, so the origin will lie in the acute angle.

Hence bisector of the acute angle is given by

$$\frac{-4x-3y+6}{\sqrt{4^2+3^2}}=\frac{5x+12y+9}{\sqrt{5^2+12^2}} \Rightarrow 7x+9y-3=0$$

Similarly bisector of obtuse angle is 9x – 7y – 41 = 0.

Illustration 32 : A ray of light is sent along the line x – 2y – 3 = 0. Upon reaching the line mirror 3x – 2y – 5 = 0, the ray is reflected from it. Find the equation of the line containing the reflected ray.

Solution : Let Q be the point of intersection of the incident ray and the line mirror, then

$$x_1 - 2y_1 - 3 = 0 \ \ \& \ \ 3x_1 - 2y_1 - 5 = 0$$

on solving these equations, we get

$$x_1 = 1 \ \ \& \ \ y_1 = -1$$

Since P(–1, –2) be a point lies on the incident ray, so we can find the image of the point P on the reflected ray about the line mirror (by property of reflection).

Let P'(h, k) be the image of point P about line mirror, then

$$\frac{h+1}{3} = \frac{k+2}{-2} = \frac{-2(-3+4-5)}{13} \Rightarrow h = \frac{11}{13} \text{ and } k = \frac{-42}{13}.$$

So $P'\left(\frac{11}{13}, \frac{-42}{13}\right)$

Then equation of reflected ray will be

$$(y+1) = \frac{\left(\frac{-42}{13}+1\right)(x-1)}{\left(\frac{11}{13}-1\right)}$$

$\Rightarrow$ 2y – 29x + 31 = 0.

23. FAMILY OF LINES :

If equation of two lines be $P \equiv a_1x + b_1y + c_1 = 0$ and $Q \equiv a_2x + b_2y + c_2 = 0$, then the equation of the lines passing through the point of intersection of these lines is : $P + \lambda Q = 0$ or $a_1x + b_1y + c_1 + \lambda(a_2x + b_2y + c_2) = 0$. The value of λ is obtained with the help of the additional informations given in the problem.

Illustration 33 : Prove that each member of the family of straight lines

$(3\sin\theta + 4\cos\theta)x + (2\sin\theta - 7\cos\theta)y + (\sin\theta + 2\cos\theta) = 0$ (θ is a parameter) passes through a fixed point.

Solution : The given family of straight lines can be rewritten as

$(3x + 2y + 1)\sin\theta + (4x - 7y + 2)\cos\theta = 0$

or, $(4x - 7y + 2) + \tan\theta(3x + 2y + 1) = 0$ which is of the form $L_1 + \lambda L_2 = 0$

Hence each member of it will pass through a fixed point which is the intersection of

$4x - 7y + 2 = 0$ and $3x + 2y + 1 = 0$ i.e. $\left(\frac{-11}{29}, \frac{2}{29}\right)$.

Do yourself - 11 :

1. Find the equations of bisectors of the angle between the lines $4x + 3y = 7$ and $24x + 7y - 31 = 0$. Also find which of them is
(a) the bisector of the angle containing origin (b) the bisector of the acute angle.

2. Find the equations of the line which pass through the point of intersection of the lines $4x - 3y = 1$ and $2x - 5y + 3 = 0$ and is equally inclined to the coordinate axes.

3. Find the equation of the line through the point of intersection of the lines $3x - 4y + 1 = 0$ & $5x + y - 1 = 0$ and perpendicular to the line $2x - 3y = 10$.

4. Consider the set of all lines $px + qy + r = 0$ such that $3p + 2q + 4r = 0$. Which one of the following statements is true ?
(A) The lines are all parallel. (B) Each line passes through the origin.
(C) The lines are not concurrent (D) The lines are concurrent at the point $\left(\frac{3}{4},\frac{1}{2}\right)$

5. If a variable line drawn through the intersection of the lines $\frac{x}{3}+\frac{y}{4} = 1$ and $\frac{x}{4}+\frac{y}{3}=1$, meets the coordinate axes at A and B, $(A \neq B)$, then the locus of the midpoint of AB is :-
(A) $14(x + y)^2 - 97(x + y) + 168 = 0$ (B) $6xy = 7(x + y)$
(C) $7xy = 6(x + y)$ (D) $4(x + y)^2 - 28 (x + y) + 49 = 0$

6. The line parallel to x-axis and passing through the point of intersection of the lines $ax + 2by + 3b = 0$ and $bx - 2ay - 3a = 0$, where $(a, b) \neq (0, 0)$ is :
(A) below x-axis at a distance 2/3 from it (B) above x-axis at a distance 3/2 from it
(C) above x-axis at a distance 2/3 from it (D) below x-axis at a distance 3/2 from it

7. The sides of a rhombus ABCD are parallel to the lines, $x - y + 2 = 0$ and $7x - y + 3 = 0$. If the diagonals of the rhombus intersect at P (1,2) and the vertex A (different from the origin) is on the y–axis, then the ordinate of A is :-
(A) 2 (B) $\frac{5}{2}$ (C) $\frac{7}{4}$ (D) $\frac{7}{2}$

8. Let $P = (-1, 0)$, $Q = (0, 0)$ and $R = \left(3, 3\sqrt{3}\right)$ be three point. The equation of the bisector of the angle PQR is
(A) $\frac{\sqrt{3}}{2}x + y = 0$ (B) $x + \sqrt{3y} = 0$ (C) $\sqrt{3}x + y = 0$ (D) $x + \frac{\sqrt{3}}{2}y = 0$

9. A square of side a lies above the x-axis and has one vertex at the origin. The side passing through the origin makes an angle $\alpha\left(0 < \alpha < \frac{\pi}{4}\right)$ with the positive direction of x-axis. The equation of its diagonal not passing through the origin is
(A) $y(\cos\alpha + \sin\alpha) + x(\cos\alpha - \sin\alpha) = a$ (B) $y(\cos\alpha - \sin\alpha) - x(\sin\alpha - \cos\alpha) = a$
(C) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha - \cos\alpha) = a$ (D) $y(\cos\alpha + \sin\alpha) + x(\sin\alpha + \cos\alpha) = a$

10. Consider the family of lines $(x - y - 6) + \lambda(2x + y + 3) = 0$ and $(x + 2y - 4) + \lambda'(3x - 2y - 4) = 0$. If the lines of these 2 families are at right angle to each other then find the locus of their point of intersection.

11. Given vertices $A(1, 1)$, $B(4, -2)$ & $C(5, 5)$ of a triangle, find the equation of the perpendicular dropped from C to the interior bisector of the angle A.

24. PAIR OF STRAIGHT LINES :

(a) Homogeneous equation of second degree :

(i) Let us consider the homogeneous equation of 2nd degree as

$ax^2 + 2hxy + by^2 = 0$ ....(i)

which represents pair of straight lines passing through the origin.

Now, we divide by x^2, we get

$$a + 2h\left(\frac{y}{x}\right) + b\left(\frac{y}{x}\right)^2 = 0$$

$$\frac{y}{x} = m \quad \text{(say)}$$

then $a + 2hm + bm^2 = 0$ ....(ii)

if m_1 & m_2 are the roots of equation (ii), then $m_1 + m_2 = -\frac{2h}{b}$, $m_1m_2 = \frac{a}{b}$

and also, $$\tan\theta = \left|\frac{m_1 - m_2}{1 + m_1m_2}\right| = \left|\frac{\sqrt{(m_1+m_2)^2 - 4m_1m_2}}{1 + m_1m_2}\right| = \left|\frac{\sqrt{\frac{4h^2}{b^2} - \frac{4a}{b}}}{1 + \frac{a}{b}}\right| = \pm\frac{2\sqrt{h^2 - ab}}{a+b}$$

These line will be :

(1) **Real and different**, if $h^2 - ab > 0$ (2) **Real and coincident**, if $h^2 - ab = 0$

(3) **Imaginary**, if $h^2 - ab < 0$

(ii) The condition that these lines are :

(1) At **right angles** to each other is $\mathbf{a + b = 0}$. i.e. coefficient of $\mathbf{x^2}$ + coefficient of $\mathbf{y^2}$ = 0.

(2) **Coincident** is $\mathbf{h^2 = ab}$.

(3) **Equally inclined** to the axes of x is $\mathbf{h = 0}$. i.e. coefficient of $\mathbf{xy = 0}$.

(iii) Homogeneous equation of 2^{nd} degree $\mathbf{ax^2 + 2hxy + by^2 = 0}$ always represent a pair of straight lines whose equations are

$$y = \left(\frac{-h \pm \sqrt{h^2 - ab}}{b}\right)x \equiv y = m_1x \;\&\; y = m_2x \quad \text{and} \quad m_1 + m_2 = -\frac{2h}{b} \;;\; m_1m_2 = \frac{a}{b}$$

These straight lines passes through the origin.

(iv) Pair of straight lines perpendicular to the lines $ax^2 + 2hxy + by^2 = 0$ and through origin are given by $bx^2 - 2hxy + ay^2 = 0$.

(v) The product of the perpendiculars drawn from the point (x_1, y_1) on the lines $ax^2 + 2hxy + by^2 = 0$ is $\left|\frac{ax_1^2 + 2hx_1y_1 + by_1^2}{\sqrt{(a-b)^2 + 4h^2}}\right|$

Note : A homogeneous equation of degree n represents n straight lines passing through **origin**.

(b) The combined equation of angle bisectors :

The combined equation of angle bisectors between the lines represented by homogeneous equation of 2nd degree is given by $\frac{x^2-y^2}{a-b}=\frac{xy}{h}$, $\mathbf{a \neq b, h \neq 0}$.

Note :

(i) If $\mathbf{a = b}$, the bisectors are $\mathbf{x^2 - y^2 = 0}$ i.e. $\mathbf{x - y = 0, x + y = 0}$

(ii) If $\mathbf{h = 0}$, the bisectors are $\mathbf{xy = 0}$ i.e. $\mathbf{x = 0, y = 0}$.

(iii) The two bisectors are always at **right angles**, since we have coefficient of $\mathbf{x^2}$ + coefficient of $\mathbf{y^2 = 0}$

(c) General Equation and Homogeneous Equation of Second Degree :

(i) The general equation of second degree $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines, if $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ i.e. $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix} = 0$

(ii) If θ be the angle between the lines, then $\tan\theta = \pm\frac{2\sqrt{h^2-ab}}{a+b}$

Obviously these lines are

(1) Parallel, if $\Delta = 0$, $h^2 = ab$ or if $h^2 = ab$ and $bg^2 = af^2$

(2) Perpendicular, if $a + b = 0$ i.e. coeff. of x^2 + coeff. of $y^2 = 0$.

(iii) The product of the perpendiculars drawn from the origin to the lines

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ is $\left|\frac{c}{\sqrt{(a-b)^2+4h^2}}\right|$

Illustration 34 : If $\lambda x^2 - 10xy + 12y^2 + 5x - 16y - 3 = 0$ represents a pair of straight lines, then λ is equal to -

(A) 4 (B) 3 (C) 2 (D) 1

Solution : Here $a = \lambda$, $b = 12$, $c = -3$, $f = -8$, $g = 5/2$, $h = -5$

Using condition $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$, we have

$\lambda(12)(-3) + 2(-8)(5/2)(-5) - \lambda(64) - 12(25/4) + 3(25) = 0$

$\Rightarrow -36\lambda + 200 - 64\lambda - 75 + 75 = 0 \quad \Rightarrow \quad 100\lambda = 200$

$\therefore \quad \lambda = 2$ **Ans. (C)**

Do yourself - 12 :

1. Prove that the equation $x^2 - 5xy + 4y^2 = 0$ represents two lines passing through the origin. Also find their equations.

2. If the equation $3x^2 + kxy - 10y^2 + 7x + 19y = 6$ represents a pair of lines, find the value of k.

3. If the equation $6x^2 - 11xy - 10y^2 - 19y - 6 = 0$ represents a pair of lines, find their equations. Also find the angle between the two lines.

4. Find the point of intersection and the angle between the lines given by the equation :
$2x^2 - 3xy - 2y^2 + 10x + 5y = 0$.

5. The equation $y = \sin x \sin(x + 2) - \sin^2(x+1)$ represents a straight line lying in :
(A) second and third quadrants only (B) third and fourth quadrants only
(C) first, third and fourth quadrants (D) first, second and fourth quadrants

6. If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is
(A) 1 (B) 2 (C) –1/2 (D) –2

7. If one of the lines given by $6x^2 - xy + 4cy^2 = 0$ is $3x + 4y = 0$, then c equals
(A) –3 (B) 1 (C) 3 (D) 1

8. If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then c has the value
(A) –2 (B) –1 (C) 2 (D) 1

9. If the pair of straight lines $x^2 - 2pxy - y^2 = 0$ and $x^2 - 2qxy - y^2 = 0$ be such that each pair bisects the angle between the other pair, then
(A) $pq = -1$ (B) $p = q$ (C) $p = -q$ (D) $pq = 1$

10. The pair of lines represented by $3ax^2 + 5xy + (a^2 - 2)y^2 = 0$ are perpendicular to each other for
(A) two values of a (B) $\forall$ a
(C) for one value of a (D) for no values of a

11. If the pair of lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ intersect on the y-axis then
(A) $2fgh = bg^2 + ch^2$ (B) $bg^2 \neq ch^2$
(C) $abc = 2fgh$ (D) none of these

12. If the line $y = mx$ bisects the angle between the lines $ax^2 + 2h\,xy + by^2 = 0$ then m is a root of the quadratic equation :
(A) $hx^2 + (a - b)x - h = 0$ (B) $x^2 + h(a - b)x - 1 = 0$
(C) $(a - b)x^2 + hx - (a - b) = 0$ (D) $(a - b)x^2 - hx - (a - b) = 0$

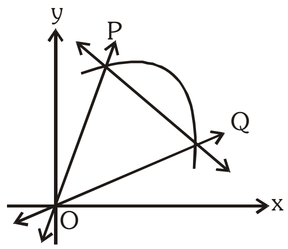
13. The greatest slope along the graph represented by the equation $4x^2 - y^2 + 2y - 1 = 0$, is-
(A) –3 (B) –2 (C) 2 (D) 3

14. Consider a line pair $2x^2 + 3xy - 2y^2 - 10x + 15y - 28 = 0$ and another line L passing through origin with gradient 3. The line pair and line L form a triangle whose vertices are A, B and C.
(a) Find the sum of the cotangents of the interior angles of the triangle ABC.
(b) Find the area of triangle ABC
(c) Find the radius of the circle touching all the 3 sides internally of the triangle.

25. EQUATIONS OF LINES JOINING THE POINTS OF INTERSECTION OF A LINE AND A CURVE TO THE ORIGIN :

(a) Let the equation of curve be :

$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ...(i)



and straight line be

$lx + my + n = 0$ ...(ii)

Now joint equation of line OP and OQ joining the origin and points of intersection P and Q can be obtained by making the equation (i) homogenous with the help of equation of the line. Thus required equation is given by

$$ax^2+2hxy+by^2+2(gx+fy)\left(\frac{\ell x+my}{-n}\right)+c\left(\frac{\ell x+my}{-n}\right)^2=0$$

$\Rightarrow$ $(an^2 + 2gln + cl^2)x^2 + 2(hn^2 + gmn + fln + clm)xy + (bn^2 + 2fmn + cm^2)y^2 = 0$ ......(iii)

All points which satisfy (i) and (ii) simultaneously, will satisfy (iii)

(b) Any second degree curve through the four points of intersection of $f(x, y) = 0$ & $xy = 0$ is given by $f(x, y) + \lambda xy = 0$ where $f(x, y) = 0$ is also a second degree curve.

Illustration 35 : The chord $\sqrt{6}y = \sqrt{8}px + \sqrt{2}$ of the curve $py^2 + 1 = 4x$ subtends a right angle at origin then find the value of p.

Solution : $\sqrt{3}y - 2px = 1$ is the given chord. Homogenizing the equation of the curve, we get,

$py^2 - 4x(\sqrt{3}y - 2px)+(\sqrt{3}y - 2px)^2 = 0$

$\Rightarrow$ $(4p^2 + 8p)x^2 + (p + 3)y^2 - 4\sqrt{3}xy - 4\sqrt{3}pxy = 0$

Now, angle at origin is 90°

$\therefore$ coefficient of x^2 + coefficient of $y^2 = 0$

$\therefore$ $4p^2 + 8p + p + 3 = 0 \Rightarrow 4p^2 + 9p + 3 = 0$

$\therefore$ $p = \dfrac{-9 \pm \sqrt{81-48}}{8} = \dfrac{-9 \pm \sqrt{33}}{8}$.

Do yourself - 13 :

1. Find the angle subtended at the origin by the intercept made on the curve $x^2 - y^2 - xy + 3x - 6y + 18 = 0$ by the line $2x - y = 3$.

2. Find the equation of the lines joining the origin to the points of intersection of the curve $2x^2 + 3xy - 4x + 1 = 0$ and the line $3x + y = 1$.

3. A straight line is drawn from the point (1, 0) to the curve $x^2 + y^2 + 6x - 10y + 1 = 0$, such that the intercept made on it by the curve subtends a right angle at the origin. Find the equations of the line.

4. Find the equation to the pair of straight lines joining the origin to the intersections of the straight line $y = mx + c$ and the curve $x^2 + y^2 = a^2$. Prove that they are at right angles if $2c^2 = a^2(1 + m^2)$.

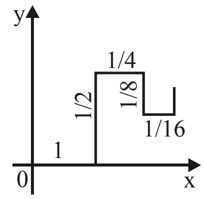
5. Prove that the straight lines joining the origin to the points of intersection of the straight line $kx + hy = 2hk$ with the curve $(x - h)^2 + (y - k)^2 = c^2$ are at right angles if $h^2 + k^2 = c^2$.

6. If line $2x - y - 1 = 0$ intersect the curve $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ in two points A and B, such that $\angle AOB = 90°$, then $\left|\frac{a+b+5c}{2g-f}\right|$ is equal to

EXERCISE (O-1)

Straight Objective Type

1. A particle begins at the origin and moves successively in the following manner as shown,



1 unit to the right, 1/2 unit up, 1/4 unit to the right,

1/8 unit down, 1/16 unit to the right etc.

The length of each move is half the length of the previous move and movement continues in the 'zigzag' manner infinitely. The co-ordinates of the point to which the 'zigzag' converges is -

(A) (4/3, 2/3) (B) (4/3,2/5) (C) (3/2,2/3) (D) (2,2/5)

SL0001

2. Suppose ABC is a triangle with 3 acute angles A,B and C. The point whose coordinates are (cosB – sinA, sinB – cosA) can be in the -

(A) first and 2nd quadrant (B) second the 3rd quadrant

(C) third and 4th quadrant (D) second quadrant only

SL0002

3. The length of a line segment AB is 10 units. If the coordinates of one extremity are (2,–3) and the abscissa of the other extremity is 10 then the sum of all possible values of the ordinate of the other extremity is -

(A) 3 (B) –4 (C) 12 (D) –6

SL0006

4. Consider the points P(2,–4); Q(4,–2) and R(7,1). The points P,Q,R -

(A) form an equilateral triangle

(B) form a right angled triangle

(C) form an isosceles triangle which is not equilateral

(D) are collinear.

SL0007

5. If in triangle ABC, A ≡ (1,10), circumcenter ≡ $\left(-\frac{1}{3},\frac{2}{3}\right)$ and orthocenter ≡ $\left(\frac{11}{3},\frac{4}{3}\right)$ then the co-ordinates of mid-point of side opposite to A is-

(A) (1,–11/3) (B) (1,5) (C) (1,–3) (D) (1,6)

SL0014

6. The co-ordinates of the orthocentre of the triangle bounded by the lines, 4x – 7y + 10 = 0; x + y = 5 and 7x + 4y = 15 is-

(A) (2,1) (B) (–1,2) (C) (1,2) (D) (1,–2)

SL0015

7. A triangle has two of its vertices at (0,1) and (2,2) in the cartesian plane. Its third vertex lies on the x-axis. If the area of the triangle is 2 square units then the sum of the possible abscissae of the third vertex, is-

(A) –4 (B) 0 (C) 5 (D) 6

SL0016

8. A line passes through (2,2) and cuts a triangle of area 9 square units from the first quadrant. The sum of all possible values for the slope of such a line, is-

(A) –2.5 (B) –2 (C) –1.5 (D) –1

SL0017

9. Let A(2,–3) and B(–2,1) be vertices of a ΔABC. If the centroid of ΔABC moves on the line $2x + 3y = 1$, then the locus of the vertex C is-

(A) $2x + 3y = 9$ (B) $2x - 3y = 7$

(C) $3x + 2y = 5$ (D) $3x - 2y = 3$

SL0018

10. A stick of length 10 units rests against the floor and a wall of a room. If the stick begins to slide on the floor then the locus of its middle point is :

(A) $x^2 + y^2 = 2.5$ (B) $x^2 + y^2 = 25$

(C) $x^2 + y^2 = 100$ (D) none

SL0019

11. A and B are any two points on the positive x and y axis respectively satisfying 2(OA) + 3(OB) = 10. If P is the middle point of AB then the locus of P is-

(A) $2x + 3y = 5$ (B) $2x + 3y = 10$

(C) $3x + 2y = 5$ (D) $3x + 2y = 10$

SL0020

12. Given the points A(0,4) and B(0,–4), the equation of the locus of the point P such that $|AP - BP| = 6$ is -

(A) $9x^2 - 7y^2 + 63 = 0$ (B) $9x^2 - 7y^2 - 63 = 0$

(C) $7x^2 - 9y^2 + 63 = 0$ (D) $7x^2 - 9y^2 - 63 = 0$

SL0022

13. If x_1,y_1 are the roots of $x^2 + 8x - 20 = 0$, x_2,y_2 are the roots of $4x^2 + 32x - 57 = 0$ and x_3,y_3 are the roots of $9x^2 + 72x - 112 = 0$, then the points, (x_1,y_1), (x_2,y_2) and (x_3,y_3) -

(A) are collinear

(B) form an equilateral triangle

(C) form a right angled isosceles triangle

(D) are concyclic

SL0027

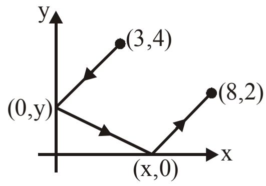
14. The area of the parallelogram formed by the lines 3x + 4y = 7a; 3x + 4y = 7b; 4x + 3y = 7c and 4x + 3y = 7d is-

(A) $\dfrac{|(a-b)(c-d)|}{7}$ (B) $|(a-b)\,(c-d)|$

(C) $\dfrac{|(a-b)(c-d)|}{49}$ (D) $7|(a-b)\,(c-d)|$

SL0029

15. Suppose that a ray of light leaves the point (3,4), reflects off the y-axis towards the x-axis, reflects off the x-axis, and finally arrives at the point (8,2). The value of x, is -



(A) $x = 4\frac{1}{2}$ (B) $x = 4\frac{1}{3}$

(C) $x = 4\frac{2}{3}$ (D) $x = 5\frac{1}{3}$

SL0032

16. m,n are integer with 0 < n < m. A is the point (m,n) on the cartesian plane. B is the reflection of A in the line y = x. C is the reflection of B in the y-axis, D is the reflection of C in the x-axis and E is the reflection of D in the y-axis. The area of the pentagon ABCDE is -

(A) 2m(m + n) (B) m(m + 3n) (C) m(2m + 3n) (D) 2m(m + 3n)

SL0033

17. If the lines $\left.\begin{array}{l}\lambda x+(\sin\alpha)y+\cos\alpha=0\\ x+(\cos\alpha)y+\sin\alpha=0\\ x-(\sin\alpha)y+\cos\alpha=0\end{array}\right]$ pass through the same point where $\alpha \in R$ then λ lies in the interval -

(A) [–1,1] (B) $\left[-\sqrt{2},\sqrt{2}\right]$ (C) [–2,2] (D) $(-\infty,\infty)$

SL0034

18. If the straight lines, ax + amy + 1 = 0, bx + (m + 1) by + 1 = 0 and cx + (m + 2)cy + 1 = 0, $m \neq 0$ are concurrent then a,b,c are in :

(A) A.P. only for m = 1 (B) A.P. for all m

(C) G.P. for all m (D) H.P. for all m

SL0035

19. Family of lines represented by the equation $(\cos\theta + \sin\theta)x + (\cos\theta - \sin\theta)y - 3(3\cos\theta + \sin\theta) = 0$ passes through a fixed point M for all real values of θ. The reflection of M in the line x – y = 0, is-

(A) (6,3) (B) (3,6) (C) (–6,3) (D) (3,–6)

SL0036

20. Consider a parallelogram whose sides are represented by the lines $2x + 3y = 0$; $2x + 3y - 5 = 0$; $3x - 4y = 0$ and $3x - 4y = 3$. The equation of the diagonal not passing through the origin, is-

(A) $21x - 11y + 15 = 0$ (B) $9x - 11y + 15 = 0$

(C) $21x - 29y - 15 = 0$ (D) $21x - 11y - 15 = 0$

SL0037

21. If the equation $ax^2 - 6xy + y^2 + 2gx + 2fy + c = 0$ represents a pair of lines whose slopes are m and m^2, then sum of all possible values of a is-

(A) 17 (B) –19 (C) 19 (D) –17

SL0040

22. Let $S = \{(x,y)|\ x^2 + 2xy + y^2 - 3x - 3y + 2 = 0\}$, then S -

(A) consists of two coincident lines.

(B) consists of two parallel lines which are not coincident.

(C) consists of two intersecting lines.

(D) is a parabola.

SL0041

23. The line x= c cuts the triangle with corners (0,0); (1,1) and (9,1) into two region. For the area of the two regions to be the same c must be equal to-

(A) 5/2 (B) 3 (C) 7/2 (D) 3 or 15

SL0062

24. If m and b are real numbers and $mb > 0$, then the line whose equation is $y = mx + b$ cannot contain the point-

(A) (0,2009) (B) (2009,0)

(C) (0,–2009) (D) (20,–100)

SL0063

25. In a triangle ABC, side AB has the equation $2x + 3y = 29$ and the side AC has the equation, $x + 2y = 16$. If the mid- point of BC is (5,6) then the equation of BC is :

(A) $x - y = -1$ (B) $5x - 2y = 13$

(C) $x + y = 11$ (D) $3x - 4y = -9$

SL0064

26. The vertex of the right angle of a right angled triangle lies on the straight line $2x - y - 10 = 0$ and the two other vertices, at points (2,–3) and (4,1) then the area of triangle in sq. units is-

(A) $\sqrt{10}$ (B) 3 (C) $\frac{33}{5}$ (D) 11

SL0065

27. A triangle ABC is formed by the lines $2x - 3y - 6 = 0$; $3x - y + 3 = 0$ and $3x + 4y - 12 = 0$. If the points $P(\alpha,0)$ and $Q(0,\beta)$ always lie on or inside the ΔABC, then :

(A) $\alpha \in [-1,2]$ and $\beta \in [-2,3]$ (B) $\alpha \in [-1,3]$ and $\beta \in [-2,4]$

(C) $\alpha \in [-2,4]$ and $\beta \in [-3,4]$ (D) $\alpha \in [-1,3]$ and $\beta \in [-2,3]$

SL0066

28. The co-ordinates of a point P on the line $2x - y + 5 = 0$ such that $|PA - PB|$ is maximum where A is $(4,-2)$ and B is $(2,-4)$ will be :

(A) $(11,27)$ (B) $(-11,-17)$ (C) $(-11,17)$ (D) $(0,5)$

SL0067

29. The line $(k + 1)^2x + ky - 2k^2 - 2 = 0$ passes through a point regardless of the value k. Which of the following is the line with slope 2 passing through the point ?

(A) $y = 2x - 8$ (B) $y = 2x - 5$ (C) $y = 2x - 4$ (D) $y = 2x + 8$

SL0068

30. If the straight lines joining the origin and the points of intersection of the curve

$5x^2 + 12xy - 6y^2 + 4x - 2y + 3 = 0$ and $x + ky - 1 = 0$

are equally inclined to the x- axis then the value of k :

(A) is equal to 1 (B) is equal to -1

(C) is equal to 2 (D) does not exist in the set of real numbers

SL0069

EXERCISE (O-2)

Multiple Correct Answer Type

1. The x-coordinates of the vertices of a square of unit area are the roots of the equation $x^2 - 3|x| + 2 = 0$ and the y-coordinates of the vertices are the roots of the equation $y^2 - 3y + 2 = 0$ then the possible vertices of the square is/are-

(A) (1,1), (2,1), (2,2), (1,2) (B) (–1,1), (–2,1), (–2,2), (–1,2)

(C) (2,1), (1,–1),(1,2),(2,2) (D) (–2,1), (–1,–1),(–1,2), (–2,2)

SL0071

2. Three vertices of a triangle are A(4,3); B(1,–1) and C(7,k). Value(s) of k for which centroid, orthocentre, incentre and circumcentre of the ΔABC lie on the same straight line is/are-

(A) 7 (B) –1 (C) –19/8 (D) none

SL0072

3. Line $\frac{x}{a}+\frac{y}{b}=1$ cuts the co-ordinate axes at A(a,0) and B(0,b) and the line $\frac{x}{a'}+\frac{y}{b'}=-1$ at A'(–a',0) and B'(0,–b'). If the points A,B,A',B' are concyclic then the orthocentre of the triangle ABA' is -

(A) (0,0) (B) (0,b') (C) $\left(0,\frac{aa'}{b}\right)$ (D) $\left(0,\frac{bb'}{a}\right)$

SL0073

4. If the vertices P,Q,R of a triangle PQR are rational points, which of the following points of the triangle PQR is/are always rational point(s) ?

(A) centroid (B) incentre (C) circumcentre (D) orthocentre

SL0074

5. The area of triangle ABC is 20 square units. The co-ordinates of vertex A are (–5,0) and B are (3,0). The vertex C lies on the line, x – y = 2. The co-ordinates of C are -

(A) (5,3) (B) (–3,–5) (C) (–5,–7) (D) (7,5)

SL0075

6. Consider the equation $y - y_1 = m(x - x_1)$. If m and x_1 are fixed and different lines are drawn for different values of y_1, then :

(A) the lines will pass through a fixed point

(B) there will be a set of parallel lines

(C) all the lines intersect the line $x = x_1$

(D) all the lines will be parallel to the line $y = x_1$.

SL0076

7. If one vertex of an equilateral triangle of side 'a' lies at the origin and the other lies on the line $x-\sqrt{3}y=0$ then the co-ordinates of the third vertex are :

(A) (0,a) (B) $\left(\frac{\sqrt{3}a}{2},-\frac{a}{2}\right)$ (C) (0,–a) (D) $\left(-\frac{\sqrt{3}a}{2},\frac{a}{2}\right)$

SL0077

8. Let B(1,–3) and D(0,4) represent two vertices of rhombus ABCD in (x,y) plane, then coordinates of vertex A if $\angle BAD = 60°$ can be equal to-

(A) $\left(\frac{1-7\sqrt{3}}{2},\frac{1-\sqrt{3}}{2}\right)$ (B) $\left(\frac{1+7\sqrt{3}}{2},\frac{1+\sqrt{3}}{2}\right)$

(C) $\left(\frac{1-14\sqrt{3}}{2},\frac{1-2\sqrt{3}}{2}\right)$ (D) $\left(\frac{1+14\sqrt{3}}{2},\frac{1+2\sqrt{3}}{2}\right)$

SL0078

9. The sides of a triangle are the straight lines x + y = 1; 7y = x and $\sqrt{3}y+x=0$. Then which of the following is an interior point of the triangle ?

(A) circumcentre (B) centroid (C) incentre (D) orthocentre

SL0079

10. A line passes through the origin and makes an angle of $\pi/4$ with the line x – y + 1 = 0. Then :

(A) equation of the line is x = 0

(B) the equation of the line is y = 0

(C) the point of intersection of the line with the given line is (–1,0)

(D) the point of intersection of the line with the given line is (0,1)

SL0080

11. Equation of a straight line passing through the point (2,3) and inclined at an angle of arc tan $\frac{1}{2}$ with the line y + 2x = 5 is-

(A) y = 3 (B) x = 2

(C) 3x + 4y – 18 = 0 (D) 4x + 3y – 17 = 0

SL0081

12. If $a^2 + 9b^2 - 4c^2 = 6ab$ then the family of lines ax + by + c = 0 are concurrent at :

(A) (1/2, 3/2) (B) (–1/2, –3/2) (C) (–1/2, 3/2) (D) (1/2, –3/2)

SL0082

13. The lines L_1 and L_2 denoted by $3x^2 + 10xy + 8y^2 + 14x + 22y + 15 = 0$ intersect at the point P and have gradients m_1 and m_2 respectively. The acute angles between them is θ. Which of the following relations hold good ?

(A) $m_1 + m_2 = 5/4$

(B) $m_1m_2 = 3/8$

(C) acute angle between L_1 and L_2 is $\sin^{-1}\left(\frac{2}{5\sqrt{5}}\right)$.

(D) sum of the abscissa and ordinate of the point P is –1.

SL0083

14. Let $y = m_1x$ & $y = m_2x$ be the equations of the lines joining the origin to the points of trisection of the portion of the line $3x + y = 12$ intercepted between the axes, then -

(A) $m_1 + m_2 = \frac{21}{2}$ (B) $|m_1 - m_2| = \frac{9}{2}$

(C) $m_1.m_2 = 9$ (D) $\frac{m_1}{m_2} = 6$

SL0167

15. If the equation $ax^2 - 6xy + y^2 + bx + cy + d = 0$ represents pair of lines whose slopes are m and m^2, then value of a is/are -

(A) $a = -8$ (B) $a = 8$ (C) $a = 27$ (D) $a = -27$

SL0168

16. The triangle formed by the lines $(5x + 12y)^2 = (12x - 5y)^2$ and $5x + 12y = 13$ will be -

(A) acute angled (B) right angled

(C) obtuse angled (D) isosceles

SL0169

17. If the slope of one of the line represent by $\lambda x^2 - 2xy + y^2 = 0$ is square of the other, then λ can be -

(A) –8 (B) 8 (C) –1 (D) 1

SL0170

18. The point $(\alpha^2, \alpha - 1)$ lie on same side of $x - 6y + 2 = 0$ as that of $(-1, 1)$, then α may be -

(A) $\frac{5}{2}$ (B) 3 (C) $\frac{9}{2}$ (D) $\frac{7}{3}$

SL0171

19. The position of the point (1, 1) w.r.t. the lines $L_1 \equiv 2x + 3y + 7 = 0$ and $L_2 \equiv 4x + 6y - 19 = 0$ is such that :

(A) it is below L_2 (B) it is above L_1

(C) it is between L_1 & L_2 (D) it is above both L_1 & L_2

SL0172

20. $2x^2 - 3xy + y^2 + 5x - 4y + 3 = 0$ represent two straight lines of slope m_1 and m_2. Then choose correct options -

(A) harmonic mean of m_1 and m_2 is $\frac{4}{3}$

(B) arithmetic mean of m_1 and m_2 is 3

(C) point of intersection of these lines is (0, –1)

(D) point of intersection of these lines is (–2, –1)

SL0173

21. Two adjacent sides of a rhombus are $2x + 3y = a - 5$ and $3x + 2y = 4 - 2a$ and its diagonals intersect at the point (1,2), then a can be -

(A) –16 (B) 16 (C) $-\frac{10}{3}$ (D) $\frac{10}{3}$

SL0174

22. Point P(2,3) lies on the line $4x + 3y = 17$. Then find the co-ordinates of points farthest from the line which are at 5 units distance from the P.

(A) (6, 6) (B) (6, –6) (C) (2, 0) (D) (–2, 0)

SL0175

23. If one side of a square is parallel to $3x - 4y = 0$ & its area being 16 while centre being (1, 1), then find equation of sides of square.

(A) $3x - 4y + 11 = 0$ (B) $3x - 4y - 9 = 0$

(C) $4x + 3y + 3 = 0$ (D) $4x + 3y - 17 = 0$

SL0176

24. Let L be the line $2x + y = 2$. If the axes are rotated by 45°, then the intercepts made by the line L on the new axes are respectively

(A) $\sqrt{2}$ and 1 (B) 1 and $\sqrt{2}$

(C) $2\sqrt{2}$ and $2\sqrt{2}/3$ (D) $2\sqrt{2}/3$ and $2\sqrt{2}$

SL0177

25. Find the equations of the sides of a triangle having (4, –1) as a vertex, if the lines $x - 1 = 0$ and $x - y - 1 = 0$ are the equations of two internal bisectors of its angles.

(A) $2x - y + 3 = 0$ (B) $x + 2y - 6 = 0$

(C) $2x + y - 7 = 0$ (D) $x - 2y - 6 = 0$

SL0178

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question Nos. 1 to 3

Consider two points $A \equiv (1,2)$ and $B \equiv (3,-1)$. Let M be a point on the straight line $L \equiv x + y = 0$.

1. If M be a point on the line L = 0 such that AM + BM is minimum, then the reflection of M in the line x = y is -

(A) (1,–1)　(B) (–1,1)　(C) (2,–2)　(D) (–2,2)

SL0086

2. If M be a point on the line L = 0 such that |AM – BM| is maximum, then the distance of M from $N \equiv (1,1)$ is-

(A) $5\sqrt{2}$　(B) 7　(C) $3\sqrt{5}$　(D) 10

SL0086

3. If M be a point on the line L = 0 such that |AM – BM| is minimum, then the area of ΔAMB equals-

(A) $\frac{13}{4}$　(B) $\frac{13}{2}$　(C) $\frac{13}{6}$　(D) $\frac{13}{8}$

SL0086

Paragraph for Question Nos. 4 to 5

Let ABCD is a square with sides of unit length. Points E and F are taken on sides AB and AD respectively so that AE= AF. Let P be a point inside the square ABCD.

4. The maximum possible area of quadrilateral CDFE is -

(A) $\frac{1}{8}$　(B) $\frac{1}{4}$　(C) $\frac{5}{8}$　(D) $\frac{3}{8}$

SL0084

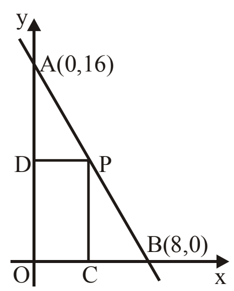
5. The value of $(PA)^2 - (PB)^2 + (PC)^2 - (PD)^2$ can not be equal to -

(A) 3　(B) 2　(C) 1　(D) 0

SL0084

Paragraph for Question Nos. 6 to 8

In the diagram, a line is drawn through the points A(0,16) and B(8,0). Point P is chosen in the first quadrant on the line through A and B. Points C and D are chosen on the x and y axis respectively, so that PDOC is a rectangle.



6. If perpendicular distance of the line AB from the point (2, 2) is $\lambda\sqrt{5}$ then value of λ is-

SL0085

7. Sum of the coordinates of the point P if PDOC is a square is -

SL0085

8. Number of possible ordered pair(s) of all positions of the point P on AB so that the area of the rectangle PDOC is 30 sq. units, is-

SL0085

Matrix Match Type

9. **List-I** **List-II**

	List-I		List-II
(I)	If the incentre of the triangle formed by the coordinate axes and the line $3x+4y=12$ is at (a, b), then $a+b$ equals	(P)	1
(II)	The line $2x-3y=6$ is shifted by 1 unit closer to the origin, the equation of line in new position is $2x-3y=6+k\sqrt{13}$, then value of $\lvert k\rvert$ is	(Q)	2
(III)	Sum of the abscissa of points on the line $x-y=3$ which lie at a unit distance from the line $4x-3y=12$ is	(R)	3
(IV)	The integral part of 'a' for which the point $(2, a)$ lies between the parallel lines $x+2y=4$ and $x+2y=5$ is	(S)	6

Which of the following is correct ?

(A) I $\to$ Q; II $\to$ R; III $\to$ P; IV $\to$ S

(B) I $\to$ P; II $\to$ Q; III $\to$ R; IV $\to$ P

(C) I $\to$ Q; II $\to$ P; III $\to$ S; IV $\to$ P

(D) I $\to$ S; II $\to$ R; III $\to$ P; IV $\to$ Q

SL0179

10. In ΔABC, let slope of side AB is $\frac{1}{3}$ & internal angle bisector AD is $3x = 4y$. Also mirror image of vertex B w.r.t. line AD is $(13, 16)$ & point D is $(12, 9)$ (where D lies on BC).

	Column-I		Column-II
(A)	Co-ordinates of vertex 'B' are (a, b), then $(a-b)$ is equal to	(P)	$\frac{13}{9}$
(B)	Slope of side AC, is	(Q)	$\frac{25}{4}$
(C)	x-intercept of external angle bisector of vertex 'A' is	(R)	11
(D)	Abscissa of point C is $\frac{a}{b}$ where a & b are coprime numbers, then $\lvert a-b\rvert$ is equal to	(S)	13
		(T)	15

SL0180

EXERCISE (O-4)

Numerical Grid Type

1. Let O(0, 0), A(6, 0) and B(3, $\sqrt{3}$) be the vertices of ΔOAB. Let R be the region consisting of all those points P inside ΔOAB which satisfy d(P, OA) $\le$ minimum (d(P,OB), d(P,AB)) where d(P,OA), d(P,OB) and d(P,AB) represent the distance of P from the sides OA,OB and AB respectively. If the area of region R is $9(a-\sqrt{b})$ where a and b are coprime, then find the value of (a + b)

SL0094

2. A triangle has side lengths 18, 24 and 30. Find the area of the triangle whose vertices are the incentre, circumcentre and centroid of the triangle. (In unit)

SL0095

3. The line 3x + 2y = 24 meets the y-axis at A & the x-axis at B. The perpendicular bisector of AB meets the line through (0, –1) parallel to x-axis at C. Find the area of the triangle ABC. (In Sq. unit)

SL0098

4. Consider a ΔABC whose sides AB, BC and CA are represented by the straight lines 2x + y = 0, x + py = q and x – y = 3 respectively. The point P is (2, 3).

(a) If P is the centroid, then find the value of (p + q).

(b) If P is the orthocentre, then find the value of (p + q).

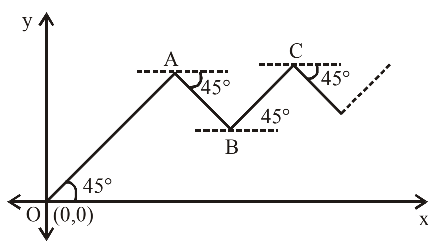
(c) If P is the circumcentre, then find the value of (p + q).

SL0117

5. The two line pairs $y^2 - 4y + 3 = 0$ and $x^2 + 4xy + 4y^2 - 5x - 10y + 4 = 0$ enclose a 4 sided convex polygon find **(i)** area of the polygon; **(ii)** length of its diagonals.

SL0123

6. Point O, A, B, C .... are shown in figure where OA = 2AB = 4BC = ... so on. Let A be the centroid of a triangle whose orthocentre and circumcenter are (2, 4) and $\left(\frac{7}{2}, \frac{5}{2}\right)$ respectively. If an insect starts moving from the point O(0,0) along the straight line in zig-zag fashion and terminates ultimately at point P(α, β) then find the value of ($\alpha + \beta$)



SL0112

7. If the straight line drawn through the point $P(\sqrt{3}, 2)$ & inclined at an angle $\frac{\pi}{6}$ with the x-axis, meets the line $\sqrt{3}x - 4y + 8 = 0$ at Q. Find the length PQ. (in unit)

SL0099

8. The sides of a triangle have the combined equation $x^2 - 3y^2 - 2xy + 8y - 4 = 0$. The third side, which is variable always passes through the point (–5, –1). If the range of values of the slope of the third line so that the origin is an interior point of the triangle, lies in the interval (a, b), then find $\left(a + \frac{1}{b^2}\right)$.

SL0122

9. The interior angle bisector of angle A for the triangle ABC whose coordinates of the vertices are A(–8, 5); B(–15, –19) and C(1, – 7) has the equation $ax + 2y + c = 0$. Find a + c.

SL0114

10. If algebraic sum of lengths of perpendiculars drawn from vertices of a hexagon ABCDEF, A(–2,–1), B(0,–2), C(4,0), D(3,2), E(2,3), F(–1,4) on a variable line $ax + by + c = 0$ is zero, then variable line always passes through a fixed point (p,q), then the value of p + q is

SL0181

EXERCISE (JM)

1. The x-coordinate of the incentre of the triangle that has the coordinates of mid points of its sides as (0, 1)(1, 1) and (1, 0) is : **[JEE-MAIN 2013]**

(1) $2+\sqrt{2}$ (2) $2-\sqrt{2}$ (3) $1+\sqrt{2}$ (4) $1-\sqrt{2}$

SL0127

2. A light ray emerging from the point source placed at P(1, 3) is reflected at a point Q in the axis of x. If the reflected ray passes through the point R(6, 7), then the abscissa of Q is :

[JEE-MAIN Online 2013]

(1) 3 (2) $\frac{7}{2}$ (3) 1 (4) $\frac{5}{2}$

SL0128

3. If the three lines $x-3y = p$, $ax + 2y = q$ and $ax + y = r$ from a right – angled triangle then:

[JEE-MAIN Online 2013]

(1) $a^2 - 6a - 12 = 0$ (2) $a^2 - 9a + 12 = 0$

(3) $a^2 - 9a + 18 = 0$ (4) $a^2 - 6a - 18 = 0$

SL0129

4. If the extremities of the base of an isosceles triangle are the points (2a, 0) and (0, a) and the equation of one of the sides isx = 2a, then the area of the triangle, in square units, is :

[JEE-MAIN Online 2013]

(1) $\frac{5}{2}a^2$ (2) $\frac{5}{4}a^2$ (3) $\frac{25a^2}{4}$ (4) $5a^2$

SL0131

5. Let θ_1 be the angle between two lines $2x + 3y + c_1 = 0$ and $-x + 5y + c_2 = 0$, andθ_2 be the angle between two lines $2x + 3y + c_1 = 0$ and $-x + 5y + c_3 = 0$, where c_1, c_2, c_3 are any real numbers :

[JEE-MAIN Online 2013]

Statement–1 : If c_2 and c_3 are proportional, then $\theta_1 = \theta_2$.

Statement–2 : $\theta_1 = \theta_2$ for all c_2 and c_3.

(1) Statement-1 is true and Statement - 2 is true, Statement-2 is not a correct explanation for Statement-1.

(2) Statement-1 is false and Statement-2 is true.

(3) Statement-1 is true and Statement-2 is false.

(4) Statement-1 is true and Statement - 2 is true, Statement-2 is a correct explanation for Statement-1.

SL0132

6. Let a, b, c and d be non-zero numbers. If the point of intersection of the lines $4ax + 2ay + c = 0$ and $5bx + 2by + d = 0$ lies in the fourth quadrant and is equidistant from the two axes then :

[JEE(Main)-2014]

(1) $2bc - 3ad = 0$ (2) $2bc + 3ad = 0$ (3) $3bc - 2ad = 0$ (4) $3bc + 2ad = 0$

SL0135

7. Let PS be the median of the triangle with vertices P (2, 2), Q (6, –1) and R (7, 3). The equation of the line passing through (1, –1) and parallel to PS is : **[JEE(Main)-2014]**

(1) $4x - 7y - 11 = 0$
(2) $2x + 9y + 7 = 0$
(3) $4x + 7y + 3 = 0$
(4) $2x - 9y - 11 = 0$

SL0136

8. Locus of the image of the point (2, 3) in the line $(2x - 3y + 4) + k(x - 2y + 3) = 0$, $k \in R$, is a

(1) circle of radius $\sqrt{2}$
(2) circle of radius $\sqrt{3}$
(3) straight line parallel to x-axis
(4) straight line parallel to y-axis

[JEE(Main)-2015]

SL0137

9. Two sides of a rhombus are along the lines, $x - y + 1 = 0$ and $7x - y - 5 = 0$. If its diagonals intersect at (–1, –2), then which one of the following is a vertex of this rhombus ? **[JEE(Main)-2016]**

(1) $\left(-\frac{10}{3}, -\frac{7}{3}\right)$
(2) (–3, –9)
(3) (–3, –8)
(4) $\left(\frac{1}{3}, -\frac{8}{3}\right)$

SL0138

10. Let k be an integer such that triangle with vertices (k, –3k), (5, k) and (–k, 2) has area 28 sq. units. Then the orthocentre of this triangle is at the point : **[JEE(Main)-2017]**

(1) $\left(2, \frac{1}{2}\right)$
(2) $\left(2, -\frac{1}{2}\right)$
(3) $\left(1, \frac{3}{4}\right)$
(4) $\left(1, -\frac{3}{4}\right)$

SL0139

11. Let the orthocentre and centroid of a triangle be A(–3, 5) and B(3, 3) respectively. If C is the circumcentre of this triangle, then the radius of the circle having line segment AC as diameter, is:

[JEE(Main)-2018]

(1) $2\sqrt{10}$
(2) $3\sqrt{\frac{5}{2}}$
(3) $\frac{3\sqrt{5}}{2}$
(4) $\sqrt{10}$

SL0140

12. A straight line through a fixed point (2, 3) intersects the coordinate axes at distinct points P and Q. If O is the origin and the rectangle OPRQ is completed, then the locus of R is : **[JEE(Main)-2018]**

(1) $2x + 3y = xy$
(2) $3x + 2y = xy$
(3) $3x + 2y = 6xy$
(4) $3x + 2y = 6$

SL0141

13. Let S be the set of all triangles in the xy-plane, each having one vertex at the origin and the other two vertices lie on coordinate axes with integral coordinates. If each triangle in S has area 50sq. units, then the number of elements in the set S is : **[JEE(Main)-2019]**

(1) 9
(2) 18
(3) 32
(4) 36

SL0142

14. If the line $3x + 4y - 24 = 0$ intersects the x-axis at the point A and the y-axis at the point B, then the incentre of the triangle OAB, where O is the origin, is **[JEE(Main)-2019]**

(1) (3, 4) (2) (2, 2) (3) (4, 4) (4) (4, 3)

SL0143

15. Two vertices of a triangle are (0,2) and (4,3). If its orthocentre is at the origin, then its third vertex lies in which quadrant ? **[JEE(Main)-2019]**

(1) Fourth (2) Second (3) Third (4) First

SL0144

16. Two sides of a parallelogram are along the lines, $x + y = 3$ and $x - y + 3 = 0$. If its diagonals intersect at (2,4), then one of its vertex is : **[JEE(Main)-2019]**

(1) (2,6) (2) (2,1) (3) (3,5) (4) (3,6)

SL0145

17. Slope of a line passing through P(2, 3) and intersecting the line, $x + y = 7$ at a distance of 4 units from P, is **[JEE(Main)-2019]**

(1) $\frac{\sqrt{5}-1}{\sqrt{5}+1}$ (2) $\frac{1-\sqrt{5}}{1+\sqrt{5}}$ (3) $\frac{1-\sqrt{7}}{1+\sqrt{7}}$ (4) $\frac{\sqrt{7}-1}{\sqrt{7}+1}$

SL0147

18. A rectangle is inscribed in a circle with a diameter lying along the line $3y = x + 7$. If the two adjacent vertices of the rectangle are (–8, 5) and (6, 5), then the area of the rectangle (in sq. units) is :-

[JEE(Main)-2019]

(1) 72 (2) 84 (3) 98 (4) 56

SL0148

19. A straight line L at a distance of 4 units from the origin makes positive intercepts on the coordinate axes and the perpendicular from the origin to this line makes an angle of 60° with the line $x + y = 0$. Then an equation of the line L is : **[JEE(Main)-2019]**

(1) $\left(\sqrt{3}+1\right)x+\left(\sqrt{3}-1\right)y=8\sqrt{2}$ (2) $\left(\sqrt{3}-1\right)x+\left(\sqrt{3}+1\right)y=8\sqrt{2}$

(3) $\sqrt{3}x+y=8$ (4) $x+\sqrt{3}y=8$

SL0149

20. The locus of the mid-points of the perpendiculars drawn from points on the line, $x = 2y$ to the line $x = y$ is : **[JEE(Main)-2020]**

(1) $2x - 3y = 0$ (2) $7x - 5y = 0$ (3) $5x - 7y = 0$ (4) $3x - 2y = 0$

SL0150

21. Let A(1, 0), B(6, 2) and $C\left(\frac{3}{2},6\right)$ be the vertices of a triangle ABC. If P is a point inside the triangle ABC such that the triangles APC, APB and BPC have equal areas, then the length of the line segment PQ, where Q is the point $\left(-\frac{7}{6},-\frac{1}{3}\right)$, is __________. **[JEE(Main)-2020]**

SL0151

22. Let two points be A(1,–1) and B(0,2). If a point P(x',y') be such that the area of $\Delta PAB = 5$ sq. units and it lies on the line, $3x + y - 4\lambda = 0$, then a value of λ is **[JEE(Main)-2020]**

(1) 1　　(2) 4　　(3) 3　　(4) –3

SL0152

23. Let C be the centroid of the triangle with vertices (3, –1), (1, 3) and (2, 4). Let P be the point of intersection of the lines $x + 3y - 1 = 0$ and $3x - y + 1 = 0$. Then the line passing through the points C and P also passes through the point : **[JEE(Main)-2020]**

(1) (7, 6)　　(2) (–9, –6)　　(3) (–9, –7)　　(4) (9, 7)

SL0153

24. Let the equation of the pair of lines, $y = px$ and $y = qx$, can be written as $(y - px)(y - qx) = 0$. Then the equation of the pair of the angle bisectors of the lines $x^2 - 4xy - 5y^2 = 0$ is : **[JEE(Main)-2021]**

(1) $x^2 - 3xy + y^2 = 0$　　(2) $x^2 + 4xy - y^2 = 0$

(3) $x^2 + 3xy - y^2 = 0$　　(4) $x^2 - 3xy - y^2 = 0$

SL0182

25. If p and q are the lengths of the perpendiculars from the origin on the lines, $x\operatorname{cosec}\alpha - y\sec\alpha = k\cot 2\alpha$ and $x\sin\alpha + y\cos\alpha = k\sin 2\alpha$ respectively, then k^2 is equal to : **[JEE(Main)-2021]**

(1) $4p^2 + q^2$　　(2) $2p^2 + q^2$　　(3) $p^2 + 2q^2$　　(4) $p^2 + 4q^2$

SL0183

26. Let the area of the triangle with vertices A(1, α), B(α, 0) and C(0, α) be 4 sq. units. If the point (α, –α), (–α, α) and (α^2, β) are collinear, then β is equal to **[JEE(Main)-2022]**

(A) 64　　(B) –8　　(C) –64　　(D) 512

SL0184

27. The equations of the sides AB, BC and CA of a triangle ABC are $2x + y = 0$, $x + py = 15a$ and $x - y = 3$ respectively. If its orthocentre is (2, a), $-\frac{1}{2} < a < 2$, then p is equal to

[JEE(Main)-2022]

SL0185

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EXERCISE (JA)

1. Consider the lines given by

$L_1 = x + 3y - 5 = 0$

$L_2 = 3x - ky - 1 = 0$

$L_3 = 5x + 2y - 12 = 0$

Match the statements / Expression in **Column-I** with the statements / Expressions in **Column-II** and indicate your answer by darkening the appropriate bubbles in the 4 × 4 matrix given in OMR.

	Column-I		Column-II
(A)	L_1, L_2, L_3 are concurrent, if	(P)	$k = -9$
(B)	One of L_1, L_2, L_3 is parallel to at least one of the other two, if	(Q)	$k = -\frac{6}{5}$
(C)	L_1, L_2, L_3 form a triangle, if	(R)	$k = \frac{5}{6}$
(D)	L_1, L_2, L_3 do not form a triangle, if	(S)	$k = 5$

[JEE 2008, 6]

SL0154

2. Let P, Q, R and S be the points on the plane with position vectors $-2\hat{i} - \hat{j}$, $4\hat{i}, 3\hat{i} + 3\hat{j}$ and $-3\hat{i} + 2\hat{j}$ respectively. The quadrilateral PQRS must be a

(A) parallelogram, which is neither a rhombus nor a rectangle

(B) square

(C) rectangle, but not a square

(D) rhombus, but not a square **[JEE 2010, 3]**

SL0155

3. A straight line L through the point (3, –2) is inclined at an angle 60° to the line $\sqrt{3}x + y = 1$. If L also intersect the x-axis, then the equation of L is **[JEE 2011, 3 (–1)]**

(A) $y + \sqrt{3}x + 2 - 3\sqrt{3} = 0$ (B) $y - \sqrt{3}x + 2 + 3\sqrt{3} = 0$

(C) $\sqrt{3}y - x + 3 + 2\sqrt{3} = 0$ (D) $\sqrt{3}y + x - 3 + 2\sqrt{3} = 0$

SL0156

4. For $a > b > c > 0$, the distance between (1, 1) and the point of intersection of the lines $ax + by + c = 0$ and $bx + ay + c = 0$ is less than $2\sqrt{2}$. Then **[JEE-Advanced 2013, 2]**

(A) $a + b - c > 0$ (B) $a - b + c < 0$

(C) $a - b + c > 0$ (D) $a + b - c < 0$

SL0157

5. For a point P in the plane, let $d_1(P)$ and $d_2(P)$ be the distances of the point P from the lines $x - y = 0$ and $x + y = 0$ respectively. The area of the region R consisting of all points P lying in the first quadrant of the plane and satisfying $2 \le d_1(P) + d_2(P) \le 4$, is

[JEE(Advanced)-2014, 3]

SL0158

Question Stem for Question Nos. 6 and 7

Question Stem

Consider the lines L_1 and L_2 defined by

$L_1 : x\sqrt{2} + y - 1 = 0$ and $L_2 : x\sqrt{2} - y + 1 = 0$

For a fixed constant λ, let C be the locus of a point P such that the product of the distance of P from L_1 and the distance of P from L_2 is λ^2. The line $y = 2x + 1$ meets C at two points R and S, where the distance between R and S is $\sqrt{270}$.

Let the perpendicular bisector of RS meet C at two distinct points R' and S'. Let D be the square of the distance between R' and S'.

[JEE(Advanced)-2021]

6. The value of λ^2 is __________. **SL0165**

7. The value of D is __________. **SL0166**

ANSWER KEY

Do yourself - 1

1. $PQ = \sqrt{34}$ units **2.** $x = 6$ or $x = 0$ **3.** 11, –7

4. A **5.** A **6.** C

7. **(a)** (2,1) ; **(b)** (7,16) **8.** **(a)** 2 : 3 (internally) ; **(b)** 9 : 4 (externally) ; **(c)** 8 : 7 (internally)

9. C **10.** 1 : 2 ; Q(–5, –3) **11.** D

Do yourself - 2

1. **(a)** $\left(\frac{8}{3}, 4\right)$; **(b)** $\left(-\frac{7}{2}, \frac{17}{2}\right)$, $\frac{5\sqrt{10}+2}{2}$, **(c)** (15, – 5) **2.** B

3. C **4.** D **5.** B **6.** A **7.** B

8. **(a)** $\left(2, \frac{8}{3}\right)$; **(b)** 4 **9.** C

Do yourself - 3

1. 25 square units **2.** 132 square units **4.** A **5.** B

6. $K = 7$ or $\frac{31}{9}$ **7.** $\left(\frac{7}{2}, \frac{13}{2}\right)$ **or** $\left(-\frac{3}{2}, \frac{3}{2}\right)$

Do yourself - 4

1. $x = \pm 2$ **2.** $y = \pm x$ **3.** B **4.** B **5.** B **6.** C

7. $y = 3x$ **8.** $x^2 + y^2 = 3$ **9.** $8x^2 + 8y^2 + 6x - 36y + 27 = 0$ **10.** $x^2 = 3y^2$

Do yourself - 5

1. D **2.** D **3.** D **4.** D **5.** A **6.** C **7.** D

8. D **9.** C **10.** C **11.** D **12.** B **13.** A **14.** D

15. B **16.** D **17.** A **18.** $83x - 35y + 92 = 0$

19. $c = -4$; B(2, 0); D(4, 4) **20.** **(a)** 5; **(b)** 2; **(c)** 3/2

21. $x - 3y - 31 = 0$ or $3x + y + 7 = 0$ **22.** $x - y = 0$

Do yourself - 6

1. **(a)** $y = \frac{2x}{3} + \frac{5}{3}, \frac{2}{3}, \frac{5}{3}$; **(b)** $\frac{x}{(-5/2)} + \frac{y}{(5/3)} = 1, -\frac{5}{2}, \frac{5}{3}$;

(c) $-\frac{2x}{\sqrt{13}} + \frac{3y}{\sqrt{13}} = \frac{5}{\sqrt{13}}, \frac{5}{\sqrt{13}}$, $\alpha = \pi - \tan^{-1}\left(\frac{3}{2}\right)$

2. $13\sqrt{2}/3$ units **3.** D **4.** D **5.** C **6.** A **7.** D

8. A **9.** D **10.** D **11.** D

Do yourself - 7

1. $\theta = 135°$ or $45°$ **2.** $3x + 4y = 18$ **3.** $2x - 3y + 12 = 0, (-6, 0)$

4. **(a)** Coincident, **(b)** Parallel, **(c)** Intersecting

5. C **6.** B **7.** C **8.** A **9.** A

10. $x + 5y + 5\sqrt{2} = 0$ or $x + 5y - 5\sqrt{2} = 0$

Do yourself - 8

1. **(a)** 2 **(b)** 33/10 **2.** $\left(\frac{a}{b}\left(b \pm \sqrt{a^2 + b^2}\right), 0\right)$ **4.** 30 **5.** C

6. B **7.** B **8.** A **9.** C

Do yourself - 9

1. opposite sides of the line **3.** $-y + x = 11$ **4.** $\lambda = -7$

5. D **6.** B **7.** C **8.** B

Do yourself - 10

1. (A) $\to$ (p), (B) $\to$ (s), (C) $\to$ (q), (D) $\to$ (t), (E) $\to$ (r) **2.** $(7, -4)$

3. $x^2 - 4y^2 = a^2$ **4.** C **5.** D **6.** C **7.** B **8.** C

9. A **10.** $14x + 23y = 40$ **11.** 15 **12.** B **13.** D **14.** C

Do yourself - 11

1. $x - 2y + 1 = 0$ & $2x + y - 3 = 0$; **(a)** $x - 2y + 1 = 0$; **(b)** $2x + y - 3 = 0$

2. $x + y = 2, x = y$ **3.** $69x + 46y - 25 = 0$ **4.** D **5.** C **6.** D

7. B **8.** C **9.** A **10.** $x^2 + y^2 - 3x + 4y - 3 = 0$ **11.** $x - 5 = 0$

Do yourself - 12

1. $x - y = 0$ & $x - 4y = 0$ **2.** $k = -1$, or $-\frac{127}{6}$

3. $2x - 5y - 2 = 0$ & $3x + 2y + 3 = 0$; $\pm \tan^{-1}\left(\frac{19}{4}\right)$ **4.** $(-1, 2)$, $90°$

5. B **6.** A **7.** A **8.** C **9.** A **10.** A **11.** A

12. A **13.** C **14.** **(a)** $\frac{50}{7}$; **(b)** $\frac{63}{10}$; **(c)** $\frac{3}{10}\left(8\sqrt{5} - 5\sqrt{10}\right)$

Do yourself - 13

1. $\theta = \pm \tan^{-1}\frac{4}{7}$; **2.** $x^2 - y^2 - 5xy = 0$ **3.** $x + y = 1$; $x + 9y = 1$ **6.** 2

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	B	D	D	D	A	C	A	A	A	B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A	A	A	D	B	B	B	D	B	D
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	B	B	B	C	B	D	B	A	B

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	B,C	B,C	A,C,D	B,D	B,C	A,B,C,D	A,B	B,C	A,B,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,C	C,D	B,C,D	B,C	B,D	B,D	A,D	A,B,D	A,B,C	A,D
Que.	21	22	23	24	25					
Ans.	A,D	A,D	A,B,C,D	C,D	A,C,D					

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	B	D	A	C	A,B,C	2.00	10.66 or 10.67	2.00	C
Q.10	A	B	C	D					
	R	P	Q	T					

EXERCISE # O-4

1. 5 **2.** 3 **3.** 91 **4.** **(a)** 74; **(b)** 50; **(c)** 47

5. **(i)** area = 6 sq. units, **(ii)** diagonals are $\sqrt{5}$ & $\sqrt{53}$ **6.** 8.00 **7.** 6.00

8. 24.00 **9.** 89.00 **10.** 2.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	4	3	1	4	3	2	1	4	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	2	2	4	2	2	4	3	2	1	3
Que.	21	22	23	24	25	26	27			
Ans.	5	3	2	3	1	C	3			

EXERCISE # JEE-ADVANCED

Que.	Q.1	A	B	C	D	
Ans.		S	P,Q	R	P,Q,S	
Que.	2	3	4	5	6	7
Ans.	A	B	A or C or A,C	6	9.00	77.14

CIRCLE

1. DEFINITION:

A circle is the locus of a point which moves in a plane in such a way that its distance from a fixed point (in the same given plane) remains constant. The fixed point is called the centre of the circle and the constant distance is called the radius of the circle.

Equation of a circle :

The curve traced by the moving point is called its circumference i.e. the equation of any circle is satisfied by co-ordinates of all points on its circumference.

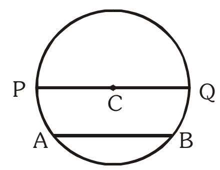
or

The equation of the circle means the equation of its circumference.

or

It is the set of all points lying on the circumference of the circle.

Chord and diameter - the line joining any two points on the circumference is called a chord. If any chord passing through its centre is called its diameter.



AB = chord, PQ = diameter

C = centre

2. STANDARD EQUATIONS OF THE CIRCLE :

(a) Central Form :

If (h, k) is the centre and r is the radius of the circle then its equation is

$$(x-h)^2 + (y-k)^2 = r^2$$

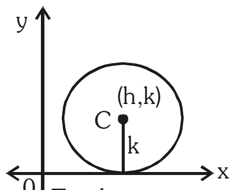
Special Cases :

(i) If centre is origin (0,0) and radius is 'r' then equation of circle is $\mathbf{x^2 + y^2 = r^2}$ and this is called the standard form.

(ii) If radius of circle is zero then equation of circle is $\mathbf{(x - h)^2 + (y - k)^2 = 0}$. Such circle is called zero circle or **point circle.**

(iii) When circle touches x-axis then equation of the circle is

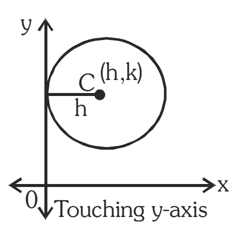
$$(x-h)^2 + (y-k)^2 = k^2.$$



Touching x-axis

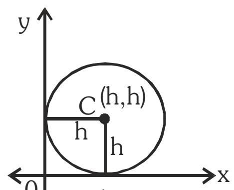
(iv) When circle touches y-axis then equation of circle is

$$(x-h)^2 + (y-k)^2 = h^2$$



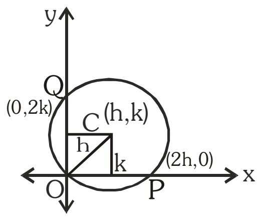
Touching y-axis

(v) When circle touches both the axes (x-axis and y-axis) then equation of circle $(x-h)^2 + (y-h)^2 = h^2$.



Touching x-axis and y-axis

(vi) When circle passes through the origin and centre of the circle is (h,k) then radius $\sqrt{h^2+k^2}=r$ and intercept cut on x-axis OP =2h, and intercept cut on y-axis is OQ = 2k and equation of circle is
$(x-h)^2 + (y-k)^2 = h^2 + k^2$ or $x^2 + y^2 - 2hx - 2ky = 0$



Note : Centre of the circle may exist in any quadrant hence for general cases use ± sign before h & k.

(b) General equation of circle

$x^2 + y^2 + 2gx + 2fy + c = 0$. where g,f,c are constants and centre is (–g,–f)

i.e. $\left(-\frac{\text{coefficient of } x}{2}, -\frac{\text{coefficient of } y}{2}\right)$ and radius $r=\sqrt{g^2+f^2-c}$

Note :

(i) If $(g^2 + f^2 - c) > 0$, then r is real and positive and the circle is a real circle.

(ii) If $(g^2 + f^2 - c) = 0$, then equation will represent a point.

(iii) If $(g^2 + f^2 - c) < 0$, then r is imaginary then circle is also an imaginary circle with real centre.

(iv) $x^2 + y^2 + 2gx + 2fy + c = 0$, has three constants and to get the equation of the circle at least three conditions should be known $\Rightarrow$ A unique circle passes through three non collinear points.

(v) **The general second degree in x and y, $ax^2 + by^2 + 2hxy + 2gx + 2fy + c = 0$ represents a real circle if :**

coefficient of x^2 = coefficient of y^2 or $a = b \neq 0$

and coefficient of xy = 0 or h = 0

and $(g^2 + f^2 - c) > 0$ (for a real circle)

$g^2 + f^2 - c = 0$ then it will represent a point.

(c) Intercepts cut by the circle on axes :

The intercepts cut by the circle $x^2 + y^2 + 2gx + 2fy + c =0$ on :

(i) x-axis = $2\sqrt{g^2-c}$ (ii) y-axis $= 2\sqrt{f^2-c}$

Note :

(i) If the circle cuts the x-axis at two distinct point, then $g^2 - c > 0$

(ii) If the circle cuts the y-axis at two distinct point, then $f^2 - c > 0$

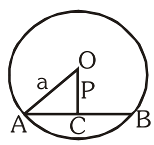
(iii) If circle touches x-axis then $g^2 = c$.

(iv) If circle touches y-axis then $f^2 = c$.

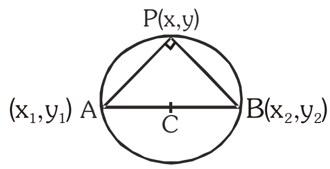
(v) Circle lies completely above or below the x-axis then $g^2 < c$.

(vi) Circle lies completely to the right or left to the y-axis, then $f^2 < c$.

(vii) Intercept cut by a line on the circle $x^2 + y^2 + 2gx + 2fy+c=0$ or length of chord of the circle $= 2\sqrt{a^2-P^2}$ where a is the radius and P is the length of perpendicular from the centre to the chord.



(d) Equation of circle in diameter form :



If $A(x_1,y_1)$ and $B(x_2,y_2)$ are the end points of the diameter of the circle and P(x,y) is the point other then A and B on the circle then from geometry we know that $\angle APB = 90°$.

$\Rightarrow$ (Slope of PA) × (Slope of PB) = –1

$\Rightarrow \quad \therefore \left(\dfrac{y-y_1}{x-x_1}\right)\left(\dfrac{y-y_2}{x-x_2}\right) = -1$

$\Rightarrow \quad \mathbf{(x-x_1)(x-x_2)+(y-y_1)(y-y_2) = 0}$

Note : This will be the circle of least radius passing through (x_1, y_1) and (x_2, y_2)

(e) Equation of circle in parametric forms :

(i) The parametric equation of the circle $x^2+y^2 = r^2$ are $\mathbf{x = r\cos\theta,\ y = r\sin\theta\ ;\ \theta \in [0, 2\pi)}$ and $(r\cos\theta,\ r\sin\theta)$ are called the parametric co-ordinates.

(ii) The parametric equation of the circle $(x-h)^2 + (y-k)^2 = r^2$ is $\mathbf{x = h + r\cos\theta,\ y = k + r\sin\theta}$ where θ is parameter.

(iii) The parametric equation of the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ are

$\mathbf{x = -g + \sqrt{g^2+f^2-c}\ \cos\theta,}$

$\mathbf{y = -f + \sqrt{g^2+f^2-c}\ \sin\theta}$ where θ is parameter.

Note : Equation of a straight line joining two point α & β on the circle $x^2 + y^2 = a^2$ is

$x\cos\dfrac{\alpha+\beta}{2} + y\sin\dfrac{\alpha+\beta}{2} = a\cos\dfrac{\alpha-\beta}{2}$.

Illustration 1 : Find the centre and the radius of the circles

(a) $3x^2 + 3y^2 - 8x - 10y + 3 = 0$

(b) $x^2 + y^2 + 2x\sin\theta + 2y\cos\theta - 8 = 0$

(c) $2x^2 + \lambda xy + 2y^2 + (\lambda - 4)x + 6y - 5 = 0$, for some λ.

Solution : (a) We rewrite the given equation as

$x^2 + y^2 - \dfrac{8}{3}x - \dfrac{10}{3}y + 1 = 0 \quad \Rightarrow \quad g = -\dfrac{4}{3},\ f = -\dfrac{5}{3},\ c = 1$

Hence the centre is $\left(\dfrac{4}{3}, \dfrac{5}{3}\right)$ and the radius is $\sqrt{\dfrac{16}{9}+\dfrac{25}{9}-1} = \sqrt{\dfrac{32}{9}} = \dfrac{4\sqrt{2}}{3}$ units

(b) $x^2 + y^2 + 2x\sin\theta + 2y\cos\theta - 8 = 0$. Centre of this circle is $(-\sin\theta, -\cos\theta)$

Radius = $\sqrt{\sin^2\theta+\cos^2\theta+8} = \sqrt{1+8} = 3$ units

(c) $2x^2 + \lambda xy + 2y^2 + (\lambda - 4)x + 6y - 5 = 0$

We rewrite the equation as

$$x^2 + \frac{\lambda}{2}xy + y^2 + \left(\frac{\lambda-4}{2}\right)x + 3y - \frac{5}{2} = 0 \qquad \text{........ (i)}$$

Since, there is no term of xy in the equation of circle $\Rightarrow \dfrac{\lambda}{2} = 0 \Rightarrow \lambda = 0$

So, equation (i) reduces to $x^2 + y^2 - 2x + 3y - \dfrac{5}{2} = 0$

$\therefore$ centre is $\left(1, -\dfrac{3}{2}\right)$ Radius = $\sqrt{1+\dfrac{9}{4}+\dfrac{5}{2}} = \dfrac{\sqrt{23}}{2}$ units.

Illustration 2 : If the lines $3x - 4y + 4 = 0$ and $6x - 8y - 7 = 0$ are tangents to a circle, then the radius of the circle is -

(A) 3/2 (B) 3/4 (C) 1/10 (D) 1/20

Solution : The diameter of the circle is perpendicular distance between the parallel lines (tangents)

$3x - 4y + 4 = 0$ and $3x - 4y - \frac{7}{2} = 0$ and so it is equal to $\frac{4+7/2}{\sqrt{9+16}} = \frac{3}{2}$.

Hence radius is $\frac{3}{4}$. **Ans. (B)**

Illustration 3 : If $y = 2x + m$ is a diameter to the circle $x^2 + y^2 + 3x + 4y - 1 = 0$, then find m

Solution : Centre of circle = $(-3/2, -2)$. This lies on diameter $y = 2x + m$

$\Rightarrow\ -2 = (-3/2) \times 2 + m \ \Rightarrow\ m = 1$

Illustration 4 : The equation of a circle which passes through the point $(1, -2)$ and $(4, -3)$ and whose centre lies on the line $3x + 4y = 7$ is

(A) $15(x^2 + y^2) - 94x + 18y - 55 = 0$ (B) $15(x^2 + y^2) - 94x + 18y + 55 = 0$

(C) $15(x^2 + y^2) + 94x - 18y + 55 = 0$ (D) none of these

Solution : Let the circle be $x^2 + y^2 + 2gx + 2fy + c = 0$ ..... (i)

Hence, substituting the points, $(1, -2)$ and $(4, -3)$ in equation (i)

$5 + 2g - 4f + c = 0$ ..... (ii)

$25 + 8g - 6f + c = 0$ ..... (iii)

centre $(-g, -f)$ lies on line $3x + 4y = 7$

Hence $-3g - 4f = 7$

solving for g, f,c, we get

Here $g = \frac{-47}{15}$, $f = \frac{9}{15}$, $c = \frac{55}{15}$

Hence the equation is $15(x^2 + y^2) - 94x + 18y + 55 = 0$ **Ans. (B)**

Illustration 5 : A circle has radius equal to 3 units and its centre lies on the line $y = x - 1$. Find the equation of the circle if it passes through (7, 3).

Solution : Let the centre of the circle be (α, β). It lies on the line $y = x - 1$

$\Rightarrow\ \beta = \alpha - 1$. Hence the centre is $(\alpha, \alpha - 1)$.

$\Rightarrow$ The equation of the circle is $(x - \alpha)^2 + (y - \alpha + 1)^2 = 9$

It passes through (7, 3) $\Rightarrow\ (7 - \alpha)^2 + (4 - \alpha)^2 = 9$

$\Rightarrow\ 2\alpha^2 - 22\alpha + 56 = 0 \ \Rightarrow\ \alpha^2 - 11\alpha + 28 = 0$

$\Rightarrow\ (\alpha - 4)(\alpha - 7) = 0 \ \Rightarrow\ \alpha = 4, 7$

Hence the required equations are

$x^2 + y^2 - 8x - 6y + 16 = 0$ and $x^2 + y^2 - 14x - 12y + 76 = 0$. **Ans.**

Illustration 6 : Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal, then which of the following equations can represent L_1?

(A) $x + y = 0$ (B) $x - y = 0$ (C) $x + 7y = 0$ (D) $x - 7y = 0$

Solution : Let L_1 be $y = mx$

lines L_1 & L_2 will be at equal distances from centre of the circle centre of the circle is $\left(\frac{1}{2}, -\frac{3}{2}\right)$

$$\Rightarrow \quad \frac{\left|\frac{1}{2}m+\frac{3}{2}\right|}{\sqrt{1+m^2}}=\frac{\left|\frac{1}{2}-\frac{3}{2}-1\right|}{\sqrt{2}} \quad \Rightarrow \quad \frac{(m+3)^2}{(1+m^2)}=8$$

$$\Rightarrow \quad 7m^2-6m-1=0 \quad \Rightarrow \quad (m-1)(7m+1)=0$$

$$\Rightarrow \quad m=1,\ m=-\frac{1}{7} \quad \Rightarrow \quad y=x,\ 7y+x=0$$

Ans. (B, C)

Do yourself - 1 :

1. Find the centre and radius of the circle $2x^2 + 2y^2 = 3x - 5y + 7$

2. Find the equation to the circle

(i) Whose radius is 10 and whose centre is $(-5, -6)$.

(ii) Whose radius is $a + b$ and whose centre is $(a, -b)$.

3. Find the coordinates of the centres and the radii of the circles whose equations are **:**

(i) $x^2 + y^2 - 4x - 8y = 41$ **(ii)** $\sqrt{1+m^2}\,(x^2 + y^2) - 2cx - 2mcy = 0$

4. Find the equation of the circle whose centre is the point of intersection of the lines $2x - 3y + 4 = 0$ & $3x + 4y - 5 = 0$ and passes through the origin.

5. Find the equation to the circle which touches the axis of :

(a) x at a distance +3 from the origin and intercepts a distance 6 on the axis of y.

(b) x, pass through the point (1, 1) and have line $x + y = 3$ as diameter.

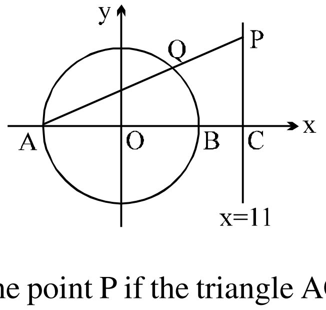
6. Find the equation of the circle the end points of whose diameter are the centres of the circles $x^2 + y^2 + 16x - 14y = 1$ & $x^2 + y^2 - 4x + 10y = 2$

7. The line $lx + my + n = 0$ intersects the curve $ax^2 + 2hxy + by^2 = 1$ at the point P and Q. The circle on PQ as diameter passes through the origin. Prove that $n^2(a + b) = l^2 + m^2$.

8. Find the equation to the circle which goes through the origin and cuts off intercepts equal to h and k from the positive parts of the axes.

9. Find the parametric form of the equation of the circle $x^2 + y^2 + px + py = 0$

10. In the given figure, the circle $x^2 + y^2 = 25$ intersects the x-axis at the point A and B. The line $x = 11$ intersects the x-axis at the point C. Point P moves along the line $x = 11$ above the x-axis and AP intersects the circle at Q. Find



(i) The coordinates of the point P if the triangle AQB has the maximum area.

(ii) The coordinates of the point P if Q is the middle point of AP.

(iii) The coordinates of P if the area of the triangle AQB is $(1/4)^{th}$ of the area of the triangle APC.

11. Find the equation to the circles which pass through the points :

(i) (0, 0), (a, 0) and (0, b) **(ii)** (1, 2), (3, –4) and (5, –6)

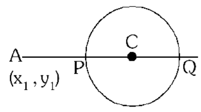
(iii) (1, 1), (2, –1) and (3, 2)

3. POSITION OF A POINT W.R.T CIRCLE :

(a) Let the circle is $x^2 + y^2 + 2gx + 2fy + c = 0$ and the point is (x_1,y_1) then -

Point (x_1,y_1) lies out side the circle or on the circle or inside the circle according as

$\Rightarrow x_1^{\,2} + y_1^{\,2} + 2gx_1 + 2fy_1 + c >, =, < 0$ or $S_1 >, =, < 0$



(b) The greatest & the least distance of a point A from a circle with centre C & radius r is $AC + r$ & $|AC - r|$ respectively.

4. POWER OF A POINT W.R.T. CIRCLE :

Theorem : The power of point $P(x_1, y_1)$ w.r.t. the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is S_1

where $S_1 = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

Note : If P outside, inside or on the circle then power of point is positive, negative or zero respectively.

If from a point $P(x_1, y_1)$, inside or outside the circle, a secant be drawn intersecting the circle in two points A & B, then PA . PB = constant. The product PA . PB is called power of point $P(x_1, y_1)$ w.r.t. the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, i.e. for number of secants PA.PB = $PA_1 . PB_1 = PA_2 . PB_2 = ... = PT^2 = S_1$

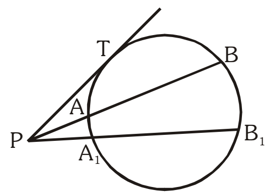


Illustration 7 : If P(2, 8) is an interior point of a circle $x^2 + y^2 - 2x + 4y - p = 0$ which neither touches nor intersects the axes, then set for p is -

(A) $p < -1$ (B) $p < -4$ (C) $p > 96$ (D) ϕ

Solution : For internal point p(2, 8), $4 + 64 - 4 + 32 - p < 0 \Rightarrow p > 96$

and x intercept = $2\sqrt{1+p}$ therefore $1 + p < 0$

$\Rightarrow \quad p < -1$ and y intercept = $2\sqrt{4+p} \quad \Rightarrow \quad p < -4$ **Ans. (D)**

Do yourself - 2 :

1. Find the position of the points (1, 2) & (6, 0) w.r.t. the circle $x^2 + y^2 - 4x + 2y - 11 = 0$

2. Find the greatest and least distance of a point P(7, 3) from circle $x^2 + y^2 - 8x - 6y + 16 = 0$. Also find the power of point P w.r.t. circle.

3. Determine the nature of the quadrilateral formed by four lines $3x + 4y - 5 = 0$; $4x - 3y - 5 = 0$; $3x + 4y + 5 = 0$ and $4x - 3y + 5 = 0$. Find the equation of the circle inscribed and circumscribing this quadrilateral.

4. (a) Find the shortest distance from the point M(–7, 2) to the circle $x^2 + y^2 - 10x - 14y - 151 = 0$.

(b) Find the co-ordinate of the point on the circle $x^2 + y^2 - 12x - 4y + 30 = 0$, which is farthest from the origin.

5. TANGENT LINE OF CIRCLE :

When a straight line meet a circle on two coincident points then it is called the tangent of the circle.

(a) Condition of Tangency :

The line L = 0 touches the circle S = 0 if P the length of the perpendicular from the centre to that line and radius of the circle r are equal i.e. P = r.

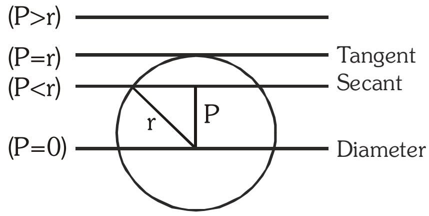


Illustration 8 : Find the range of parameter 'a' for which the variable line y = 2x + a lies between the circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 16x - 2y + 61 = 0$ without intersecting or touching either circle.

Solution : The given circles are $C_1 : (x-1)^2 + (y-1)^2 = 1$ and $C_2 : (x-8)^2 + (y-1)^2 = 4$

The line y – 2x – a = 0 will lie between these circle if centre of the circles lie on opposite sides of the line, i.e. $(1-2-a)(1-16-a) < 0 \Rightarrow a \in (-15, -1)$

Line wouldn't touch or intersect the circles if, $\frac{|1-2-a|}{\sqrt{5}} > 1$, $\frac{|1-16-a|}{\sqrt{5}} > 2$

$\Rightarrow \quad |1 + a| > \sqrt{5}, |15 + a| > 2\sqrt{5}$

$\Rightarrow \quad a > \sqrt{5} - 1$ or $a < -\sqrt{5} - 1$, $a > 2\sqrt{5} - 15$ or $a < -2\sqrt{5} - 15$

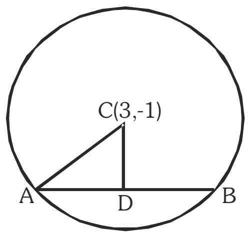
Hence common values of 'a' are $(2\sqrt{5} - 15, -\sqrt{5} - 1)$.

Illustration 9 : The equation of a circle whose centre is (3, –1) and which cuts off a chord of length 6 on the line 2x– 5y+ 18 = 0

(A) $(x-3)^2 + (y+1)^2 = 38$ (B) $(x+3)^2 + (y-1)^2 = 38$

(C) $(x-3)^2 + (y+1)^2 = \sqrt{38}$ (D) none of these

Solution : Let AB(= 6) be the chord intercepted by the line 2x – 5y + 18 = 0 from the circle and let CD be the perpendicular drawn from centre (3, –1) to the chord AB.



i.e., AD = 3, CD = $\frac{2.3 - 5(-1) + 18}{\sqrt{2^2 + 5^2}} = \sqrt{29}$

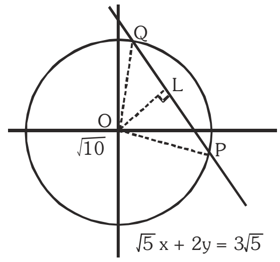
Therefore, $CA^2 = 3^2 + (\sqrt{29})^2 = 38$

Hence required equation is $(x-3)^2 + (y+1)^2 = 38$ **Ans. (A)**

Illustration 10 : The area of the triangle formed by line joining the origin to the points of intersection(s) of the line $x\sqrt{5}+2y=3\sqrt{5}$ and circle $x^2+y^2=10$ is

(A) 3 (B) 4 (C) 5 (D) 6

Solution : Length of perpendicular from origin to the line $x\sqrt{5}+2y=3\sqrt{5}$ is



$$OL=\frac{3\sqrt{5}}{\sqrt{(\sqrt{5})^2+2^2}}=\frac{3\sqrt{5}}{\sqrt{9}}=\sqrt{5}$$

Radius of the given circle = $\sqrt{10}$ = OQ = OP

PQ = 2QL = $2\sqrt{OQ^2-OL^2}=2\sqrt{10-5}=2\sqrt{5}$

Thus area of $\Delta OPQ=\frac{1}{2}\times PQ\times OL=\frac{1}{2}\times 2\sqrt{5}\times\sqrt{5}=5$ **Ans. (C)**

(b) Equation of the tangent (T = 0) :

(i) Tangent at the point (x_1,y_1) on the circle $x^2+y^2=a^2$ is $\mathbf{xx_1+yy_1=a^2}$.

(ii) (1) The tangent at the point $(a\cos t, a\sin t)$ on the circle $x^2+y^2=a^2$ is $\mathbf{x\cos t+y\sin t=a}$

(2) The point of intersection of the tangents at the points $P(\alpha)$ and $Q(\beta)$

is $\left(\frac{a\cos\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}, \frac{a\sin\frac{\alpha+\beta}{2}}{\cos\frac{\alpha-\beta}{2}}\right)$.

(iii) The equation of tangent at the point (x_1,y_1) on the circle $x^2+y^2+2gx+2fy+c=0$ is $\mathbf{xx_1+yy_1+g(x+x_1)+f(y+y_1)+c=0}$

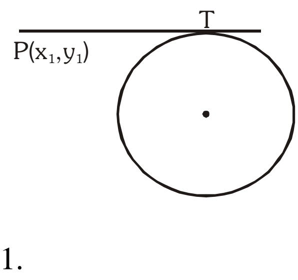
(iv) If line $y=mx+c$ is a straight line touching the circle $x^2+y^2=a^2$, then $c=\pm a\sqrt{1+m^2}$ and contact **points are** $\left(\mp\frac{\mathbf{am}}{\sqrt{\mathbf{1+m^2}}}, \pm\frac{\mathbf{a}}{\sqrt{\mathbf{1+m^2}}}\right)$ **or** $\left(\mp\frac{\mathbf{a^2m}}{\mathbf{c}}, \pm\frac{\mathbf{a^2}}{\mathbf{c}}\right)$ and equation of tangent is $\mathbf{y=mx\pm a\sqrt{1+m^2}}$

(v) The equation of tangent with slope m of the circle $(x-h)^2+(y-k)^2=a^2$ is

$\mathbf{(y-k)=m(x-h)\pm a\sqrt{1+m^2}}$

Note : To get the equation of tangent at the point $(x_1\ y_1)$ on any second degree curve we replace xx_1 in place of x^2, yy_1 in place of y^2, $\frac{x+x_1}{2}$ in place of x, $\frac{y+y_1}{2}$ in place of y, $\frac{xy_1+yx_1}{2}$ in place of xy and c in place of c.

(c) Length of tangent ($\sqrt{S_1}$) :



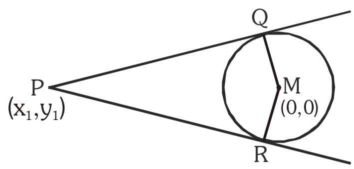
The length of tangent drawn from point (x_1,y_1) out side the circle $S\equiv x^2+y^2+2gx+2fy+c=0$ is,

$PT=\sqrt{S_1}=\sqrt{x_1^2+y_1^2+2gx_1+2fy_1+c}$

Note : When we use this formula the coefficient of x^2 and y^2 must be 1.

(d) Equation of Pair of tangents ($SS_1 = T^2$) :

Let the equation of circle $S \equiv x^2 + y^2 = a^2$ and $P(x_1,y_1)$ is any point outside the circle. From the point we can draw two real and distinct tangent PQ & PR and combine equation of pair of tangents is -



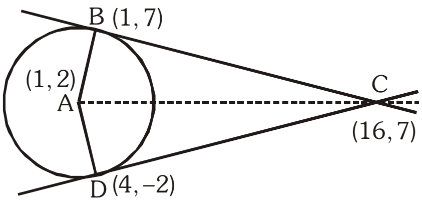
$$(x^2 + y^2 - a^2)(x_1^2 + y_1^2 - a^2) = (xx_1 + yy_1 - a^2)^2 \quad \textbf{or}$$

$$SS_1 = T^2$$

Illustration 11 : Let A be the centre of the circle $x^2 + y^2 - 2x - 4y - 20 = 0$ and B(1, 7) and D(4, –2) are points on the circle then, if tangents be drawn at B and D, which meet at C, then area of quadrilateral ABCD is -

(A) 150 (B) 75 (C) 75/2 (D) none of these

Solution :



Here centre A(1, 2) and Tangent at (1, 7) is

$x.1 + y.7 - 1(x + 1) - 2(y + 7) - 20 = 0$ or $y = 7$ .... (i)

Tangent at D(4, –2) is $3x - 4y - 20 = 0$ .... (ii)

Solving (i) and (ii), C is (16, 7)

Area ABCD = AB × BC = 5 × 15 = 75 units. **Ans. (B)**

Do yourself - 3 :

1. Find the equation of tangent to the circle $x^2 + y^2 - 2ax = 0$ at the point $(a(1 + \cos\alpha), a\sin\alpha)$.

2. Find the equations of tangents to the circle $x^2 + y^2 - 6x + 4y - 12 = 0$ which are parallel to the line $4x - 3y + 6 = 0$

3. Find the equation of the tangents to the circle $x^2 + y^2 = 4$ which are perpendicular to the line $12x - 5y + 9 = 0$. Also find the points of contact.

4. Find the value of 'c' if the line $y = c$ is a tangent to the circle $x^2 + y^2 - 2x + 2y - 2 = 0$ at the point (1, 1)

5. **(i)** Write down the equation of the tangent to the circle $x^2 + y^2 - 3x + 10y = 15$ at the point (4, –11)

(ii) Find the condition that the straight line $3x + 4y = k$ may touch the circle $x^2 + y^2 = 10x$.

6. Find the locus of the middle points of portions of the tangents to the circle $x^2 + y^2 = a^2$ terminated by the coordinate axes.

7. If M and m are the maximum and minimum values of $\frac{y}{x}$ for pair of real number (x,y) which satisfy the equation $(x-3)^2 + (y-3)^2 = 6$, then find the value of (M + m).

8. Find the equation of the tangent to the circle

(a) $x^2 + y^2 - 6x + 4y = 12$, which are parallel to the straight line $4x + 3y + 5 = 0$.

(b) $x^2 + y^2 - 22x - 4y + 25 = 0$, which are perpendicular to the straight line $5x + 12y + 9 = 0$

(c) $x^2 + y^2 = 25$, which are inclined at 30° to the axis of x.

9. A circle passes through the points (–1, 1), (0, 6) and (5, 5). Find the points on the circle the tangents at which are parallel to the straight line joining origin to the centre.

10. Tangents are drawn to the concentric circles $x^2 + y^2 = a^2$ and $x^2 + y^2 = b^2$ at right angle to one another. Show that the locus of their point of intersection is a 3rd concentric circle. Find its radius.

11. A variable circle passes through the point A (a, b) & touches the x-axis. Show that the locus of the other end of the diameter through A is $(x-a)^2 = 4by$.

12. A circle is drawn with its centre on the line $x + y = 2$ to touch the line $4x - 3y + 4 = 0$ and pass through the point (0, 1). Find its equation.

13. Show that the line $3x - 4y - c = 0$ will meet the circle having centre at (2, 4) and the radius 5 in real and distinct points if $-35 < c < 15$.

14. Let L_1 be a straight line through the origin and L_2 be the straight line $x + y = 1$. If the intercepts made by the circle $x^2 + y^2 - x + 3y = 0$ on L_1 & L_2 are equal, then find the equation(s) which represent L_1.

15. Find the equation of a line with gradient 1 such that the two circles $x^2 + y^2 = 4$ and $x^2 + y^2 - 10x - 14y + 65 = 0$ intercept equal length on it.

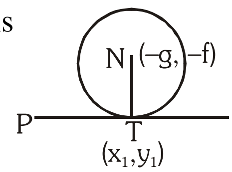
16. Find the equations of straight lines which pass through the intersection of the lines $x - 2y - 5 = 0$, $7x + y = 50$ & divide the circumference of the circle $x^2 + y^2 = 100$ into two arcs whose lengths are in the ratio 2 : 1.

6. NORMAL OF CIRCLE :

Normal at a point is the straight line which is perpendicular to the tangent at the point of contact.

Note : Normal at point of the circle passes through the centre of the circle.

(a) Equation of normal at point (x_1,y_1) of circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is



$$\mathbf{y - y_1 = \left(\frac{y_1 + f}{x_1 + g}\right)(x - x_1)}$$

(b) The equation of normal on any point (x_1,y_1) of circle $x^2 + y^2 = a^2$ is $\frac{y}{x} = \frac{y_1}{x_1}$.

(c) If $x^2 + y^2 = a^2$ is the equation of the circle then at any point 't' of this circle (a cos t, a sint), the equation of normal is **xsint – ycost = 0.**

Illustration 12 : Find the equation of the normal to the circle $x^2 + y^2 - 5x + 2y - 48 = 0$ at the point (5, 6).

Solution : Since normal to the circle always passes through the centre so equation of the normal will be the line passing through (5, 6) & $\left(\frac{5}{2}, -1\right)$

i.e. $y + 1 = \frac{7}{5/2}\left(x - \frac{5}{2}\right) \Rightarrow 5y + 5 = 14x - 35$

$\Rightarrow \quad 14x - 5y - 40 = 0$ **Ans.**

Illustration 13 : If the straight line $ax + by = 2$; $a, b \neq 0$ touches the circle $x^2 + y^2 - 2x = 3$ and is normal to the circle $x^2 + y^2 - 4y = 6$, then the values of a and b are respectively

(A) 1, –1 (B) 1, 2 (C) $-\frac{4}{3}$, 1 (D) 2, 1

Solution : Given $x^2 + y^2 - 2x = 3$

$\therefore$ centre is (1, 0) and radius is 2

Given $x^2 + y^2 - 4y = 6$

$\therefore$ centre is (0, 2) and radius is $\sqrt{10}$. Since line $ax + by = 2$ touches the first circle

$\therefore \quad \frac{|a(1) + b(0) - 2|}{\sqrt{a^2 + b^2}} = 2$ or $|(a - 2)| = [2\sqrt{a^2 + b^2}]$ .... (i)

Also the given line is normal to the second circle. Hence it will pass through the centre of the second circle.

$\therefore \quad a(0) + b(2) = 2$ or $2b = 2$ or $b = 1$

Putting this value in equation (i) we get $|a - 2| = 2\sqrt{a^2 + 1^2}$ or $(a - 2)^2 = 4(a^2 + 1)$

or $a^2 + 4 - 4a = 4a^2 + 4$ or $3a^2 + 4a = 0$ or $a(3a + 4) = 0$ or $a = 0, -\frac{4}{3}$ $(a \neq 0)$

$\therefore$ values of a and b are $\left(-\frac{4}{3}, 1\right)$. **Ans. (C)**

Illustration 14 : Find the equation of a circle having the lines $x^2 + 2xy + 3x + 6y = 0$ as its normal and having size just sufficient to contain the circle $x(x - 4) + y(y - 3) = 0$.

Solution : Pair of normals are $(x + 2y)(x + 3) = 0$

$\therefore$ Normals are $x + 2y = 0$, $x + 3 = 0$.

Point of intersection of normals is the centre of required circle i.e. $C_1(-3, 3/2)$ and centre of given circle is $C_2(2, 3/2)$ and radius $r_2 = \sqrt{4 + \frac{9}{4}} = \frac{5}{2}$

Let r_1 be the radius of required circle

$\Rightarrow \quad r_1 = C_1C_2 + r_2 = \sqrt{(-3-2)^2 + \left(\frac{3}{2} - \frac{3}{2}\right)^2} + \frac{5}{2} = \frac{15}{2}$

Hence equation of required circle is $x^2 + y^2 + 6x - 3y - 45 = 0$

Do yourself - 4 :

1. Find the equation of the normal to the circle $x^2 + y^2 = 2x$, which is parallel to the line $x + 2y = 3$.

2. Statement I : The only circle having radius $\sqrt{10}$ and a diameter along line $2x + y = 5$ is $x^2 + y^2 - 6x + 2y = 0$.

Statement II : $2x + y = 5$ is a normal to the circle $x^2 + y^2 - 6x + 2y = 0$.

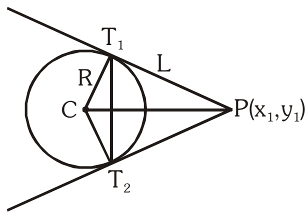
(1) Statement I is false, Statement II is true

(2) Statement I is true ; Statement II is false.

(3) Statement I is true, Statement II is true, Statement II is not a correct explanation for Statement I.

(4) Statement I is true : Statement II is true ; Statement II is a correct explanation for Statement I.

7. CHORD OF CONTACT (T = 0) :



A line joining the two points of contacts of two tangents drawn from a point out side the circle, is called chord of contact of that point.

If two tangents PT_1 & PT_2 are drawn from the point $P(x_1, y_1)$ to the circle

$S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$, then the equation of the chord of contact T_1T_2 is :

$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$ (i.e. T = 0 same as equation of tangent).

Remember :

(a) Length of chord of contact $T_1T_2 = \dfrac{2LR}{\sqrt{R^2 + L^2}}$.

(b) Area of the triangle formed by the pair of the tangents & its chord of contact $= \dfrac{RL^3}{R^2 + L^2}$,

where R is the radius of the circle & L is the length of the tangent from (x_1, y_1) on $S = 0$.

(c) Angle between the pair of tangents from $P(x_1, y_1) = \tan^{-1}\left(\dfrac{2RL}{L^2 - R^2}\right)$

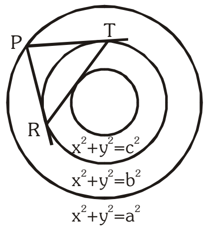
(d) Equation of the circle circumscribing the triangle PT_1T_2 or quadrilateral CT_1PT_2 is :
$(x - x_1)(x + g) + (y - y_1)(y + f) = 0$.

(e) The joint equation of a pair of tangents drawn from the point $A(x_1, y_1)$ to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is : $SS_1 = T^2$.

Where $S \equiv x^2 + y^2 + 2gx + 2fy + c$; $S_1 \equiv x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$

$T \equiv xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c$.

Illustration 15 : The chord of contact of tangents drawn from a point on the circle $x^2 + y^2 = a^2$ to the circle $x^2 + y^2 = b^2$ touches the circle $x^2 + y^2 = c^2$. Show that a, b, c are in GP.



Solution : Let P(acosθ, asinθ) be a point on the circle $x^2 + y^2 = a^2$.

Then equation of chord of contact of tangents drawn from P(acosθ, asinθ) to the circle $x^2 + y^2 = b^2$ is $ax\cos\theta + ay\sin\theta = b^2$ ..... (i)

This touches the circle $x^2 + y^2 = c^2$ ..... (ii)

∴ Length of perpendicular from (0, 0) to (i) = radius of (ii)

$$\therefore \quad \frac{|0+0-b^2|}{\sqrt{(a^2\cos^2\theta + a^2\sin^2\theta)}} = c$$

or $b^2 = ac \Rightarrow$ a, b, c are in GP.

Do yourself - 5 :

1. Find the equation of the chord of contact of the point (1, 2) with respect to the circle $x^2 + y^2 + 2x + 3y + 1 = 0$

2. Tangents are drawn from the point P(4, 6) to the circle $x^2 + y^2 = 25$. Find the area of the triangle formed by them and their chord of contact.

3. A point moving around circle $(x + 4)^2 + (y + 2)^2 = 25$ with centre C broke away from it either at the point A or point B on the circle and moved along a tangent to the circle passing through the point D (3, –3). Find the following.

(i) Equation of the tangents at A and B.

(ii) Coordinates of the points A and B.

(iii) Angle ADB and the maximum and minimum distances of the point D from the circle.

(iv) Area of quadrilateral ADBC and the ΔDAB.

(v) Equation of the circle circumscribing the ΔDAB and also the length of the intercepts made by this circle on the coordinate axes.

4. Tangents OP and OQ are drawn from the origin O to the circle $x^2 + y^2 + 2gx + 2fy + c = 0$. Find the equation of the circumcircle of the triangle OPQ.

5. The straight line $x - 2y + 1 = 0$ intersects the circle $x^2 + y^2 = 25$ in points T and T', find the co-ordinates of a point of intersection of tangents drawn at T and T' to the circle.

8. EQUATION OF THE CHORD WITH A GIVEN MIDDLE POINT ($T = S_1$) :

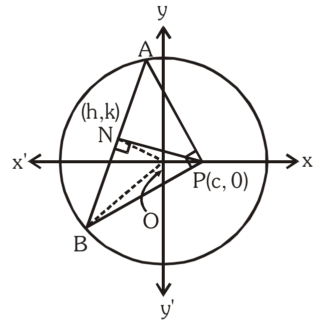
The equation of the chord of the circle $S \equiv x^2 + y^2 + 2gx + 2fy + c = 0$ in terms of its mid point M (x_1, y_1) is $\mathbf{y - y_1 = -\frac{x_1+g}{y_1+f}(x - x_1)}$. This on simplification can be put in the form

$xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = x_1^2 + y_1^2 + 2gx_1 + 2fy_1 + c$ which is designated by $\mathbf{T = S_1}$.

Note that : The shortest chord of a circle passing through a point 'M' inside the circle, is one chord whose middle point is M.

Illustration 16 : Find the locus of middle points of chords of the circle $x^2 + y^2 = a^2$, which subtend right angle at the point (c, 0).

Solution : Let N(h, k) be the middle point of any chord AB, which subtend a right angle at P(c, 0).



Since $\angle APB = 90°$

$\therefore$ NA = NB = NP

(since distance of the vertices from middle point of the hypotenuse are equal)

or $(NA)^2 = (NB)^2 = (h - c)^2 + (k - 0)^2$ ..... (i)

But also $\angle BNO = 90°$

$\therefore$ $(OB)^2 = (ON)^2 + (NB)^2$

$\Rightarrow$ $-(NB)^2 = (ON)^2 - (OB)^2$ $\Rightarrow$ $-[(h - c)^2 + (k - 0)^2] = (h^2 + k^2) - a^2$

or $2(h^2 + k^2) - 2ch + c^2 - a^2 = 0$

$\therefore$ Locus of N(h, k) is $2(x^2 + y^2) - 2cx + c^2 - a^2 = 0$ **Ans.**

Illustration 17 : Let a circle be given by $2x(x - a) + y(2y - b) = 0$ $(a \neq 0, b \neq 0)$

Find the condition on a and b if two chords, each bisected by the x-axis, can be drawn to the circle from (a, b/2).

Solution : The given circle is $2x(x - a) + y(2y - b) = 0$

or $x^2 + y^2 - ax - by/2 = 0$

Let AB be the chord which is bisected by x-axis at a point M. Let its co-ordinates be M(h, 0).

and $S \equiv x^2 + y^2 - ax - by/2 = 0$

$\therefore$ Equation of chord AB is $T = S_1$

$$hx + 0 - \frac{a}{2}(x+h) - \frac{b}{4}(y+0) = h^2 + 0 - ah - 0$$

Since its passes through (a, b/2) we have $ah - \frac{a}{2}(a+h) - \frac{b^2}{8} = h^2 - ah \Rightarrow h^2 - \frac{3ah}{2} + \frac{a^2}{2} + \frac{b^2}{8} = 0$

Now there are two chords bisected by the x-axis, so there must be two distinct real roots of h.

$\therefore$ $B^2 - 4AC > 0$

$$\Rightarrow \left(\frac{-3a}{2}\right)^2 - 4.1.\left(\frac{a^2}{2} + \frac{b^2}{8}\right) > 0 \Rightarrow a^2 > 2b^2.$$ **Ans.**

Do yourself - 6 :

1. Find the equation of the chord of $x^2 + y^2 - 6x + 10 - a = 0$ which is bisected at $(-2, 4)$.
2. Find the locus of mid point of chord of $x^2 + y^2 + 2gx + 2fy + c = 0$ that pass through the origin.
3. Find the co-ordinates of the middle point of the chord which the circle $x^2 + y^2 - 2x + 2y - 2 = 0$ cuts off on the line $y = x - 1$.

   Find also the equation of the locus of the middle point of all chords of the circle which are parallel to the line $y = x - 1$.
4. Find the locus of the mid point of all chords of the circle $x^2 + y^2 - 2x - 2y = 0$ such that the pair of lines joining $(0, 0)$ & the point of intersection of the chords with the circles make equal angle with axis of x.

9. DIRECTOR CIRCLE :

The locus of point of intersection of two perpendicular tangents to a circle is called director circle. Let P(h,k) is the point of intersection of two tangents drawn on the circle $x^2 + y^2 = a^2$. Then the equation of the pair of tangents is $SS_1 = T^2$

i.e. $(x^2 + y^2 - a^2)(h^2 + k^2 - a^2) = (hx + ky - a^2)^2$

As lines are perpendicular to each other then, coefficient of x^2 + coefficient of $y^2 = 0$

$\Rightarrow$ $[(h^2 + k^2 - a^2) - h^2] + [(h^2 + k^2 - a^2) - k^2] = 0$

$\Rightarrow$ $\mathbf{h^2 + k^2 = 2a^2}$

$\therefore$ locus of (h,k) is $x^2 + y^2 = 2a^2$ which is the equation of the director circle.

$\therefore$ director circle is a concentric circle whose radius is $\sqrt{2}$ times the radius of the circle.

Note : The director circle of $x^2 + y^2 + 2gx + 2fy + c = 0$ is $x^2 + y^2 + 2gx + 2fy + 2c - g^2 - f^2 = 0$

Illustration 18 : Let P be any moving point on the circle $x^2 + y^2 - 2x = 1$, from this point chord of contact is drawn w.r.t. the circle $x^2 + y^2 - 2x = 0$. Find the locus of the circumcentre of the triangle CAB, C being centre of the circle and A, B are the points of contact.

Solution : The two circles are

$(x - 1)^2 + y^2 = 1$ .... (i)

$(x - 1)^2 + y^2 = 2$ .... (ii)

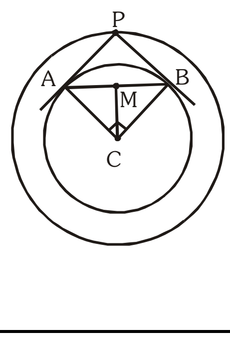
So the second circle is the director circle of the first. So $\angle APB = \pi/2$

Also $\angle ACB = \pi/2$

Now circumcentre of the right angled triangle CAB would lie on the mid point of AB

So let the point be $M \equiv (h, k)$

Now, $CM = CB\sin 45° = \dfrac{1}{\sqrt{2}}$



So, $(h - 1)^2 + k^2 = \left(\dfrac{1}{\sqrt{2}}\right)^2$

So, locus of M is $(x - 1)^2 + y^2 = \dfrac{1}{2}$.

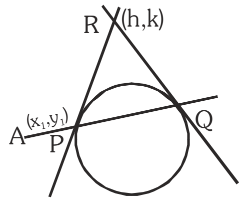
Do yourself - 7 :

1. Find the equation of the director circle of the circle $(x-h)^2 + (y-k)^2 = a^2$.

2. If the angle between the tangents drawn to $x^2 + y^2 + 4x + 8y + c = 0$ from $(0, 0)$ is $\frac{\pi}{2}$, then find value of 'c'

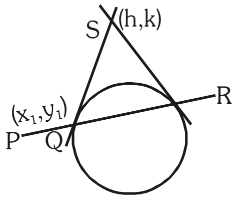
3. If two tangents are drawn from a point on the circle $x^2 + y^2 = 50$ to the circle $x^2 + y^2 = 25$, then find the angle between the tangents.

10. POLE AND POLAR :



Let any straight line through the given point $A(x_1,y_1)$ intersect the given circle S =0 in two points P and Q and if the tangent of the circle at P and Q meet at the point R then locus of point R is called polar of the point A and point A is called the pole, with respect to the given circle.

(a) The equation of the polar of point (x_1,y_1) w.r.t. circle $x^2 + y^2 = a^2$ (T = 0).



Let PQR is a chord which passes through the point $P(x_1,y_1)$ which intersects the circle at points Q and R and the tangents are drawn at points Q and R meet at point S(h,k) then equation of QR the chord of contact is $x_1h + y_1k = a^2$ $\therefore$ locus of point S(h,k) is $\mathbf{xx_1 + yy_1 = a^2}$ which is the equation of the polar.

Note :

(i) The equation of the polar is the T=0, so the polar of point (x_1,y_1) w.r.t circle $x^2 + y^2 + 2gx + 2fy + c = 0$ is $xx_1 + yy_1 + g(x + x_1) + f(y + y_1) + c = 0$

(ii) If point is outside the circle then equation of polar and chord of contact is same. So the chord of contact is polar.

(iii) If point is inside the circle then chord of contact does not exist but polar exists.

(iv) If point lies on the circle then polar, chord of contact and tangent on that point are same.

(v) If the polar of P w.r.t. a circle passes through the point Q, then the polar of point Q will pass through P and hence P & Q are conjugate points of each other w.r.t. the given circle.

(vi) If pole of a line w.r.t. a circle lies on second line. Then pole of second line lies on first line and hence both lines are conjugate lines of each other w.r.t. the given circle.

(vii) If O be the centre of a circle and P be any point, then OP is perpendicular to the polar of P.

(viii) If O be the centre of a circle and P any point, then if OP (produce, if necessary) meet the polar of P in Q, then OP. OQ = (radius)^2

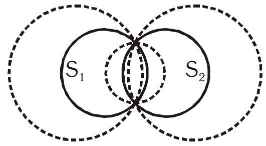
(b) Pole of a given line with respect to a circle

To find the pole of a line we assume the coordinates of the pole then from these coordinates we find the polar. This polar and given line represent the same line. Then by comparing the coefficients of similar terms we can get the coordinates of the pole. The pole of $\ell x + my + n = 0$

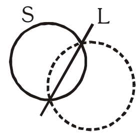
w.r.t. circle $x^2 + y^2 = a^2$ will be $\left(\frac{-\ell a^2}{n}, \frac{-ma^2}{n}\right)$

11. FAMILY OF CIRCLES :

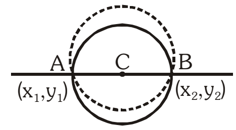
(a) The equation of the family of circles passing through the points of intersection of two circles $S_1 = 0$ & $S_2 = 0$ is : $S_1 + K\,S_2 = 0 \quad (K \neq -1)$.



(b) The equation of the family of circles passing through the point of intersection of a circle $S = 0$ & a line $L = 0$ is given by $S + KL = 0$.

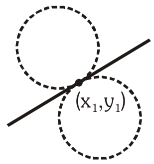


(c) The equation of a family of circles passing through two given points (x_1, y_1) & (x_2, y_2) can be written in the form :

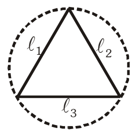


$$(x-x_1)(x-x_2) + (y-y_1)(y-y_2) + K\begin{vmatrix} x & y & 1 \\ x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \end{vmatrix} = 0 \text{ where K is a parameter.}$$

(d) The equation of a family of circles touching a fixed line $y - y_1 = m(x - x_1)$ at the fixed point (x_1, y_1) is $(x - x_1)^2 + (y - y_1)^2 + K[y - y_1 - m(x - x_1)] = 0$, where K is a parameter.



(e) Family of circles circumscribing a triangle whose sides are given by $L_1 = 0$; $L_2 = 0$ & $L_3 = 0$ is given by ; $L_1L_2 + \lambda L_2L_3 + \mu L_3L_1 = 0$ provided coefficient of $xy = 0$ & coefficient of x^2 = coefficient of y^2.



(f) Equation of circle circumscribing a quadrilateral whose sides in order are represented by the lines $L_1 = 0, L_2 = 0, L_3 = 0$ & $L_4 = 0$ is $L_1L_3 + \lambda L_2L_4 = 0$ provided coefficient of x^2 = coefficient of y^2 and coefficient of $xy = 0$.

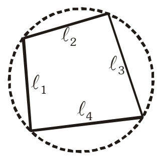


Illustration 19 : The equation of the circle through the points of intersection of $x^2 + y^2 - 1 = 0$, $x^2 + y^2 - 2x - 4y + 1 = 0$ and touching the line $x + 2y = 0$, is -

(A) $x^2 + y^2 + x + 2y = 0$ (B) $x^2 + y^2 - x + 20 = 0$

(C) $x^2 + y^2 - x - 2y = 0$ (D) $2(x^2 + y^2) - x - 2y = 0$

Solution : Family of circles is $x^2 + y^2 - 2x - 4y + 1 + \lambda(x^2 + y^2 - 1) = 0$

$(1 + \lambda)x^2 + (1 + \lambda)y^2 - 2x - 4y + (1 - \lambda) = 0$

$$x^2 + y^2 - \frac{2}{1+\lambda}x - \frac{4}{1+\lambda}y + \frac{1-\lambda}{1+\lambda} = 0$$

Centre is $\left(\frac{1}{1+\lambda}, \frac{2}{1+\lambda}\right)$ and radius = $\sqrt{\left(\frac{1}{1+\lambda}\right)^2+\left(\frac{2}{1+\lambda}\right)^2-\frac{1-\lambda}{1+\lambda}}=\frac{\sqrt{4+\lambda^2}}{|1+\lambda|}$.

Since it touches the line x + 2y = 0, hence

Radius = Perpendicular distance from centre to the line.

i.e., $\left|\frac{\frac{1}{1+\lambda}+2\frac{2}{1+\lambda}}{\sqrt{1^2+2^2}}\right|=\frac{\sqrt{4+\lambda^2}}{|1+\lambda|} \Rightarrow \sqrt{5}=\sqrt{4+\lambda^2} \Rightarrow \lambda = \pm 1$

$\lambda = -1$ cannot be possible in case of circle. So $\lambda = 1$.

Thus, we get the equation of circle. **Ans. (C)**

Do yourself - 8 :

1. Prove that the polar of a given point with respect to any one of circles $x^2 + y^2 - 2kx + c^2 = 0$, where k is a variable, always passes through a fixed point, whatever be the value of k.

2. Find the equation of the circle through the points of intersection of the circles $x^2 + y^2 + 2x + 3y - 7 = 0$ and $x^2 + y^2 + 3x - 2y - 1 = 0$ and passing through the point (1, 2).

3. Find the equation of the circle passing through the point of intersection of the circles $x^2 + y^2 - 6x + 2y + 4 = 0$, $x^2 + y^2 + 2x - 4y - 6 = 0$ and with its centre on the line y = x.

4. Find the equation of the circle passing through the points of intersection of the circles $x^2 + y^2 - 2x - 4y - 4 = 0$ and $x^2 + y^2 - 10x - 12y + 40 = 0$ and whose radius is 4.

5. Find the equation of the circle through points of intersection of the circle $x^2 + y^2 - 2x - 4y + 4 = 0$ and the line x + 2y = 4 which touches the line x + 2y = 0.

6. Find the equations of the circles which pass through the common points of the following pair of circles.

(a) $x^2 + y^2 + 2x + 3y - 7 = 0$ and $x^2 + y^2 + 3x - 2y - 1 = 0$ through the point (1,2)

(b) $x^2 + y^2 + 4x - 6y - 12 = 0$ and $x^2 + y^2 - 5x + 17y = 19$ and having its centre on x + y = 0.

7. The line 2x – 3y + 1 = 0 is tangent to a circle S = 0 at (1, 1). If the radius of the circle is $\sqrt{13}$. Find the equation of the circle S.

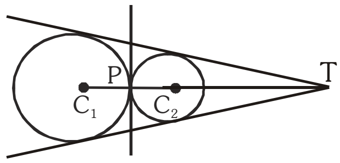
8. Find the equation of the circle which passes through the point (1, 1) & which touches the circle $x^2 + y^2 + 4x - 6y - 3 = 0$ at the point (2, 3) on it.

9. Consider a family of circles passing through two fixed points A(3, 7) & B(6, 5). The chords in which the circle $x^2 + y^2 - 4x - 6y - 3 = 0$ cuts the members of the family are concurrent at a point. Find the coordinates of this point.

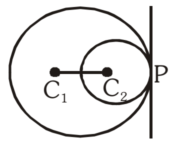
12. DIRECT AND TRANSVERSE COMMON TANGENTS :

Let two circles having centre C_1 and C_2 and radii, r_1 and r_2 and C_1C_2 is the distance between their centres then :

(a) Both circles will touch :

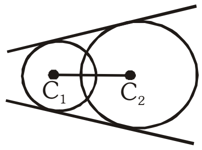


(i) Externally if $C_1C_2 = r_1 + r_2$ i.e. the distance between their centres is equal to sum of their radii and point P & T divides C_1C_2 in the ratio $r_1 : r_2$ (internally & externally respectively). In this case there are **three common tangents.**



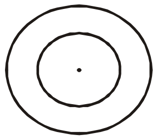
(ii) Internally if $C_1C_2 = |r_1-r_2|$ i.e. the distance between their centres is equal to difference between their radii and point P divides C_1C_2 in the ratio $r_1 : r_2$ **externally** and in this case there will be only **one common tangent.**

(b) The circles will intersect :



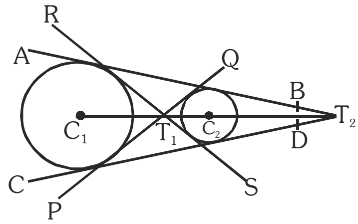
when $|r_1 - r_2| < C_1C_2 < r_1 + r_2$ in this case there are **two common tangents.**

(c) The circles will not intersect :



(i) One circle will lie inside the other circle if $C_1C_2 < |r_1-r_2|$ In this case there will be no common tangent.

(ii) When circle are apart from each other then $C_1C_2>r_1+r_2$ and in this case there will be **four common tangents.** Lines PQ and RS are called **transverse** or **indirect** or **internal** common tangents and these lines meet line C_1C_2 on $T_1$and T_1 divides the line C_1C_2 in the ratio $r_1 : r_2$ internally and lines AB & CD are called **direct** or **external** common tangents. These lines meet C_1C_2 produced on T_2. Thus T_2 divides C_1C_2 externally in the ratio $r_1 : r_2$.



Note : Length of direct common tangent $= \sqrt{(C_1C_2)^2 - (r_1 - r_2)^2}$

Length of transverse common tangent $= \sqrt{(C_1C_2)^2 - (r_1 + r_2)^2}$

Illustration 20 : Prove that the circles $x^2 + y^2 + 2ax + c^2 = 0$ and $x^2 + y^2 + 2by + c^2 = 0$ touch each other, if $\frac{1}{a^2} + \frac{1}{b^2} = \frac{1}{c^2}$.

Solution : Given circles are $x^2 + y^2 + 2ax + c^2 = 0$ .... (i)

and $x^2 + y^2 + 2by + c^2 = 0$ .... (ii)

Let C_1 and C_2 be the centres of circles (i) and (ii), respectively and r_1 and r_2 be their radii, then

$C_1 = (-a, 0)$, $C_2 = (0, -b)$, $r_1 = \sqrt{a^2 - c^2}$, $r_2 = \sqrt{b^2 - c^2}$

Here we find the two circles touch each other internally or externally.

For touch, $|C_1C_2| = |r_1 \pm r_2|$

or $\quad \sqrt{(a^2 + b^2)} = \left|\sqrt{(a^2 - c^2)} \pm \sqrt{(b^2 - c^2)}\right|$

On squaring $\;a^2 + b^2 = a^2 - c^2 + b^2 - c^2 \pm 2\sqrt{(a^2 - c^2)}\sqrt{(b^2 - c^2)}$

or $\quad c^2 = \pm\sqrt{a^2b^2 - c^2(a^2 + b^2) + c^4}$

Again squaring, $\quad c^4 = a^2b^2 - c^2(a^2 + b^2) + c^4$

or $\quad c^2(a^2 + b^2) = a^2b^2$

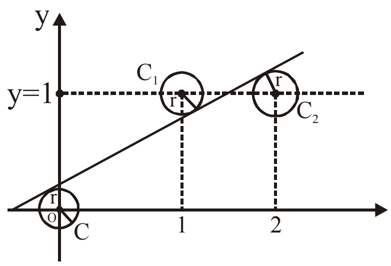
or $\quad \dfrac{1}{a^2} + \dfrac{1}{b^2} = \dfrac{1}{c^2}$

Do yourself - 9 :

1. Two circles with radius 5 touches at the point (1, 2). If the equation of common tangent is $4x + 3y = 10$ and one of the circle is $x^2 + y^2 + 6x + 2y - 15 = 0$. Find the equation of other circle.

2. Find the number of common tangents to the circles $x^2 + y^2 = 1$ and $x^2 + y^2 - 2x - 6y + 6 = 0$.

3. As shown in the figure, three circles which have the same radius r, have centres at (0,0) ; (1,1) and (2,1). If they have a common tangent line, as shown then, their radius 'r' is -



(A) $\dfrac{\sqrt{5}-1}{2}$ (B) $\dfrac{\sqrt{5}}{10}$ (C) $\dfrac{1}{2}$ (D) $\dfrac{\sqrt{3}-1}{2}$

4. Circle K is circle of largest radius, inscribed in the first quadrant touching the circle $x^2 + y^2 = 36$ internally. The length of the radius of the circle K, is-

(A) $\dfrac{6-\sqrt{2}}{2}$ (B) $\dfrac{3\sqrt{2}}{2}$ (C) 3 (D) $6\left(\sqrt{2}-1\right)$

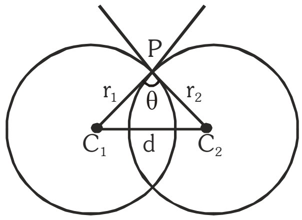
5. A circle S = 0 is drawn with its centre at (–1, 1) so as to touch the circle $x^2 + y^2 - 4x + 6y - 3 = 0$ externally. Find the intercept made by the circle S = 0 on the coordinate axes.

6. Find the equation of the circle whose radius is 3 and which touches the circle $x^2 + y^2 - 4x - 6y - 12 = 0$ internally at the point (–1, – 1).

13. THE ANGLE OF INTERSECTION OF TWO CIRCLES :

Definition : The angle between the tangents of two circles at the point of intersection of the two circles is called angle of intersection of two circles. If two circles are $S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$

$S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$ and θ is the acute angle between them



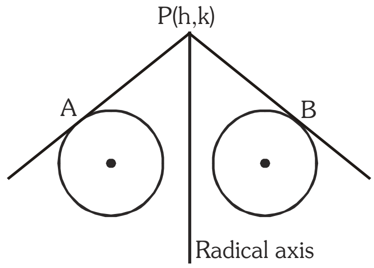
then $\cos\theta = \left|\left(\dfrac{r_1^2 + r_2^2 - d^2}{2r_1r_2}\right)\right| = \left|\dfrac{2g_1g_2 + 2f_1f_2 - c_1 - c_2}{2\sqrt{g_1^2 + f_1^2 - c_1}\ \sqrt{g_2^2 + f_2^2 - c_2}}\right|$

Here r_1 and r_2 are the radii of the circles and d is the distance between their centres.

If the angle of intersection of the two circles is a right angle then such circles are called **"Orthogonal circles"** and conditions for the circles to be orthogonal is -

$\mathbf{2g_1g_2 + 2f_1f_2 = c_1 + c_2}$

14. RADICAL AXIS OF THE TWO CIRCLES ($S_1 - S_2 = 0$) :



(a) Definition : The locus of a point, which moves in such a way that the length of tangents drawn from it to the circles are equal and is called the radical axis. If two circles are -

$S_1 \equiv x^2 + y^2 + 2g_1x + 2f_1\ y + c_1 = 0$

$S_2 \equiv x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$

Let P(h,k) is a point and PA,PB are length of two tangents on the circles from point P, Then from definition -

$\sqrt{h^2 + k^2 + 2g_1h + 2f_1k + c_1} = \sqrt{h^2 + k^2 + 2g_2h + 2f_2k + c_2}$ or $2(g_1 - g_2)\,h + 2(f_1 - f_2)\,k + c_1 - c_2 = 0$

$\therefore$ locus of (h,k)

$2x(g_1 - g_2) + 2y(f_1 - f_2)k + c_1 - c_2 = 0$

$\mathbf{S_1 - S_2 = 0}$

which is the equation of radical axis.

Note :

(i) To get the equation of the radical axis first of all make the coefficient of x^2 and $y^2 = 1$

(ii) If circles touch each other then radical axis is the common tangent to both the circles.

(iii) When the two circles intersect on real points then common chord is the radical axis of the two circles.

(iv) The radical axis of the two circles is perpendicular to the line joining the centre of two circles but not always pass through mid point of it.

(v) Radical axis (if exist) bisects common tangent to two circles.

(vi) The radical axes of three circles (taking two at a time) meet at a point.

(vii) If circles are concentric then the radical axis does not exist but if circles are not concentric then radical axis always exists.

(viii) If two circles are orthogonal to the third circle then radical axis of both circle passes through the centre of the third circle.

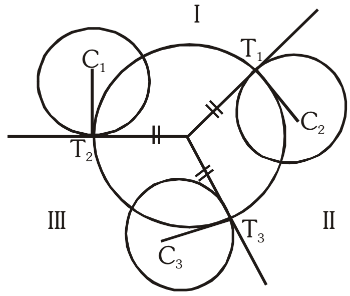
(ix) A system of circle, every pair of which have the same radical axis, is called a **coaxial system of circles.**

(b) Radical centre :

The radical centre of three circles is the point from which length of tangents on three circles are equal i.e. the point of intersection of radical axis of the circles is the radical centre of the circles. To get the radical axis of three circles $S_1 = 0$, $S_2=0$, $S_3=0$ we have to solve any two

$S_1 - S_2 = 0$, $S_2 - S_3 = 0$, $S_3 - S_1 = 0$

Note :



(i) The circle with centre as radical centre and radius equal to the length of tangent from radical centre to any of the circle, will cut the three circles orthogonally.

(ii) If three circles are drawn on three sides of a triangle taking them as diameter then its orthocenter will be its radical centre.

(iii) Locus of the centre of a variable circle orthogonal to two fixed circles is the radical axis between the two fixed circles.

(iv) If two circles are orthogonal, then the polar of a point 'P' on first circle w.r.t. the second circle passes through the point Q which is the other end of the diameter through P. Hence locus of a point which moves such that its polars w.r.t. the circles $S_1 = 0$, $S_2 = 0$ & $S_3 = 0$ are concurrent is a circle which is orthogonal to all the three circles.

Illustration 21 : A and B are two fixed points and P moves such that PA = nPB where $n \neq 1$. Show that locus of P is a circle and for different values of n all the circles have a common radical axis.

Solution : Let $A \equiv (a, 0)$, $B \equiv (-a, 0)$ and $P(h, k)$

so PA = nPB

$\Rightarrow \quad (h - a)^2 + k^2 = n^2[(h + a)^2 + k^2]$

$\Rightarrow \quad (1 - n^2)h^2 + (1 - n^2)k^2 - 2ah(1 + n^2) + (1 - n^2)a^2 = 0$

$\Rightarrow \quad h^2 + k^2 - 2ah\left(\frac{1+n^2}{1-n^2}\right) + a^2 = 0$

Hence locus of P is

$x^2 + y^2 - 2ax\left(\frac{1+n^2}{1-n^2}\right) + a^2 = 0$, which is a circle of different values of n.

Let n_1 and n_2 are two different values of n so their radical axis is x = 0 i.e. y-axis. Hence for different values of n the circles have a common radical axis.

Illustration 22 : Find the equation of the circle through the points of intersection of the circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ and cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

Solution : The equation of the circle through the intersection of the given circles is

$x^2 + y^2 - 4x - 6y - 12 + \lambda(-10x - 10y) = 0$ .... (i)

where $(-10x - 10y = 0)$ is the equation of radical axis for the circle

$x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$.

Equation (i) can be re-arranged as

$x^2 + y^2 - x(10\lambda + 4) - y(10\lambda + 6) - 12 = 0$

It cuts the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

Hence $2gg_1 + 2ff_1 = c + c_1$

$\Rightarrow \quad 2(5\lambda + 2)(1) + 2(5\lambda + 3)(0) = -12 - 4 \Rightarrow \lambda = -2$

Hence the required circle is

$x^2 + y^2 - 4x - 6y - 12 - 2(-10x - 10y) = 0$

i.e., $x^2 + y^2 + 16x + 14y - 12 = 0$

Illustration 23 : Find the radical centre of circles $x^2 + y^2 + 3x + 2y + 1 = 0$, $x^2 + y^2 - x + 6y + 5 = 0$ and $x^2 + y^2 + 5x - 8y + 15 = 0$. Also find the equation of the circle cutting them orthogonally.

Solution : Given circles are $S_1 \equiv x^2 + y^2 + 3x + 2y + 1 = 0$

$S_2 \equiv x^2 + y^2 - x + 6y + 5 = 0$

$S_3 \equiv x^2 + y^2 + 5x - 8y + 15 = 0$

Equations of two radical axes are $S_1 - S_2 \equiv 4x - 4y - 4 = 0$ or $x - y - 1 = 0$

and $S_2 - S_3 \equiv -6x + 14y - 10 = 0$ or $3x - 7y + 5 = 0$

Solving them the radical centre is (3, 2). Also, if r is the length of the tangent drawn from the radical centre (3, 2) to any one of the given circles, say S_1, we have

$r = \sqrt{S_1} = \sqrt{3^2 + 2^2 + 3.3 + 2.2 + 1} = \sqrt{27}$

Hence (3, 2) is the centre and $\sqrt{27}$ is the radius of the circle intersecting them orthogonally.

$\therefore \quad$ Its equation is $(x - 3)^2 + (y - 2)^2 = r^2 = 27 \Rightarrow x^2 + y^2 - 6x - 4y - 14 = 0$

Alternative Method :

Let $x^2 + y^2 + 2gx + 2fy + c = 0$ be the equation of the circle cutting the given circles orthogonally.

$\therefore \quad 2g\left(\frac{3}{2}\right) + 2f(1) = c + 1$ or $3g + 2f = c + 1$ .... (i)

$$2g\left(-\frac{1}{2}\right)+2f(3) = c + 5 \text{ or } \quad -g + 6f = c + 5 \qquad \text{........ (ii)}$$

and $$2g\left(\frac{5}{2}\right)+2f(-4) = c + 15 \text{ or } \quad 5g - 8f = c + 15 \qquad \text{........ (iii)}$$

Solving (i), (ii) and (iii) we get $g = -3, f = -2$ and $c = -14$

$\therefore$ equation of required circle is $x^2 + y^2 - 6x - 4y - 14 = 0$ **Ans.**

Do yourself - 10 :

1. Find the angle of intersection of two circles
$S : x^2 + y^2 - 4x + 6y + 11 = 0$ & $S' : x^2 + y^2 - 2x + 8y + 13 = 0$

2. Find the equation of the radical axis of the circle $x^2 + y^2 - 3x - 4y + 5 = 0$ and $3x^2 + 3y^2 - 7x - 8y + 11 = 0$

3. Find the radical centre of three circles described on the three sides $4x - 7y + 10 = 0$, $x + y - 5 = 0$ and $7x + 4y - 15 = 0$ of a triangle as diameters.

4. The angle at which the circle $(x-1)^2 + y^2 = 10$ and $x^2 + (y - 2)^2 = 5$ intersect is -

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) $\frac{\pi}{2}$

5. Find the radical centre of the following set of circles
$x^2 + y^2 - 3x - 6y + 14 = 0$; $x^2 + y^2 - x - 4y + 8 = 0$; $x^2 + y^2 + 2x - 6y + 9 = 0$

6. Find the equation of a circle which is co-axial with circles $2x^2 + 2y^2 - 2x + 6y - 3 = 0$ & $x^2 + y^2 + 4x + 2y + 1 = 0$. It is given that the centre of the circle to be determined lies on the radical axis of these two circles.

7. Write the condition so that circles $x^2 + y^2 + 2ax + c = 0$ and $x^2 + y^2 + 2by + c = 0$ touch externally.

15. SOME IMPORTANT RESULTS TO REMEMBER :

(a) If the circle $S_1 = 0$, bisects the circumference of the circle $S_2 = 0$, then their common chord will be the diameter of the circle $S_2 = 0$.

(b) The radius of the director circle of a given circle is $\sqrt{2}$ times the radius of the given circle.

(c) The locus of the middle point of a chord of a circle subtend a right angle at a given point will be a circle.

(d) The length of side of an equilateral triangle inscribed in the circle $x^2 + y^2 = a^2$ is $\sqrt{3}\,a$

(e) If the lengths of tangents from the points A and B to a circle are ℓ_1 and ℓ_2 respectively, then if the points A and B are conjugate to each other, then $(AB)^2 = \ell_1^2 + \ell_2^2$.

(f) Length of transverse common tangent is less than the length of direct common tangent.

Do yourself - 11 :

1. When the circles $x^2 + y^2 + 4x + 6y + 3 = 0$ and $2(x^2 + y^2) + 6x + 4y + c = 0$ intersect orthogonally, then find the value of c is

2. Two circles whose radii are equal to 4 and 8 intersect at right angles. The length of their common chord is -

(A) $\frac{16}{\sqrt{5}}$ (B) 8 (C) $4\sqrt{6}$ (D) $\frac{8\sqrt{5}}{5}$

3. Two congruent circles with centres at (2,3) and (5,6) which intersect at right angles has radius equal to -

(A) $2\sqrt{2}$ (B) 3 (C) 4 (D) none

4. The equation of a circle which touches the line $x + y = 5$ at N(–2,7) and cuts the circle $x^2 + y^2 + 4x - 6y + 9 = 0$ orthogonally, is -

(A) $x^2 + y^2 + 7x - 11y + 38 = 0$ (B) $x^2 + y^2 = 53$

(C) $x^2 + y^2 + x - y - 44 = 0$ (D) $x^2 + y^2 - x + y - 62 = 0$

5. Find the equation to the circle orthogonal to the two circles

$x^2 + y^2 - 4x - 6y + 11 = 0$; $x^2 + y^2 - 10x - 4y + 21 = 0$ and has $2x + 3y = 7$ as diameter.

6. Find the equation of the circle through the points of intersection of circles $x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 4y - 12 = 0$ & cutting the circle $x^2 + y^2 - 2x - 4 = 0$ orthogonally.

7. The centre of the circle S = 0 lie on the line $2x - 2y + 9 = 0$ & S = 0 cuts orthogonally the circle $x^2 + y^2 = 4$. Show that circle S = 0 passes through two fixed points & find their coordinates.

8. (a) Find the equation of a circle passing through the origin if the line pair, $xy - 3x + 2y - 6 = 0$ is orthogonal to it. If this circle is orthogonal to the circle $x^2 + y^2 - kx + 2ky - 8=0$ then find the value of k.

(b) Find the equation of the circle which cuts the circle $x^2 + y^2 - 14x - 8y + 64 = 0$ and the coordinate axes orthogonally.

9. Find the equation to the circle, cutting orthogonally each of the following circles :

$x^2 + y^2 - 2x + 3y - 7 = 0$; $x^2 + y^2 + 5x - 5y + 9 = 0$; $x^2 + y^2 + 7x - 9y + 29 = 0$.

<u>Miscellaneous Illustrations</u> :

Illustration 24 : Find the equation of a circle which passes through the point (2, 0) and whose centre is the limit of the point of intersection of the lines $3x + 5y = 1$ and $(2 + c)x + 5c^2y = 1$ as $c \to 1$.

Solution : Solving the equations $(2 + c)x + 5c^2y = 1$ and $3x + 5y = 1$

then $(2 + c)x + 5c^2\left(\frac{1-3x}{5}\right) = 1$ or $(2 + c)x + c^2(1 - 3x) = 1$

$\therefore \quad x = \dfrac{1-c^2}{2+c-3c^2}$ or $x = \dfrac{(1+c)(1-c)}{(3c+2)(1-c)} = \dfrac{1+c}{3c+2}$

$\therefore \quad x = \lim\limits_{c \to 1} \dfrac{1+c}{3c+2}$ or $x = \dfrac{2}{5}$

$\therefore \quad y = \dfrac{1-3x}{5} = \dfrac{1-\frac{6}{5}}{5} = -\dfrac{1}{25}$

Therefore the centre of the required circle is $\left(\dfrac{2}{5}, \dfrac{-1}{25}\right)$ but circle passes through (2, 0)

$\therefore$ Radius of the required circle $= \sqrt{\left(\dfrac{2}{5}-2\right)^2 + \left(-\dfrac{1}{25}-0\right)^2} = \sqrt{\dfrac{64}{25}+\dfrac{1}{625}} = \sqrt{\dfrac{1601}{625}}$

Hence the required equation of the circle is $\left(x-\dfrac{2}{5}\right)^2 + \left(y+\dfrac{1}{25}\right)^2 = \dfrac{1601}{625}$

or $25x^2 + 25y^2 - 20x + 2y - 60 = 0$ **Ans.**

Illustration 25 : Two straight lines rotate about two fixed points. If they start from their position of coincidence such that one rotates at the rate double that of the other. Prove that the locus of their point of intersection is a circle.

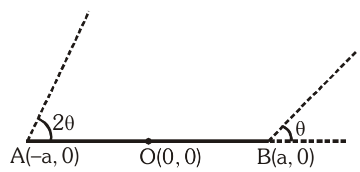
Solution : Let $A \equiv (-a, 0)$ and $B \equiv (a, 0)$ be two fixed points.

Let one line which rotates about B an angle θ with the x-axis at any time t and at that time the second line which rotates about A make an angle 2θ with x-axis.

Now equation of line through B and A are respectively

$y - 0 = \tan\theta(x - a)$ ...... (i)

and $y - 0 = \tan 2\theta(x + a)$ ...... (ii)



From (ii), $y = \dfrac{2\tan\theta}{1-\tan^2\theta}(x+a)$

$= \left\{\dfrac{\dfrac{2y}{(x-a)}}{1-\dfrac{y^2}{(x-a)^2}}\right\}(x+a)$ (from (i))

$\Rightarrow \quad y = \dfrac{2y(x-a)(x+a)}{(x-a)^2-y^2} \quad \Rightarrow \quad (x-a)^2 - y^2 = 2(x^2 - a^2)$

or $x^2 + y^2 + 2ax - 3a^2 = 0$ which is the required locus.

Illustration 26 : If the circle $x^2 + y^2 + 6x - 2y + k = 0$ bisects the circumference of the circle $x^2 + y^2 + 2x - 6y - 15 = 0$, then k =

(A) 21 (B) –21 (C) 23 (D) –23

Solution : $2g_2(g_1 - g_2) + 2f_2(f_1 - f_2) = c_1 - c_2$

$2(1)(3 - 1) + 2(-3)(-1 + 3) = k + 15$

$4 - 12 = k + 15$ or $-8 = k + 15 \Rightarrow k = -23$ **Ans. (D)**

Illustration 27 : Find the equation of the circle of minimum radius which contains the three circles.

$S_1 \equiv x^2 + y^2 - 4y - 5 = 0$

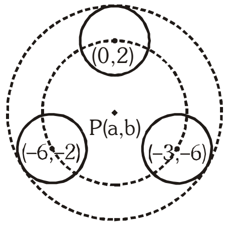
$S_2 \equiv x^2 + y^2 + 12x + 4y + 31 = 0$

$S_3 \equiv x^2 + y^2 + 6x + 12y + 36 = 0$

Solution : For S_1, centre = (0, 2) and radius = 3

For S_2, centre = (–6, –2) and radius = 3

For S_3, centre = (–3, –6) and radius = 3



let P(a, b) be the centre of the circle passing through the centres of the three given circles, then

$(a - 0)^2 + (b - 2)^2 = (a + 6)^2 + (b + 2)^2$

$\Rightarrow \quad (a + 6)^2 - a^2 = (b - 2)^2 - (b + 2)^2$

$(2a + 6)6 = 2b(-4)$

$$b = \frac{2 \times 6(a+3)}{-8} = -\frac{3}{2}(a+3)$$

again $(a - 0)^2 + (b - 2)^2 = (a + 3)^2 + (b + 6)^2$

$\Rightarrow \quad (a + 3)^2 - a^2 = (b - 2)^2 - (b + 6)^2$

$(2a + 3)3 = (2b + 4)(-8)$

$$(2a + 3)3 = -16\left[-\frac{3}{2}(a+3) + 2\right]$$

$6a + 9 = -8(-3a - 5)$

$6a + 9 = 24a + 40$

$18a = -31$

$$a = -\frac{31}{18}, \; b = -\frac{23}{12}$$

radius of the required circle $= 3 + \sqrt{\left(-\frac{31}{18}\right)^2 + \left(-\frac{23}{12} - 2\right)^2} = 3 + \frac{5}{36}\sqrt{949}$

$\therefore$ equation of the required circle is $\left(x + \frac{31}{18}\right)^2 + \left(y + \frac{23}{12}\right)^2 = \left(3 + \frac{5}{36}\sqrt{949}\right)^2$

Illustration 28 : Find the equation of the image of the circle $x^2 + y^2 + 16x - 24y + 183 = 0$ by the line mirror $4x + 7y + 13 = 0$.

Solution : Centre of given circle = (–8, 12), radius = 5

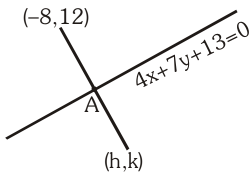
the given line is $4x + 7y + 13 = 0$

let the centre of required circle is (h, k)

since radius will not change. so radius of required circle is 5.

Now (h, k) is the reflection of centre (–8, 12) in the line $4x + 7y + 13 = 0$

Co-ordinates of A = $\left(\dfrac{-8+h}{2}, \dfrac{12+k}{2}\right)$



$\Rightarrow \quad \dfrac{4(-8+h)}{2} + \dfrac{7(12+k)}{2} + 13 = 0$

$-32 + 4h + 84 + 7k + 26 = 0$

$4h + 7k + 78 = 0$ ....(i)

Also $\dfrac{k-12}{h+8} = \dfrac{7}{4}$

$4k - 48 = 7h + 56$

$4k = 7h + 104$ ....(ii)

solving (i) & (ii)

$h = -16, k = -2$

$\therefore$ required circle is $(x + 16)^2 + (y + 2)^2 = 5^2$

Illustration 29 : The circle $x^2 + y^2 - 6x - 10y + k = 0$ does not touch or intersect the coordinate axes and the point (1, 4) is inside the circle. Find the range of the value of k.

Solution : Since (1, 4) lies inside the circle

$\Rightarrow \quad S_1 < 0$

$\Rightarrow \quad (1)^2 + (4)^2 - 6(1) - 10(4) + k < 0$

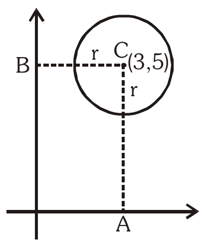
$\Rightarrow \quad k < 29$

Also centre of given circle is (3, 5) and circle does not touch or intersect the coordinate axes

$\Rightarrow \quad r < CA \quad \& \; r < CB$

$CA = 5$

$CB = 3$



$\Rightarrow \quad r < 5 \quad \& \quad r < 3$

$\Rightarrow \quad r < 3 \quad \text{or} \quad r^2 < 9$

$r^2 = 9 + 25 - k$

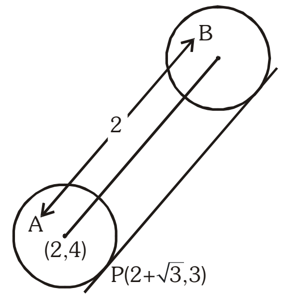
$r^2 = 34 - k \qquad \Rightarrow \qquad 34 - k < 9$

$k > 25$

$\Rightarrow \quad k \in (25, 29)$

Illustration 30 : The circle $x^2 + y^2 - 4x - 8y + 16 = 0$ rolls up the tangent to it at $(2 + \sqrt{3}, 3)$ by 2 units, find the equation of the circle in the new position.

Solution : Given circle is $x^2 + y^2 - 4x - 8y + 16 = 0$



let $P \equiv (2 + \sqrt{3}, 3)$

Equation of tangent to the circle at $P(2 + \sqrt{3}, 3)$ will be

$(2 + \sqrt{3})x + 3y - 2(x + 2 + \sqrt{3}) - 4(y + 3) + 16 = 0$

or $\quad \sqrt{3}x - y - 2\sqrt{3} = 0$

slope $= \sqrt{3} \quad \Rightarrow \quad \tan\theta = \sqrt{3}$

$\theta = 60°$

line AB is parallel to the tangent at P

$\Rightarrow$ coordinates of point B = $(2 + 2\cos 60°, 4 + 2\sin 60°)$

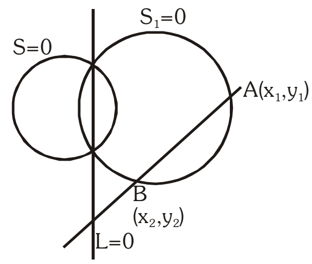
thus $B = (3, 4 + \sqrt{3})$

radius of circle $= \sqrt{2^2 + 4^2 - 16} = 2$

$\therefore$ equation of required circle is $(x - 3)^2 + (y - 4 - \sqrt{3})^2 = 2^2$

Illustration 31 : A fixed circle is cut by a family of circles all of which, pass through two given points $A(x_1, y_1)$ and $B(x_2, y_2)$. Prove that the chord of intersection of the fixed circle with any circle of the family passes through a fixed point.

Solution : Let S = 0 be the equation of fixed circle



let $S_1 = 0$ be the equation of any circle through A and B

which intersect S = 0 in two points.

$L \equiv S - S_1 = 0$ is the equation of the chord of intersection

of S = 0 and $S_1 = 0$

let $L_1 = 0$ be the equation of line AB

let S_2 be the equation of the circle whose diametrical ends are $A(x_1, y_1)$ & $B(x_2, y_2)$

then $S_1 \equiv S_2 - \lambda L_1 = 0$

$\Rightarrow \quad L \equiv S - (S_2 - \lambda L_1) = 0 \quad$ or $\quad L \equiv (S - S_2) + \lambda L_1 = 0$

or $\quad L \equiv L' + \lambda L_1 = 0 \qquad$ ....(i)

(i) implies each chord of intersection passes through the fixed point, which is the point of intersection of lines L' = 0 & $L_1 = 0$. Hence proved.

EXERCISE (O-1)

Straight Objective Type

1. Centres of the three circles $x^2 + y^2 - 4x - 6y - 14 = 0$, $x^2 + y^2 + 2x + 4y - 5 = 0$ and $x^2 + y^2 - 10x - 16y + 7 = 0$

(A) are the vertices of a right triangle

(B) the vertices of an isosceles triangle which is not regular

(C) vertices of a regular triangle

(D) are collinear

CR0001

2. $y - 1 = m_1(x - 3)$ and $y - 3 = m_2(x - 1)$ are two family of straight lines, at right angled to each other. The locus of their point of intersection is

(A) $x^2 + y^2 - 2x - 6y + 10 = 0$ (B) $x^2 + y^2 - 4x - 4y + 6 = 0$

(C) $x^2 + y^2 - 2x - 6y + 6 = 0$ (D) $x^2 + y^2 - 4x - 4y - 6 = 0$

CR0002

3. Suppose that the equation of the circle having (–3, 5) and (5, –1) as end points of a diameter is $(x - a)^2 + (y - b)^2 = r^2$. Then $a + b + r$, $(r > 0)$ is

(A) 8 (B) 9 (C) 10 (D) 11

CR0003

4. The area of an equilateral triangle inscribed in the circle $x^2 + y^2 - 2x = 0$ is

(A) $\frac{3\sqrt{3}}{4}$ (B) $\frac{3\sqrt{3}}{2}$ (C) $\frac{3\sqrt{3}}{8}$ (D) none

CR0004

5. The smallest distance between the circle $(x - 5)^2 + (y + 3)^2 = 1$ and the line $5x + 12y - 4 = 0$, is

(A) 1/13 (B) 2/13 (C) 3/15 (D) 4/15

CR0005

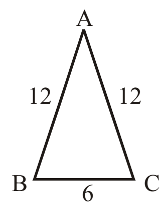
6. The equation of the image of the circle $x^2 + y^2 + 16x - 24y + 183 = 0$ by the line mirror $4x + 7y + 13 = 0$ is

(A) $x^2 + y^2 + 32x - 4y + 235 = 0$ (B) $x^2 + y^2 + 32x + 4y - 235 = 0$

(C) $x^2 + y^2 + 32x - 4y - 235 = 0$ (D) $x^2 + y^2 + 32x + 4y + 235 = 0$

CR0006

7. The radius of the circle passing through the vertices of the triangle ABC, is



(A) $\frac{8\sqrt{15}}{5}$ (B) $\frac{3\sqrt{15}}{5}$

(C) $3\sqrt{5}$ (D) $3\sqrt{2}$

CR0007

8. (6, 0), (0, 6) and (7, 7) are the vertices of a triangle. The circle inscribed in the triangle has the equation

(A) $x^2 + y^2 - 9x + 9y + 36 = 0$ (B) $x^2 + y^2 - 9x - 9y + 36 = 0$

(C) $x^2 + y^2 + 9x - 9y + 36 = 0$ (D) $x^2 + y^2 - 9x - 9y - 36 = 0$

CR0008

9. The line joining (5, 0) to $(10\cos\theta, 10\sin\theta)$ is divided internally in the ratio 2 : 3 at P. If θ varies then the locus of P is :

(A) a pair of straight lines (B) a circle

(C) a straight line (D) a second degree curve which is not a circle

CR0009

10. The locus of the center of the circles such that the point (2, 3) is the mid point of the chord $5x + 2y = 16$ is

(A) $2x - 5y + 11 = 0$ (B) $2x + 5y - 11 = 0$

(C) $2x + 5y + 11 = 0$ (D) none

CR0010

11. In the xy-plane, the length of the shortest path from (0, 0) to (12, 16) that does not go inside the circle $(x - 6)^2 + (y - 8)^2 = 25$ is

(A) $10\sqrt{3}$ (B) $10\sqrt{5}$ (C) $10\sqrt{3}+\frac{5\pi}{3}$ (D) $10+5\pi$

CR0011

12. Tangents PA and PB are drawn to the circle $x^2 + y^2 = 4$, then the locus of the point P if the triangle PAB is equilateral, is equal to-

(A) $x^2 + y^2 = 16$ (B) $x^2 + y^2 = 8$ (C) $x^2 + y^2 = 64$ (D) $x^2 + y^2 = 32$

CR0012

13. The points $(x_1, y_1), (x_2, y_2), (x_1, y_2)$ and (x_2, y_1) are always

(A) collinear (B) concyclic

(C) vertices of a square (D) vertices of a rhombus

CR0013

14. Locus of all point P(x, y) satisfying $x^3 + y^3 + 3xy = 1$ consists of union of

(A) a line and an isolated point (B) a line pair and an isolated point

(C) a line and a circle (D) a circle and a isolated point.

CR0014

15. In the xy plane, the segment with end points (3, 8) and (–5, 2) is the diameter of the circle. The point (k, 10) lies on the circle for

(A) no value of k (B) exactly one integral k

(C) exactly one non integral k (D) two real values of k

CR0015

16. If a circle of constant radius 3k passes through the origin 'O' and meets co-ordinate axes at A and B then the locus of the centroid of the triangle OAB is -

(A) $x^2 + y^2 = (2k)^2$ (B) $x^2 + y^2 = (3k)^2$ (C) $x^2 + y^2 = (4k)^2$ (D) $x^2 + y^2 = (6k)^2$

CR0016

17. Consider the points P(2, 1); Q(0, 0); R(4, –3) and the circle S : $x^2 + y^2 - 5x + 2y - 5 = 0$

(A) exactly one point lies outside S (B) exactly two points lie outside S

(C) all the three points lie outside S (D) none of the point lies outside S

CR0017

18. B and C are fixed points having co-ordinates (3, 0) and (–3, 0) respectively. If the vertical angle BAC is 90°, then the locus of the centroid of the ΔABC has the equation :

(A) $x^2 + y^2 = 1$ (B) $x^2 + y^2 = 2$ (C) $9(x^2 + y^2) = 1$ (D) $9(x^2 + y^2) = 4$

CR0018

19. The angle between the two tangents from the origin to the circle $(x - 7)^2 + (y + 1)^2 = 25$ equals

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{3}$ (C) $\frac{\pi}{2}$ (D) $\frac{\pi}{4}$

CR0019

20. Tangents are drawn from (4, 4) to the circle $x^2 + y^2 - 2x - 2y - 7 = 0$ to meet the circle at A and B. The length of the chord AB is

(A) $2\sqrt{3}$ (B) $3\sqrt{2}$ (C) $2\sqrt{6}$ (D) $6\sqrt{2}$

CR0020

21. The area of the quadrilateral formed by the tangents from the point (4, 5) to the circle $x^2 + y^2 - 4x - 2y - 11 = 0$ with the pair of radii through the points of contact of the tangents is :

(A) 4 sq. units (B) 8 sq. units (C) 6 sq. units (D) none

CR0021

22. If L_1 and L_2 are the length of the tangent from (0, 5) to the circles $x^2 + y^2 + 2x - 4 = 0$ and $x^2 + y^2 - 2x - y + 1 = 0$ then

(A) $L_1 = 2L_2$ (B) $L_2 = 2L_1$ (C) $L_1 = L_2$ (D) $L_1^2 = L_2$

CR0022

23. From (3, 4) chords are drawn to the circle $x^2 + y^2 - 4x = 0$. The locus of the mid points of the chords is :

(A) $x^2 + y^2 - 5x - 4y + 6 = 0$ (B) $x^2 + y^2 + 5x - 4y + 6 = 0$

(C) $x^2 + y^2 - 5x + 4y + 6 = 0$ (D) $x^2 + y^2 - 5x - 4y - 6 = 0$

CR0023

24. Tangents are drawn to a unit circle with centre at the origin from each point on the line $2x + y = 4$. Then the equation to the locus of the middle point of the chord of contact is -

(A) $2(x^2 + y^2) = x + y$ (B) $2(x^2 + y^2) = x + 2y$ (C) $4(x^2 + y^2) = 2x + y$ (D) none

CR0024

25. Chord AB of the circle $x^2 + y^2 = 100$ passes through the point (7, 1) and subtends an angle of 60° at the circumference of the circle. If m_1 and m_2 are the slopes of two such chords then the value of m_1m_2, is

(A) –1 (B) 1 (C) 7/12 (D) –3

CR0025

26. Combined equation to the pair of tangents drawn from the origin to the circle $x^2 + y^2 + 4x + 6y + 9 = 0$ is

(A) $3(x^2 + y^2) = (x + 2y)^2$ (B) $2(x^2 + y^2) = (3x + y)^2$

(C) $9(x^2 + y^2) = (2x + 3y)^2$ (D) $x^2 + y^2 = (2x + 3y)^2$

CR0026

27. Sum of the abscissa and ordinate of the centre of the circle touching the line $3x + y + 2 = 0$ at the point (–1,1) and passing through the point (3,5) is-

(A) 2 (B) 3 (C) 4 (D) 5

CR0027

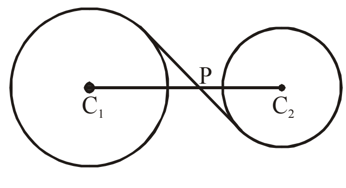
28. A circle of radius 5 is tangent to the line $4x – 3y = 18$ at M(3,–2) and lies above the line. The equation of the circle, is-

(A) $x^2 + y^2 – 6x + 4y – 12 = 0$ (B) $x^2 + y^2 + 2x – 2y – 3 = 0$

(C) $x^2 + y^2 + 2x – 2y – 23 = 0$ (D) $x^2 + y^2 + 6x + 4y – 12 = 0$

CR0028

29. In the figure given, two circles with centres C_1 and C_2 are 35 units apart, i.e. $C_1C_2 = 35$. The radii of the circles with centres C_1 and C_2 are 12 and 9 respectively. If P is the intersection of C_1C_2 and a common internal tangent to the circles, then $l(C_1P)$ equals-



(A) 18 (B) 20 (C) 12 (D) 15

CR0029

30. Let C_1 and C_2 are circles defined by $x^2 + y^2 – 20x + 64 = 0$ and $x^2 + y^2 + 30x + 144 = 0$.

The length of the shortest line segment PQ that is tangent to C_1 at P and to C_2 at Q is -

(A) 15 (B) 18 (C) 20 (D) 24

CR0030

EXERCISE (O-2)

Multiple Correct Answer Type

1. $\dfrac{x-x_1}{\cos\theta}=\dfrac{y-y_1}{\sin\theta}=r$, represents : (Where x_1, y_1 are constant)

(A) equation of a straight line, if θ is constant and r is variable

(B) equation of a circle, if r is constant and θ is a variable

(C) a straight line passing through a fixed point and having a known slope

(D) a circle with a known centre and a given radius.

CR0039

2. Which of the following lines have the intercepts of equal lengths on the circle, $x^2+y^2-2x+4y=0$?

(A) $3x-y=0$ (B) $x+3y=0$

(C) $x+3y+10=0$ (D) $3x-y-10=0$

CR0038

3. Consider two circles $C_1 : x^2+y^2-1=0$ and $C_2 : x^2+y^2-2=0$. Let A(1,0) be a fixed point on the circle C_1 and B be any variable point on the circle C_2. The line BA meets the curve C_2 again at C. Which of the following alternative(s) is/are correct ?

(A) $OA^2+OB^2+BC^2 \in [7, 11]$, where O is the origin.

(B) $OA^2+OB^2+BC^2 \in [4, 7]$, where O is the origin.

(C) Locus of midpoint of AB is a circle of radius $\dfrac{1}{\sqrt{2}}$.

(D) Locus of midpoint of AB is a circle of area $\dfrac{\pi}{2}$.

CR0163

4. One of the diameter of the circle circumscribing the rectangle ABCD is $x-3y+1=0$.

If two verticles of rectangle are the points (–2, 5) and (6, 5) respectively, then which of the following hold(s) good?

(A) Area of rectangle ABCD is 64 square units.

(B) Centre of circle is (2, 1)

(C) The other two vertices of the rectangle are (–2, –3) and (6, –3)

(D) Equation of sides are $x=-2$, $y=-3$, $x=5$ and $y=6$.

CR0164

5. Three concentric circles of which the biggest is $x^2+y^2=1$, have their radii in A.P. If the line $y=x+1$ cuts all the circles in real and distinct points. The permissible values of common difference of A.P. is/are

(A) 0.4 (B) 0.6 (C) 0.01 (D) 0.1

R0165

6. If $4\ell^2 - 5m^2 + 6\ell + 1 = 0$, then line $\ell x + my + 1 = 0$ touches a definite circle, then which of the following is/are true.

(A) Centre (0, 3) (B) centre (3, 0) (C) Radius $\sqrt{5}$ (D) Radius 5

CR0166

7. The equation of circles passing through (3, –6) touching both the axes is

(A) $x^2 + y^2 - 6x + 6y + 9 = 0$ (B) $x^2 + y^2 + 6x - 6y + 9 = 0$

(C) $x^2 + y^2 + 30x - 30y + 225 = 0$ (D) $x^2 + y^2 - 30x + 30y + 225 = 0$

CR0167

8. Tangents are drawn to the circle $x^2 + y^2 = 50$ from a point 'P' lying on the x-axis. These tangents meet the y-axis at points 'P_1' and 'P_2'. Possible coordinates of 'P' so that area of triangle PP_1P_2 is minimum, is/are

(A) (10, 0) (B) $(10\sqrt{2}, 0)$ (C) (–10, 0) (D) $(-10\sqrt{2}, 0)$

CR0168

9. $x^2 + y^2 = a^2$ and $(x - 2a)^2 + y^2 = a^2$ are two equal circles touching each other. Find the equation of circle (or circles) of the same radius touching both the circles

(A) $x^2 + y^2 + 2ax + 2\sqrt{3}\,ay + 3a^2 = 0$ (B) $x^2 + y^2 - 2ax + 2\sqrt{3}\,ay + 3a^2 = 0$

(C) $x^2 + y^2 + 2ax - 2\sqrt{3}\,ay + 3a^2 = 0$ (D) $x^2 + y^2 - 2ax - 2\sqrt{3}\,ay + 3a^2 = 0$

CR0169

10. A family of linear functions is given by $f(x) = 1 + c(x + 3)$ where $c \in R$. If a member of this family meets a unit circle centred at origin in two coincident points then 'c' can be equal to

(A) –3/4 (B) 0 (C) 3/4 (D) 1

CR0040

11. The equations of the tangents drawn from the origin to the circle, $x^2 + y^2 - 2rx - 2hy + h^2 = 0$ are :

(A) $x = 0$ (B) $y = 0$

(C) $(h^2 - r^2)x - 2rhy = 0$ (D) $(h^2 - r^2)x + 2rhy = 0$

CR0041

12. Tangents PA and PB are drawn to the circle $S \equiv x^2 + y^2 - 2y - 3 = 0$ from the point P(3,4). Which of the following alternative(s) is/are correct ?

(A) The power of point P(3,4) with respect to circle S = 0 is 14.

(B) The angle between tangents from P(3,4) to the circle S = 0 is $\frac{\pi}{3}$

(C) The equation of circumcircle of ΔPAB is x^2 +7\` $y^2 - 3x - 5y + 4 = 0$

(D) The area of quadrilateral PACB is $3\sqrt{7}$ square units where C is the centre of circle S = 0.

CR0042

13. Consider the circles $C_1 : x^2 + y^2 = 16$ and $C_2 : x^2 + y^2 - 12x + 32 = 0$. Which of the following statement is/are correct ?

(A) Number of common tangent to these circles is 3.

(B) The point P with coordinates (4,1) lies outside the circle C_1 and inside the circle C_2.

(C) Their direct common tangent intersect at (12,0).

(D) Slope of their radical axis is not defined.

CR0043

14. Which of the following is/are True ?

The circles $x^2 + y^2 - 6x - 6y + 9 = 0$ and $x^2 + y^2 + 6x + 6y + 9 = 0$ are such that -

(A) they do not intersect

(B) they touch each other

(C) their exterior common tangents are parallel.

(D) their interior common tangents are perpendicular.

CR0044

15. Consider the circles $S_1 : x^2 + y^2 = 4$ and $S_2 : x^2 + y^2 - 2x - 4y + 4 = 0$ which of the following statements are correct ?

(A) Number of common tangents to these circles is 2.

(B) If the power of a variable point P w.r.t. these two circles is same then P moves on the line $x + 2y - 4 = 0$

(C) Sum of the y-intercepts of both the circles is 6.

(D) The circles S_1 and S_2 are orthogonal.

CR0045

16. For the circles $x^2 + y^2 - 10x + 16y + 89 - r^2 = 0$ and $x^2 + y^2 + 6x - 14y + 42 = 0$ which of the following is/are true.

(A) Number of integral values of r are 14 for which circles are intersecting.

(B) Number of integral values of r are 9 for which circles are intersecting.

(C) For r equal to 13 number of common tangents are 3.

(D) For r equal to 21 number of common tangents are 2.

CR0170

17. Which of the following statement(s) is/are correct with respect to the circles $S_1 \equiv x^2 + y^2 - 4 = 0$ and $S_2 \equiv x^2 + y^2 - 2x - 4y + 4 = 0$?

(A) S_1 and S_2 intersect at an angle of 90°.

(B) The point of intersection of the two circle are (2, 0) and $\left(\frac{6}{5}, \frac{8}{5}\right)$.

(C) Length of the common of chord of S_1 and S_2 is $\frac{4}{\sqrt{5}}$.

(D) The point (2, 3) lies outside the circles S_1 and S_2.

CR0171

18. Two circles of radii r_1 and r_2 are both touching the coordinate axes and intersecting each other orthogonally. The value of r_1/r_2 equals -

(A) $2+\sqrt{3}$ (B) $\sqrt{3}+1$ (C) $2-\sqrt{3}$ (D) $2+\sqrt{5}$

CR0035

19. Two circles $x^2 + y^2 + px + py - 7 = 0$ and $x^2 + y^2 - 10x + 2py + 1 = 0$ intersect each other orthogonally then the value of p is -

(A) 1 (B) 2 (C) 3 (D) 5

CR0046

20. If the circle C_1: $x^2 + y^2 = 16$ intersects another circle C_2 of radius 5 in such a manner that the common chord is of maximum length and has a slope equal to 3/4, then the co ordinates of the centre of C_2 are:

(A) $\left(\frac{9}{5},\frac{12}{5}\right)$ (B) $\left(\frac{9}{5},\frac{-12}{5}\right)$ (C) $\left(\frac{-9}{5},\frac{-12}{5}\right)$ (D) $\left(\frac{-9}{5},\frac{+12}{5}\right)$

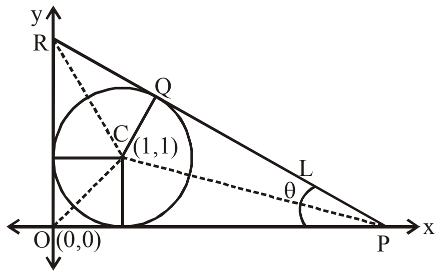
CR0172

EXERCISE (O-3)

Linked Comprehension Type

Paragraph for Question nos. 1 to 3

In the diagram as shown, a circle is drawn with centre C(1, 1) and radius 1 and a line L. The line L is tangential to the circle at Q. Further L meet the y-axis at R and the x-axis at P is such a way that the angle OPQ equals θ where $0 < \theta < \frac{\pi}{2}$.



1. The coordinates of Q are

(A) $(1 + \cos\theta, 1 + \sin\theta)$ (B) $(\sin\theta, \cos\theta)$

(C) $(1 + \sin\theta, \cos\theta)$ (D) $(1 + \sin\theta, 1 + \cos\theta)$

CR0047

2. Equation of the line PR is

(A) $x\cos\theta + y\sin\theta = \sin\theta + \cos\theta + 1$ (B) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta - 1$

(C) $x\sin\theta + y\cos\theta = \cos\theta + \sin\theta + 1$ (D) $x\tan\theta + y = 1 + \cot\left(\frac{\theta}{2}\right)$

CR0047

3. If the area bounded by the circle, the x-axis and PQ is $A(\theta)$, then $A\left(\frac{\pi}{4}\right)$ equals

(A) $\sqrt{2} + 1 - \frac{3\pi}{8}$ (B) $\sqrt{2} - 1 + \frac{3\pi}{8}$ (C) $\sqrt{2} + 1 + \frac{\pi}{8}$ (D) $\sqrt{2} - 1 + \frac{\pi}{8}$

CR0047

Paragraph for question Nos. 4 and 5

Consider the circle $S : x^2 + y^2 - 4x - 1 = 0$ and the line $L : y = 3x - 1$. If the line L cuts the circle at A & B.

4. Which of the following option(s) is/are **CORRECT** ?

(A) Length of the chord AB equal is $\sqrt{10}$ unit

(B) The angle subtended by the chord AB in the minor arc of S is $\frac{3\pi}{4}$

(C) Acute angle between the line L and the circle S is $\frac{\pi}{4}$

(D) Area bounded by L = 0 and minor arc of S = 0 is $\frac{5}{4}(\pi - 2)$ sq. unit

CR0173

5. If the equation of the circle on AB as diameter is of the form $x^2 + y^2 + ax + by + c = 0$ then

(A) magnitude of the vector $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$ has the value equal to $\sqrt{6}$

(B) $|a + b + c| = 4$ (C) $|a| + |b| + |c| = 5$ (D) $a^3 + b^3 + c^3 = 10$

CR0174

Paragraph for question Nos. 6 to 8

Let S_1, S_2, S_3 be the circles $x^2 + y^2 + 3x + 2y + 1 = 0$, $x^2 + y^2 - x + 6y + 5 = 0$ and $x^2 + y^2 + 5x - 8y + 15 = 0$, respectively then

6. Point from which length of tangents to these three circles is same is (α, β) then $(\alpha + \beta)$ is

CR0175

7. Equation of circle S_4 which cut orthogonally to all given circle is $x^2 + y^2 + ax + by + c = 0$ then $|a + b + c|$ is

CR0176

8. Radical centre of circles S_1, S_2, & S_4 is (p, q) then $|p + q|$ is

CR0177

9. Consider two circles C_1 of radius 'a' and C_2 of radius 'b' (b > a) both lying in the first quadrant and touching the coordinate axes. In each of the conditions listed in **List-I**, the ratio of b/a is given in **List-II.**

List-I		**List-II**	
(I)	C_1 and C_2 touch each other	(P)	$2+\sqrt{2}$
(II)	C_1 and C_2 are orthogonal	(Q)	3
(III)	C_1 and C_2 intersect so that the common chord is longest	(R)	$2+\sqrt{3}$
(IV)	C_2 passes through the centre of C_1	(S)	$3+2\sqrt{2}$
		(T)	$3-2\sqrt{2}$

(A) (I) → (S), (II) → (R), (III) → (Q), (IV) → (P)

(B) (I) → (S), (II) → (R), (III) → (T), (IV) → (Q)

(C) (I) → (R), (II) → (T), (III) → (Q), (IV) → (P)

(D) (I) → (P), (II) → (R), (III) → (S), (IV) → (T)

CR0114

10.

	Column - I		**Column - II**
(A)	If director circle of two given circles C_1 and C_2 of equal radii touches each other, then ratio of length of internal common tangent of C_1 and C_2 to their radii equals to	(P)	13
(B)	Let two circles having radii r_1 and r_2 are orthogonal to each other. If length of their common chord is k times the square root of hormonic mean between squares of their radii, then k^4 equals to	(Q)	7
(C)	The axes are translated so that the new equation of the circle $x^2 + y^2 - 5x + 2y - 5 = 0$ has no first degree terms and the new equation $x^2 + y^2 = \frac{\lambda^2}{4}$, then a value of λ is	(R)	4
(D)	The number of integral points which lie on or inside the circle $x^2 + y^2 = 4$ is	(S)	2

CR0178

EXERCISE (O-4)

Numerical Grid Type

1. A circle with center in the first quadrant is tangent to $y = x + 10$, $y = x - 6$, and the y-axis. Let (h, k) be the center of the circle. If the value of $(h + k) = a + b\sqrt{a}$ where $\sqrt{a}$ is a surd, find the value of a + b.

CR0105

2. If the points $(\lambda, -\lambda)$ lies inside the circle $x^2 + y^2 - 4x + 2y - 8 = 0$, then number of integers in the range of λ is

CR0068

3. One of the diameters of the circle circumscribing the rectangle ABCD is $4y = x + 7$. If A & B are the points (–3, 4) & (5,4) respectively, then find the area of the rectangle. (in square unit)

CR0058

4. Circles C_1 and C_2 are externally tangent and they are both internally tangent to the circle C_3. The radii of C_1 and C_2 are 4 and 10, respectively and the centres of the three circles are collinear. A chord of C_3 is also a common internal tangent of C_1 and C_2. Given that the length of the chord is $\frac{m\sqrt{n}}{p}$ where *m*, *n* and *p* are positive integers, *m* and *p* are relatively prime and *n* is not divisible by the square of any prime, find the value of $(m + n + p)$.

CR0083

5. Consider a circle S with centre at the origin and radius 4. Four circles A, B, C and D each with radius unity and centres (–3, 0), (–1, 0), (1, 0) and (3, 0) respectively are drawn. A chord PQ of the circle S touches the circle B and passes through the centre of the circle C. If the length of this chord can be expressed as $\sqrt{x}$, find x.

CR0084

6. A line with gradient 2 is passing through the point P(1, 7) and touches the circle $x^2 + y^2 + 16x + 12y + c = 0$ at the point Q. If (a, b) are the coordinates of the point Q, then find the value of $(7a + 7b + c)$.

CR0079

7. Real number x, y satisfies $x^2 + y^2 = 1$. If the maximum and minimum value of the expression $z = \frac{4-y}{7-x}$ are M and m respectively, then find the value $(2M + 6m)$.

CR0103

8. Let $S_1 = 0$ and $S_2 = 0$ be two circles intersecting at P (6, 4) and both are tangent to x-axis and line $y = mx$ (where $m > 0$). If product of radii of the circles $S_1 = 0$ and $S_2 = 0$ is $\frac{52}{3}$, then the value of m^2 is

CR0106

9. Through a given point P(5, 2), secants are drawn to cut the circle $x^2 + y^2 = 25$ at points $A_1(B_1)$, $A_2(B_2)$, $A_3(B_3)$, $A_4(B_4)$ and $A_5(B_5)$ such that $PA_1 + PB_1 = 5$, $PA_2 + PB_2 = 6$, $PA_3 + PB_3 = 7$, $PA_4 + PB_4 = 8$ and $PA_5 + PB_5 = 9$. Find the value of $\sum_{i=1}^{5} PA_i^2 + \sum_{i=1}^{5} PB_i^2$.

[**Note :** $A_r(B_r)$ denotes that the line passing through P(5, 2) meets the circle $x^2 + y^2 = 25$ at two points A_r and B_r.]

CR0107

10. If the circle $x^2 + y^2 + 4x + 22y + a = 0$ bisects the circumference of the circle $x^2 + y^2 - 2x + 8y - b = 0$ (where a, b > 0), then find the maximum value of (ab).

CR0102

EXERCISE (JM)

1. The number of common tangents to the circle
$x^2 + y^2 - 4x - 6y - 12 = 0$ and $x^2 + y^2 + 6x + 18y + 26 = 0$, is : **[JEE(Main)-2015]**

(1) 3 (2) 4 (3) 1 (4) 2

CR0125

2. If one of the diameters of the circle, given by the equation, $x^2 + y^2 - 4x + 6y - 12 = 0$, is a chord of a circle S, whose centre is at (–3, 2), then the radius of S is :- **[JEE(Main)-2016]**

(1) 10 (2) $5\sqrt{2}$ (3) $5\sqrt{3}$ (4) 5

CR0126

3. The centres of those circles which touch the circle, $x^2 + y^2 - 8x - 8y - 4 = 0$, externally and also touch the x-axis, lie on :- **[JEE(Main)-2016]**

(1) A parabola (2) A circle
(3) An ellipse which is not a circle (4) A hyperbola

CR0127

4. Three circles of radii a, b, c(a < b < c) touch each other externally. If they have x-axis as a common tangent, then : **[JEE(Main)-2019]**

(1) $\frac{1}{\sqrt{a}} = \frac{1}{\sqrt{b}} + \frac{1}{\sqrt{c}}$ (2) a, b, c are in A.P.

(3) $\sqrt{a}, \sqrt{b}, \sqrt{c}$ are in A.P. (4) $\frac{1}{\sqrt{b}} = \frac{1}{\sqrt{a}} + \frac{1}{\sqrt{c}}$

CR0128

5. If a circle C passing through the point (4,0) touches the circle $x^2 + y^2 + 4x - 6y = 12$ externally at the point (1, –1), then the radius of C is : **[JEE(Main)-2019]**

(1) $\sqrt{57}$ (2) 4 (3) $2\sqrt{5}$ (4) 5

CR0129

6. If the area of an equilateral triangle inscribed in the circle, $x^2 + y^2 + 10x + 12y + c = 0$ is $27\sqrt{3}$ sq. units then c is equal to : **[JEE(Main)-2019]**

(1) 20 (2) 25 (3) 13 (4) –25

CR0130

7. A square is inscribed in the circle $x^2 + y^2 - 6x + 8y - 103 = 0$ with its sides parallel to the corrdinate axes. Then the distance of the vertex of this square which is nearest to the origin is :-**[JEE(Main)-2019]**

(1) 13 (2) $\sqrt{137}$ (3) 6 (4) $\sqrt{41}$

CR0131

8. A circle cuts a chord of length 4a on the x-axis and passes through a point on the y-axis, distant 2b from the origin. Then the locus of the centre of this circle, is :- **[JEE(Main)-2019]**

(1) A hyperbola (2) A parabola (3) A straight line (4) An ellipse

CR0132

9. If a variable line, $3x + 4y - \lambda = 0$ is such that the two circles $x^2 + y^2 - 2x - 2y + 1 = 0$ and $x^2 + y^2 - 18x - 2y + 78 = 0$ are on its opposite sides, then the set of all values of λ is the interval :- **[JEE(Main)-2019]**

(1) [12, 21] (2) (2, 17) (3) (23, 31) (4) [13, 23]

CR0133

10. The sum of the squares of the lengths of the chords intercepted on the circle, $x^2 + y^2 = 16$, by the lines, $x + y = n$, $n \in N$, where N is the set of all natural numbers, is : **[JEE(Main)-2019]**

(1) 320 (2) 160 (3) 105 (4) 210

CR0134

11. If a tangent to the circle $x^2 + y^2 = 1$ intersects the coordinate axes at distinct points P and Q, then the locus of the mid-point of PQ is **[JEE(Main)-2019]**

(1) $x^2 + y^2 - 2xy = 0$ (2) $x^2 + y^2 - 16x^2y^2 = 0$

(3) $x^2 + y^2 - 4x^2y^2 = 0$ (4) $x^2 + y^2 - 2x^2y^2 = 0$

CR0135

12. The common tangent to the circles $x^2 + y^2 = 4$ and $x^2 + y^2 + 6x + 8y - 24 = 0$ also passes through the point :- **[JEE(Main)-2019]**

(1) (–4, 6) (2) (6, –2) (3) (–6, 4) (4) (4, –2)

CR0136

13. Let the tangents drawn from the origin to the circle, $x^2 + y^2 - 8x - 4y + 16 = 0$ touch it at the points A and B. The $(AB)^2$ is equal to : **[JEE(Main)-2020]**

(1) $\frac{52}{5}$ (2) $\frac{32}{5}$ (3) $\frac{56}{5}$ (4) $\frac{64}{5}$

CR0137

14. If a line, $y = mx + c$ is a tangent to the circle, $(x - 3)^2 + y^2 = 1$ and it is perpendicular to a line L_1, where L_1 is the tangent to the circle, $x^2 + y^2 = 1$ at the point $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$, then **[JEE(Main)-2020]**

(1) $c^2 - 6c + 7 = 0$ (2) $c^2 + 6c + 7 = 0$ (3) $c^2 + 7c + 6 = 0$ (4) $c^2 - 7c + 6 = 0$

CR0138

15. If the curves, $x^2 - 6x + y^2 + 8 = 0$ and $x^2 - 8y + y^2 + 16 - k = 0$, $(k > 0)$ touch each other at a point, then the largest value of k is ______. **[JEE(Main)-2020]**

CR0139

16. If one of the diameters of the circle $x^2 + y^2 - 2x - 6y + 6 = 0$ is a chord of another circle 'C', whose center is at (2, 1), then its radius is ______ . **[JEE(Main)-2021]**

CR0179

17. If the locus of the mid-point of the line segment from the point (3, 2) to a point on the circle, $x^2 + y^2 = 1$ is a circle of radius r, then r is equal to : **[JEE(Main)-2021]**

(1) 1 (2) $\frac{1}{2}$ (3) $\frac{1}{3}$ (4) $\frac{1}{4}$

CR0180

18. Let the normals at all the points on a given curve pass through a fixed point (a, b). If the curve passes through (3, –3) and $\left(4,-2\sqrt{2}\right)$, and given that $a-2\sqrt{2}\,b=3$, then $(a^2 + b^2 + ab)$ is equal to ______.

[JEE(Main)-2021]

CR0181

19. Let the lengths of intercepts on x-axis and y-axis made by the circle $x^2 + y^2 + ax + 2ay + c = 0$, $(a < 0)$ be $2\sqrt{2}$ and $2\sqrt{5}$, respectively. Then the shortest distance from origin to a tangent to this circle which is perpendicular to the line $x + 2y = 0$, is euqal to : **[JEE(Main)-2021]**

(1) $\sqrt{11}$ (2) $\sqrt{7}$ (3) $\sqrt{6}$ (4) $\sqrt{10}$

CR0182

20. The minimum distance between any two points P_1 and P_2 while considering point P_1 on one circle and point P_2 on the other circle for the given circles' equations

$x^2 + y^2 -10x - 10y + 41 = 0$

$x^2 + y^2 - 24x - 10y + 160 = 0$ is ______ . **[JEE(Main)-2021]**

CR0183

21. Let a circle $C : (x - h)^2 + (y - k)^2 = r^2$, $k > 0$, touch the x-axis at (1, 0). If the line $x + y = 0$ intersects the circle C at P and Q such that the length of the chord PQ is 2, then the value of $h + k + r$ is equal to _____. **[JEE(Main)-2022]**

CR0184

22. The set of values of k for which the circle $C : 4x^2+4y^2-12x+8y+k=0$ lies inside the fourth quadrant and the point $\left(1,-\frac{1}{3}\right)$ lies on or inside the circle C is : **[JEE(Main)-2022]**

(A) An empty set (B) $\left(6,\frac{95}{9}\right]$ (C) $\left[\frac{80}{9},10\right)$ (D) $\left(9,\frac{92}{9}\right]$

CR0185

23. A rectangle R with end points of the one of its sides as (1, 2) and (3, 6) is inscribed in a circle. If the equation of a diameter of the circle is $2x - y + 4 = 0$, then the area of R is ________.

[JEE(Main)-2022]

CR0186

24. Let the tangent to the circle $C_1 : x^2 + y^2 = 2$ at the point M(–1, 1) intersect the circle $C_2 : (x - 3)^2 + (y - 2)^2 = 5$, at two distinct points A and B. If the tangents to C_2 at the points A and B intersect at N, then the area of the triangle ANB is equal to : **[JEE(Main)-2022]**

(A) $\frac{1}{2}$ (B) $\frac{2}{3}$ (C) $\frac{1}{6}$ (D) $\frac{5}{3}$

CR0187

25. Let the abscissae of the two points P and Q on a circle be the roots of $x^2 - 4x - 6 = 0$ and the ordinates of P and Q be the roots of $y^2 + 2y - 7 = 0$. If PQ is a diameter of the circle $x^2 + y^2 + 2ax + 2by + c = 0$, then the value of $(a + b - c)$ is **[JEE(Main)-2022]**

(A) 12 (B) 13 (C) 14 (D) 16

CR0188

EXERCISE (JA)

1. Circle(s) touching x-axis at a distance 3 from the origin and having an intercept of length $2\sqrt{7}$ on y-axis is (are) **[JEE(Advanced) 2013, 3, (–1)]**

(A) $x^2 + y^2 - 6x + 8y + 9 = 0$ (B) $x^2 + y^2 - 6x + 7y + 9 = 0$

(C) $x^2 + y^2 - 6x - 8y + 9 = 0$ (D) $x^2 + y^2 - 6x - 7y + 9 = 0$

CR0145

2. A circle S passes through the point (0, 1) and is orthogonal to the circles $(x - 1)^2 + y^2 = 16$ and $x^2 + y^2 = 1$. Then :- **[JEE(Advanced)-2014, 3]**

(1) radius of S is 8 (B) radius of S is 7

(3) centre of S is (–7, 1) (D) centre is S is (–8, 1)

CR0146

3. Let RS be the diameter of the circle $x^2 + y^2 = 1$, where S is the point (1,0). Let P be a variable point (other than R and S) on the circle and tangents to the circle at S and P meet at the point Q. The normal to the circle at P intersects a line drawn through Q parallel to RS at point E. then the locus of E passes through the point(s)- **[JEE(Advanced)-2016, 4(–2)]**

(A) $\left(\frac{1}{3}, \frac{1}{\sqrt{3}}\right)$ (B) $\left(\frac{1}{4}, \frac{1}{2}\right)$ (C) $\left(\frac{1}{3}, -\frac{1}{\sqrt{3}}\right)$ (D) $\left(\frac{1}{4}, -\frac{1}{2}\right)$

CR0147

4. For how many values of p, the circle $x^2 + y^2 + 2x + 4y - p = 0$ and the coordinate axes have exactly three common points ? **[JEE(Advanced)-2017, 3]**

CR0148

Paragraph "X"

Let S be the circle in the xy-plane defined by the equation $x^2 + y^2 = 4$.

(*There are two question based on Paragraph "X", the question given below is one of them*)

5. Let E_1E_2 and F_1F_2 be the chord of S passing through the point $P_0(1, 1)$ and parallel to the x-axis and the y-axis, respectively. Let G_1G_2 be the chord of S passing through P_0 and having slop –1. Let the tangents to S at E_1 and E_2 meet at E_3, the tangents of S at F_1 and F_2 meet at F_3, and the tangents to S at G_1 and G_2 meet at G_3. Then, the points E_3, F_3 and G_3 lie on the curve **[JEE(Advanced)-2018,3(–1)]**

(A) $x + y = 4$ (B) $(x - 4)^2 + (y - 4)^2 = 16$

(C) $(x - 4)(y - 4) = 4$ (D) $xy = 4$

CR0149

Paragraph "X"

Let S be the circle in the xy-plane defined by the equation $x^2 + y^2 = 4$.

(*There are two question based on Paragraph "X", the question given below is one of them*)

6. Let P be a point on the circle S with both coordinates being positive. Let the tangent to S at P intersect the coordinate axes at the points M and N. Then, the mid-point of the line segment MN must lie on the curve - **[JEE(Advanced)-2018, 3(–1)]**

(A) $(x + y)^2 = 3xy$ (B) $x^{2/3} + y^{2/3} = 2^{4/3}$ (C) $x^2 + y^2 = 2xy$ (D) $x^2 + y^2 = x^2y^2$

CR0150

7. Let T be the line passing through the points P(–2, 7) and Q(2, –5). Let F_1 be the set of all pairs of circles (S_1, S_2) such that T is tangents to S_1 at P and tangent to S_2 at Q, and also such that S_1 and S_2 touch each other at a point, say, M. Let E_1 be the set representing the locus of M as the pair (S_1, S_2) varies in F_1. Let the set of all straight line segments joining a pair of distinct points of E_1 and passing through the point R(1, 1) be F_2. Let E_2 be the set of the mid-points of the line segments in the set F_2. Then, which of the following statement(s) is (are) TRUE ? **[JEE(Advanced)-2018, 4(–2)]**

(A) The point (–2, 7) lies in E_1

(B) The point $\left(\frac{4}{5}, \frac{7}{5}\right)$ does **NOT** lie in E_2

(C) The point $\left(\frac{1}{2}, 1\right)$ lies in E_2

(D) The point $\left(0, \frac{3}{2}\right)$ does **NOT** lie in E_1

CR0151

8. A line y = mx + 1 intersects the circle $(x - 3)^2 + (y + 2)^2 = 25$ at the points P and Q. If the midpoint of the line segment PQ has x-coordinate $-\frac{3}{5}$, then which one of the following options is correct ?

[JEE(Advanced)-2019, 3(–1)]

(1) $6 \le m < 8$ (2) $2 \le m < 4$

(3) $4 \le m < 6$ (4) $-3 \le m < -1$

CR0152

9. Let the point B be the reflection of the point A(2, 3) with respect to the line 8x – 6y – 23 = 0. Let Γ_A and Γ_B be circles of radii 2 and 1 with centres A and B respectively. Let T be a common tangent to the circles Γ_A and Γ_B such that both the circles are on the same side of T. If C is the point of intersection of T and the line passing through A and B, then the length of the line segment AC is _____ **[JEE(Advanced)-2019, 3(0)]**

CR0153

Answer the following by appropriately matching the lists based on the information given in the paragraph

Let the circles $C_1 : x^2 + y^2 = 9$ and $C_2 : (x-3)^2 + (y-4)^2 = 16$, intersect at the points X and Y. Suppose that another circle $C_3 : (x-h)^2 + (y-k)^2 = r^2$ satisfies the following conditions :

(i) centre of C_3 is collinear with the centres of C_1 and C_2

(ii) C_1 and C_2 both lie inside C_3, and

(iii) C_3 touches C_1 at M and C_2 at N.

Let the line through X and Y intersect C_3 at Z and W, and let a common tangent of C_1 and C_3 be a tangent to the parabola $x^2 = 8\alpha y$.

There are some expression given in the List-I whose values are given in List-II below :

	List-I		**List-II**
(I)	$2h + k$	(P)	6
(II)	$\dfrac{\text{Length of ZW}}{\text{Length of XY}}$	(Q)	$\sqrt{6}$
(III)	$\dfrac{\text{Area of triangle MZN}}{\text{Area of triangle ZMW}}$	(R)	$\dfrac{5}{4}$
(IV)	α	(S)	$\dfrac{21}{5}$
		(T)	$2\sqrt{6}$
		(U)	$\dfrac{10}{3}$

10. Which of the following is the only INCORRECT combination ? **[JEE(Advanced)-2019, 3(–1)]**

(1) (IV), (S) (2) (IV), (U) (3) (III), (R) (4) (I), (P)

CR0154

11. Which of the following is the only CORRECT combination ? **[JEE(Advanced)-2019, 3(–1)]**

(1) (II), (T) (2) (I), (S) (3) (I), (U) (4) (II), (Q)

CR0154

12. Let O be the centre of the circle $x^2 + y^2 = r^2$, where $r > \dfrac{\sqrt{5}}{2}$. Suppose PQ is a chord of this circle and the equation of the line passing through P and Q is $2x + 4y = 5$. If the centre of the circumcircle of the triangle OPQ lies on the line $x + 2y = 4$, then the value of r is _____ **[JEE(Advanced)-2020]**

CR0159

13. Consider a triangle whose two sides lie on the x-axis and the line $x + y + 1 = 0$. If the orthocenter of is (1, 1), then the equation of the circle passing through the vertices of the triangle is

[JEE(Advanced)-2021]

(A) $x^2 + y^2 - 3x + y = 0$ (B) $x^2 + y^2 + x + 3y = 0$

(C) $x^2 + y^2 + 2y - 1 = 0$ (D) $x^2 + y^2 + x + y = 0$

CR0160

Paragraph for Question 14 and 15

Let $M = \{(x, y) \in \mathbb{R} \times \mathbb{R} : x^2 + y^2 \le r^2\}$,

where $r > 0$. Consider the geometric progression $a_n = \frac{1}{2^{n-1}}$, $n = 1, 2, 3, \ldots$. Let $S_0 = 0$ and, for $n \ge 1$, let S_n denote the sum of the first n terms of this progression. For $n \ge 1$, let C_n denote the circle with center $(S_{n-1}, 0)$ and radius a_n, and D_n denote the circle with center (S_{n-1}, S_{n-1}) and radius a_n.

[JEE(Advanced)-2021]

14. Consider M with $r = \frac{1025}{513}$. Let k be the number of all those circles C_n that are inside M. Let l be the maximum possible number of circles among these k circles such that no two circles intersect. Then

(A) $k + 2l = 22$ (B) $2k + l = 26$ (C) $2k + 3l = 34$ (D) $3k + 2l = 40$

CR0161

15. Consider M with $r = \frac{(2^{199} - 1)\sqrt{2}}{2^{198}}$. The number of all those circles D_n that are inside M is

(A) 198 (B) 199 (C) 200 (D) 201

CR0162

16. Let ABC be the triangle with AB = 1, AC = 3 and $\angle BAC = \frac{\pi}{2}$. If a circle of radius $r > 0$ touches the sides AB, AC and also touches internally the circumcircle of the triangle ABC, then the value of r is ____________.

[JEE(Advanced)-2022]

CR0189

17. Let G be a circle of radius $R > 0$. Let $G_1, G_2, \ldots, G_n$ be n circles of equal radius $r > 0$. Suppose each of the n circles $G_1, G_2, \ldots, G_n$ touches the circle G externally. Also, for $i = 1, 2, \ldots, n-1$, the circle G_i touches G_{i+1} externally, and G_n touches G_1 externally. Then, which of the following statements is/are TRUE?

[JEE(Advanced)-2022]

(A) If $n = 4$, then $(\sqrt{2} - 1)r < R$ (B) If $n = 5$, then $r < R$

(C) If $n = 8$, then $(\sqrt{2} - 1)\, r < R$ (D) If $n = 12$, then $\sqrt{2}(\sqrt{3} + 1)\, r > R$

CR0190

ANSWER KEY

Do yourself - 1

1. Centre $\left(\frac{3}{4}, -\frac{5}{4}\right)$, Radius $\frac{3\sqrt{10}}{4}$

2. **(i)** $x^2 + y^2 + 10x + 12y = 39$; **(ii)** $x^2 + y^2 - 2ax + 2by = 2ab$

3. **(i)** $(2, 4); \sqrt{61}$; **(ii)** $\left(\frac{c}{\sqrt{1+m^2}}, \frac{mc}{\sqrt{1+m^2}}\right)$; c

4. $17(x^2 + y^2) + 2x - 44y = 0$

5. (a) $x^2 + y^2 - 6x \pm 6\sqrt{2}y + 9 = 0$; (b) $x^2 + y^2 + 4x - 10y + 4 = 0$; $x^2 + y^2 - 4x - 2y + 4 = 0$

6. $x^2 + y^2 + 6x - 2y - 51 = 0$ **8.** $x^2 + y^2 - hx - ky = 0$

9. $x = \frac{p}{2}(-1+\sqrt{2}\cos\theta)$; $y = \frac{p}{2}(-1+\sqrt{2}\sin\theta)$

10. **(i)** (11, 16), **(ii)** (11, 8), **(iii)** (11, 12)

11. **(i)** $x^2 + y^2 - ax - by = 0$; **(ii)** $x^2 + y^2 - 22x - 4y + 25 = 0$; **(iii)** $x^2 + y^2 - 5x - y + 4 = 0$

Do yourself - 2

1. (1, 2) lie inside the circle and the point (6, 0) lies outside the circle

2. min = 0, max = 6, power = 0

3. square of side 2; $x^2 + y^2 = 1$; $x^2 + y^2 = 2$

4. (a) 2; (b) (9, 3)

Do yourself - 3

1. $x\cos\alpha + y\sin\alpha = a(1 + \cos\alpha)$ **2.** $4x - 3y + 7 = 0$ & $4x - 3y - 43 = 0$

3. $5x + 12y = \pm 26$; $\left(\mp\frac{10}{13}, \mp\frac{24}{13}\right)$ **4.** 1

5. **(i)** $5x - 12y = 152$, **(ii)** $k = 40$ or -10 **6.** $a^2(x^2 + y^2) = 4x^2y^2$ **7.** 6

8. **(a)** $4x + 3y + 19 = 0$ and $4x + 3y - 31 = 0$;

(b) $12x - 5y + 8 = 0$ and $12x - 5y - 252 = 0$

(c) $x - \sqrt{3}y \pm 10 = 0$

9. (5, 1) & (–1, 5) **10.** $x^2 + y^2 = a^2 + b^2$; $r = \sqrt{a^2 + b^2}$

12. $x^2 + y^2 - 2x - 2y + 1 = 0$ **OR** $x^2 + y^2 - 42x + 38y - 39 = 0$

14. $x - y = 0$; $x + 7y = 0$ **15.** $2x - 2y - 3 = 0$

16. $4x - 3y - 25 = 0$ **OR** $3x + 4y - 25 = 0$

Do yourself - 4

1. $x + 2y = 1$ **2.** 1

Do yourself - 5

1. $4x + 7y + 10 = 0$ **2.** $\dfrac{405\sqrt{3}}{52}$ sq. units

3. (i) $3x - 4y = 21$; $4x + 3y = 3$; (ii) A(0, 1) and B (–1, – 6); (iii) 90°, $5\left(\sqrt{2} \pm 1\right)$ units

(iv) 25 sq. units, 12.5 sq. units; (v) $x^2 + y^2 + x + 5y - 6$, x-intercept = 5; y-intercept = 7

4. $x^2 + y^2 + gx + fy = 0$ **5.** (–25, 50)

Do yourself - 6

1. $5x - 4y + 26 = 0$ **2.** $x^2 + y^2 + gx + fy = 0$

3. $\left(\dfrac{1}{2}, -\dfrac{1}{2}\right)$, $x + y = 0$ **4.** $x + y = 2$

Do yourself - 7

1. $(x - h)^2 + (y - k)^2 = 2a^2$ **2.** 10 **3.** angle between the tangents = 90°

Do yourself - 8

2. $x^2 + y^2 + 4x - 7y + 5 = 0$

3. $7x^2 + 7y^2 - 10x - 10y - 12 = 0$

4. $2x^2 + 2y^2 - 18x - 22y + 69 = 0$ and $x^2 + y^2 - 2y - 15 = 0$

5. $x^2 + y^2 - x - 2y = 0$

6. (a) $x^2 + y^2 + 4x - 7y + 5 = 0$, (b) $7(x^2 + y^2) + 19x - 19y - 91 = 0$

7. $x^2 + y^2 - 6x + 4y = 0$ **OR** $x^2 + y^2 + 2x - 8y + 4 = 0$

8. $x^2 + y^2 + x - 6y + 3 = 0$

9. $\left(2, \dfrac{23}{3}\right)$

Do yourself - 9

1. $(x - 5)^2 + (y - 5)^2 = 25$ **2.** 4 **3.** B **4.** D

5. zero, zero **6.** $5x^2 + 5y^2 - 8x - 14y - 32 = 0$

Do yourself - 10

1. 135° or 45° **2.** $x + 2y = 2$ **3.** (1, 2) **4.** B

5. (1, 2) **6.** $4x^2 + 4y^2 + 6x + 10y - 1 = 0$ **7.** $a^{-2} + b^{-2} = c^{-1}$

Do yourself - 11

1. 18 **2.** A **3.** B **4.** A

5. $x^2 + y^2 - 4x - 2y + 3 = 0$ **6.** $x^2 + y^2 + 16x + 14y - 12 = 0$

7. (–4, 4) ; (– 1/2, 1/2) **8.** (a) $x^2 + y^2 + 4x - 6y = 0$; k = 1; (b) $x^2 + y^2 = 64$

9. $x^2 + y^2 - 16x - 18y - 4 = 0$

EXERCISE # O-1

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	D	B	A	A	B	D	A	B	B	A
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	C	A	B	A	B	A	D	A	C	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	C	A	C	A	C	C	C	B	C

EXERCISE # O-2

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B	A,B,C,D	A,C,D	A,B,C	C,D	B,C	A,D	A,C	B,D	A,B
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	A,C	A,C	A,C,D	A,C,D	A,B,D	A,C	A,C,D	A,C	B,C	B,D

EXERCISE # O-3

Que.	1	2	3	4	5	6	7	8	9
Ans.	D	C	A	A,B,C,D	A,B	5.00	24.00	2.20	A

Q.10	A	B	C	D
	S	R	Q	P

EXERCISE # O-4

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	10	4	32	19	63	4.00	4.00	3.00	215.00	625.00

EXERCISE # JEE-MAIN

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	1	3	1	1	4	2	4	2	1	4
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	3	2	4	2	36	3	2	9	3	1
Que.	21	22	23	24	25					
Ans.	7	D	16	C	A					

EXERCISE # JEE-ADVANCED

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,C	B,C	A,C	2	A	D	B,D	2	10.00	1
Que.	11	12	13	14	15	16	17			
Ans.	4	2	B	D	B	0.83 or 0.84	C,D			

Important Notes